DNA/RNA dynamics

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Module 2

Practical overview of microarray data analysis

1. Theoretical introduction
2. Practical tutorial
3. Up to you!

Module 2

Examination
Students will be divided into teams. Each team must prepare a **self-made team report** on **the analysis of a DNA methylation dataset** provided during the Course (**up to 10 scores, adding + in case of outstanding**).

Teams will be composed of 5-7 members

Each team will receive a different data filtering/pipeline.

Students cannot refuse the score assigned to the self-made team report.

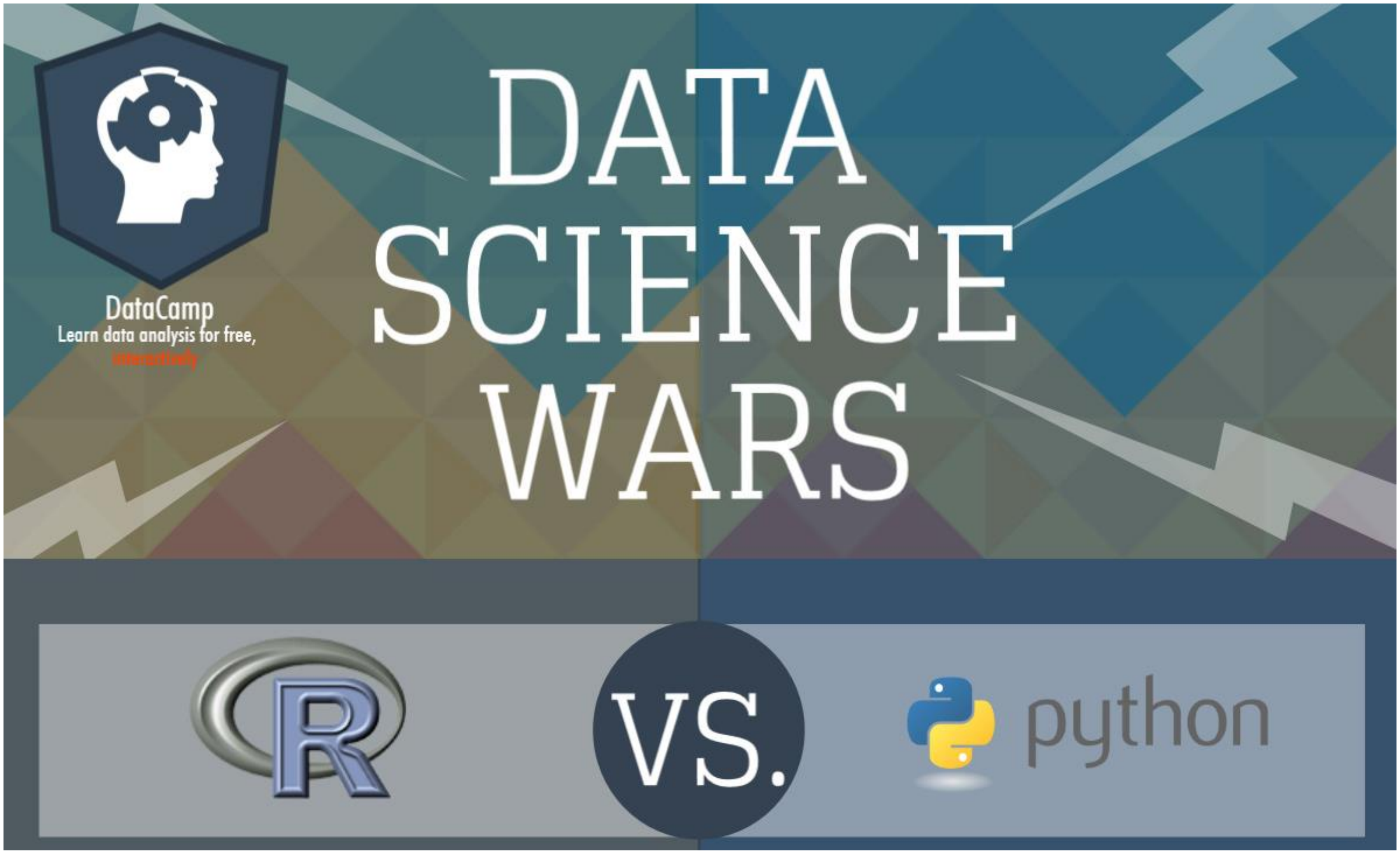
Module 2

TO BE DELIVERED ONE DAY BEFORE THE EXAM DATE.

DATES	Program	Classroom	Module II- team-made report
June 17 th -10.a.m	All module I program except scRNA-seq	Via Beverara, 121/1-UE1	**July 1 st**
July 2 nd – 12 p.m.	All module I program	to be determined	**July 1 st**
July 18th -12 p.m.	All module I program	to be determined	**July 17th**
Other dates	In agreement with students	to be determined	**one day before the date**

R vs python

A Data science war!



A few things about R

Why R?

- R is cross-platform and runs on Windows, Mac, and Linux (as well as more obscure systems).
- R provides a vast number of useful statistical tools, many of which have been painstakingly tested.
- R produces publication-quality graphics in a variety of formats.
- R plays well with FORTRAN, C, and scripts in many languages.
- R scales, making it useful for small and large projects. It is NOT Excel.
- R eschews the GUI.

Anecdote

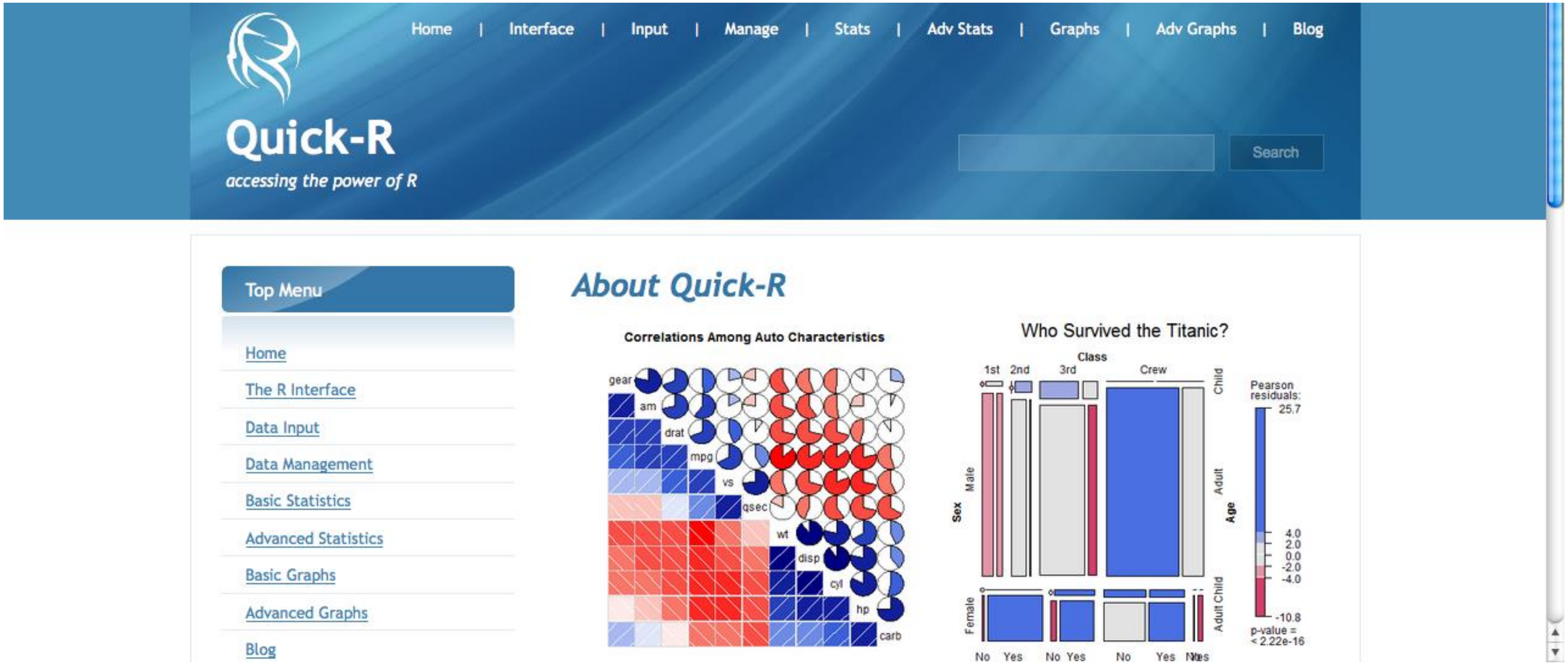
I can develop code for analysis on my Mac laptop. I can then install the *same* code on our 20k core cluster and run it in parallel on 100 samples, monitor the process, and then update a database with R when complete.

Why Not R?

- R cannot do everything.
- R is not always the "best" tool for the job.
- R will *not* hold your hand.
- The documentation can be opaque.
- R can drive you crazy (on a good day) or age you prematurely (on a bad one).
- Finding the right package to do the job you want to do can be challenging; worse, some contributed packages are unreliable.
- R eschews the GUI.

A few things about R

- http://www.statmethods.net



- **A Not-so-brief Introduction to R (Sean Davis)**

- **Intro2R (Alex Douglas)**
  https://alexd106.github.io/intro2R/index.html
  https://intro2r.com/

R and R packages

R is a free programming language and software environment for statistical computing and graphics.

A function is a set of statements organized together to perform a specific task

1 2 3 n

```
read.table(’’$PWD/myData’’, sep=’’\t’’, …)
```

function name

function arguments

R and R packages

R is a free programming language and software environment for statistical computing and graphics.

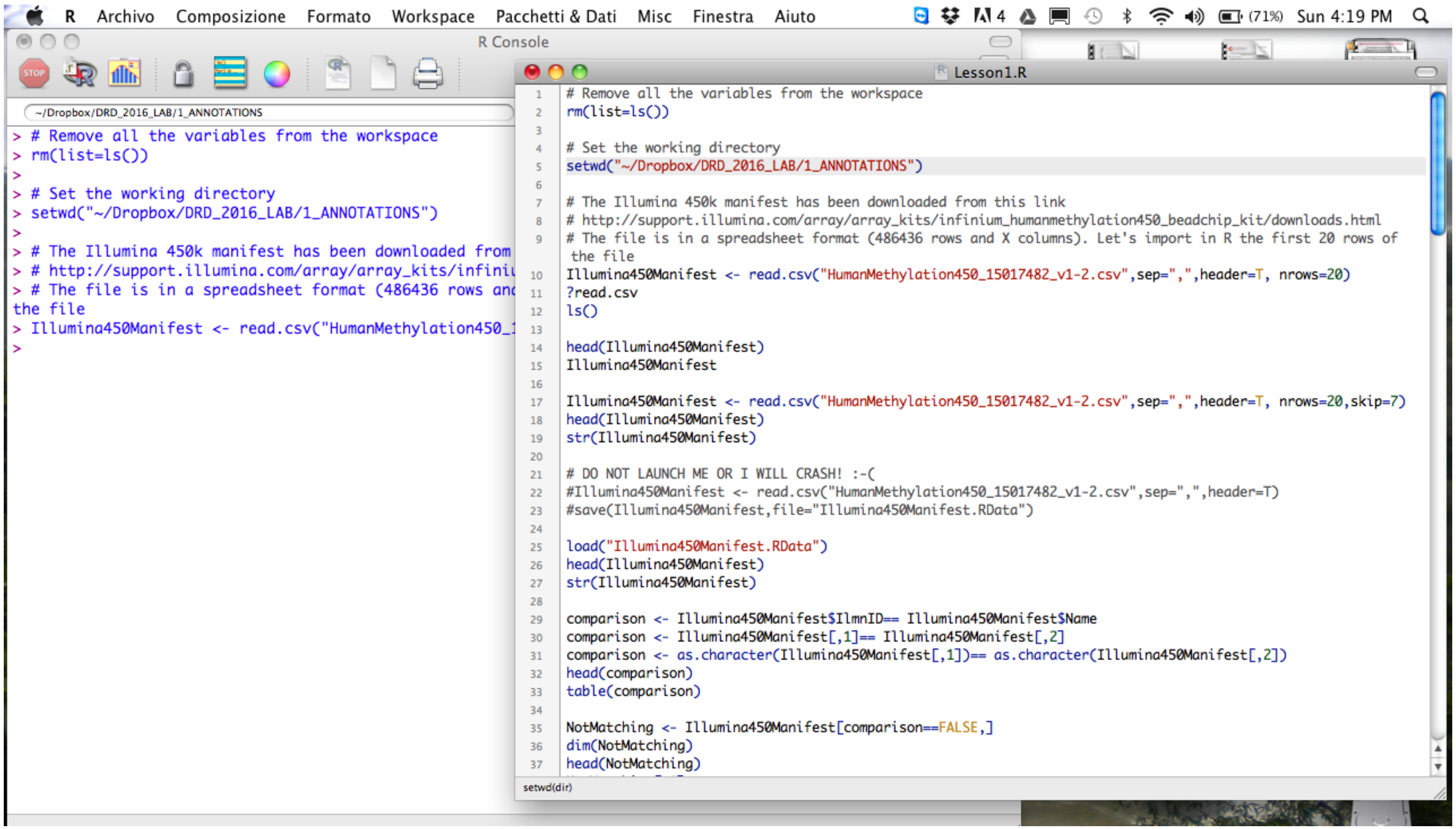
R Packages are add-on modules for R. They are collections of R functions, data, and compiled code in a well-defined format. The directory where packages are stored is called the **library**. R comes with a standard set of packages. Others are available for download and installation.

R packages are available at:
CRAN (Comprehensive R Archive Network) package repository

Bioconductor is a free, open source and open development software project which provides tools for the analysis and comprehension of high-throughput genomic data. Bioconductor is primarily based on R.

GitHub

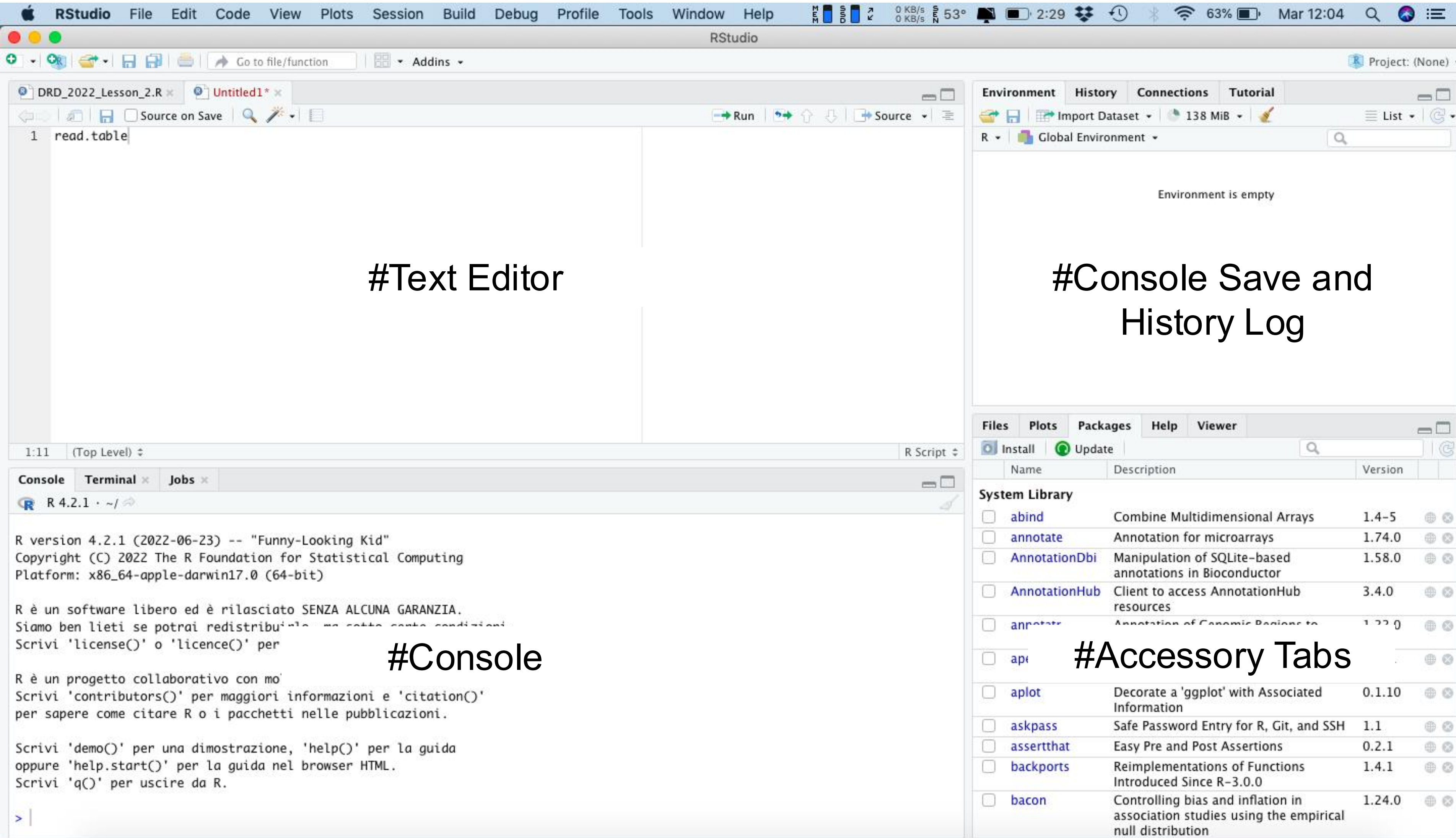
Rbase Console



For this course, I suggest you to type the code into a text editor/IDE

RStudio: https://www.rstudio.com/

R Editors and IDE



A few things about R: the workspace

- The workspace is your current R working environment
- It contains the R objects that you create: R objects can be vectors, matrices, data frames, lists, functions

- It can be saved using the command `save.image()`
- It can be removed using the command `rm(name_of_the_object)`
- `ls()` # list the objects in the current workspace
- `getwd()` # print the current working directory
- `setwd("~/Dropbox/mydirectory")` # change to mydirectory

Important Note to Windows Users:
R gets confused if you use a path in your code like:
`c:\mydocuments\myfile.txt`
This is because R sees "\" as an escape character.
Instead, use: `c:/mydocuments/myfile.txt` or
`c:\\mydocuments\\myfile.txt`

A few things about R: R objects

To create a new R object:

- Choose the name of the object, following the following rules:
  Any combination of letters, numbers, underscore, and "."
  Must not start with special characters (like $,%,*,+,-,#, etc),
  or alphabetic symbols with diacritical marks (like à,è,è,ò,
  etc)
  Must not start with numbers, underscore.

  R is case-sensitive. Spacing in your commands is not relevant.
  Text after the symbol # is ignored.

  For example:
  My_data, my_data, My_data1
  But Not 1_my_data

- Use the assignment operator **<-** to indicate what should be stored in the object; assignement is right to left

Note: The "<-" and "=" are both assignment operators, but it is clearer to use <-
(= can be confused with the equal operator ==)

A few things about R: R data types

Simple data types:

Numbers

Characters: for example, “ciao”

Logical: TRUE and FALSE

Factors: categorical variables

Missing: NA

Aggregated data types

Vector: one-dimensional data, single data type; In R, even a single value is a vector with length=1.

To create a vector, use the c() command

```
> v <- c(1,2,3,4,5)
```

List: list of data types, not all need to be the same type

Matrix: two-dimensional data, single data type

Data frame: two-dimensional data, multiple data types. In dataframes, row and columns have names

R factors

Subject	Treatment
subject_01	treatm_B
subject_02	treatm_A
subject_03	treatm_A
subject_04	treatm_C
subject_05	treatm_B
subject_06	treatm_C
subject_07	treatm_A
subject_08	treatm_B

Factors are nominal values

- Grouping
- Categorical variables

```
> # create a character vector of 8 subjects
> subjects_id <- c
("subject_01","subject_02","subject_03","subject_04","subject_05","subject_06","subject_07","subject_08")
> subjects_id
[1] "subject_01" "subject_02" "subject_03" "subject_04" "subject_05" "subject_06" "subject_07" "subject_08"
>
> # create a character vector of 3 treatments
> treatm <- c("treatm_B","treatm_A","treatm_A","treatm_C","treatm_B","treatm_C","treatm_A","treatm_B")
> treatm
[1] "treatm_B" "treatm_A" "treatm_A" "treatm_C" "treatm_B" "treatm_C" "treatm_A" "treatm_B"
>
> # convert to factor
> treatmf<-factor(treatm)
> treatmf
[1] treatm_B treatm_A treatm_A treatm_C treatm_B treatm_C treatm_A treatm_B
Levels: treatm_A treatm_B treatm_C
>
> levels(treatmf)
[1] "treatm_A" "treatm_B" "treatm_C"
> # convert factor back to character vector
> as.character(treatmf)
[1] "treatm_B" "treatm_A" "treatm_A" "treatm_C" "treatm_B" "treatm_C" "treatm_A" "treatm_B"
> # convert to numeric vector
> as.numeric(treatmf)
[1] 2 1 1 3 2 3 1 2
```

A few things about R: operators

Arithmetic Operators

Operator	Description
+	addition
-	subtraction
*	multiplication
/	division
^ or **	exponentiation

Logical Operators

Operator	Description
<	less than
<=	less than or equal to
>	greater than
>=	greater than or equal to
==	exactly equal to
!=	not equal to
!x	Not x
x | y	x OR y
x & y	x AND y
isTRUE(x)	test if X is TRUE

`x %in% y` returns TRUE if the left operand (x) occurs in the right operand (y); FALSE otherwise

A few things about R: accessors

When you want to subset of an object use []

Subsetting a vector: vector[index]

Subsetting a matrix/dataframe:

dataframe[index_row,index_column]

- A subset is usually the same type as the original object.
- You can use negative integers to indicate exclusion.

When you want to extract an element from many use [[]]

Vectors yield vectors with a single value; data frames give a column vector; for list, one element

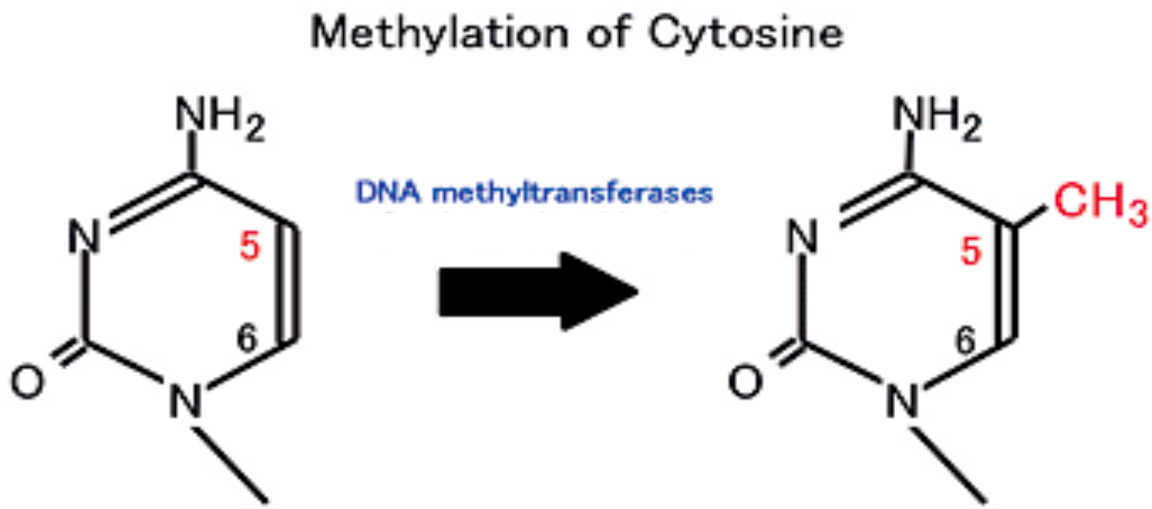
When you want to extract by name use $

You can indicate a column of a dataframe by its name by the $ operator

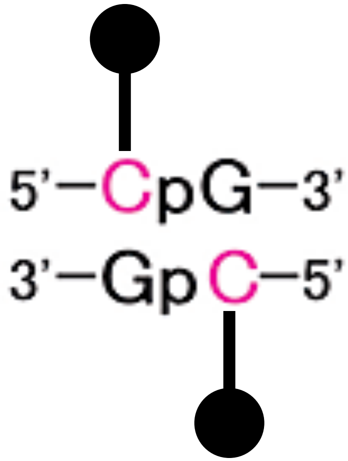
Exploratory functions for dataframes

dim() → dimensions of the data frame (number of rows and number of columns). The output is a vector.
nrow() and ncol() → number of rows and number of columns,
head() → first 6 observations
tail() → last 6 observations
str() → types of data for each column.
levels() → all of the categories or levels of a categorical variable
summary() → for each column reports a 5-number summary (continuous variables) or the class and the number of data in each level. (categorical variable). In addition, it provides the count of missing values (onyl for continuous variables).
table() → creates tabular results of categorical variables (also two-way table of co-occurrences)

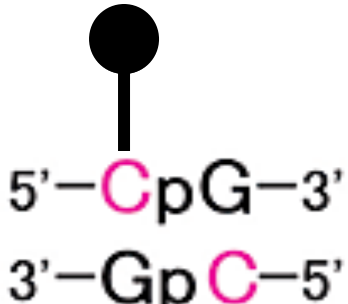
DNA methylation



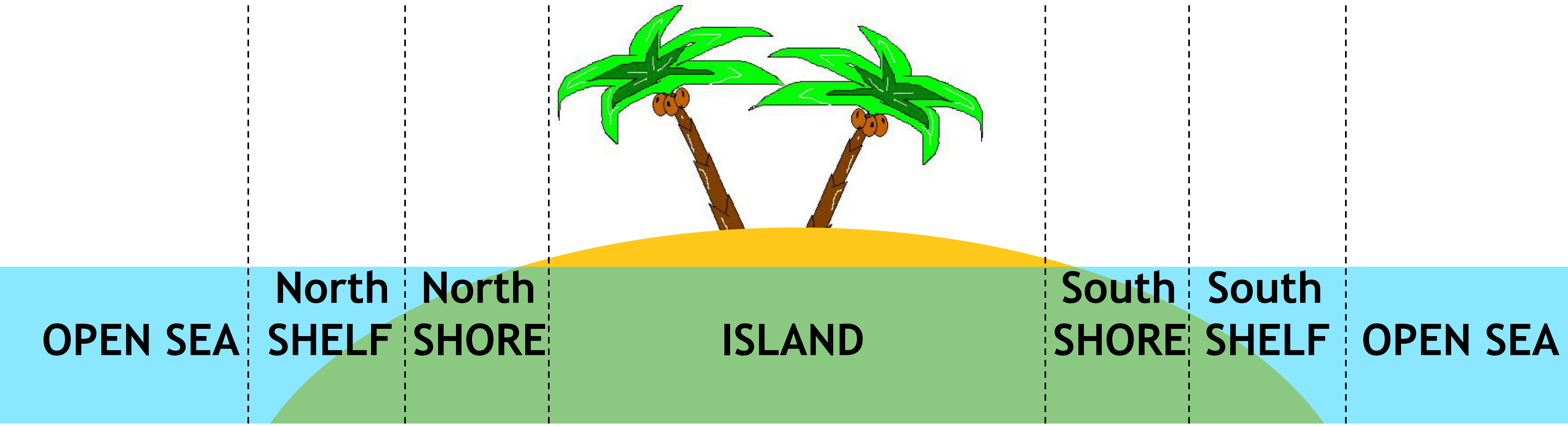
Both strands are methylated



Only one strand is methylated (hemimethylated DNA)



CpG islands, shores, shelves and open-sea CpGs



CpG islands (according to UCSC):

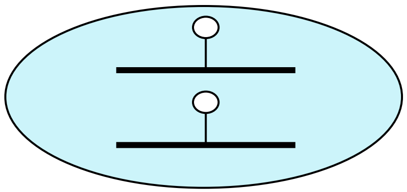
- GC content of 50% or greater
- length greater than 200 bp
- Observed CpG/expected CpG > 0.6

CpG shores: 0-2 kb from CpG island

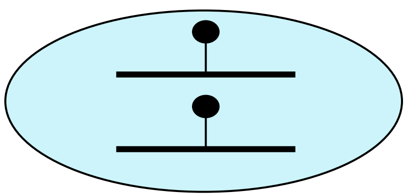
CpG shelves: 2-4 kb from CpG island

28 milion CpGs in human genome

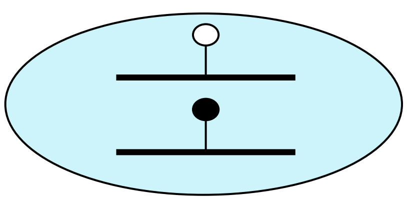
Measuring DNA methylation…



0% methylation

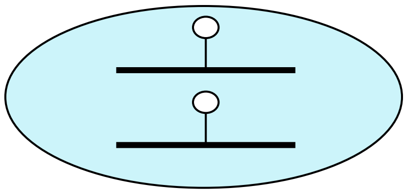


100% methylation

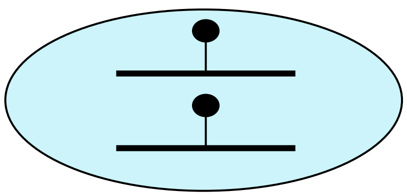


50% methylation

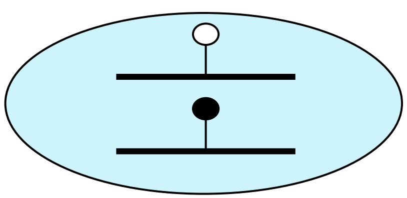
Measuring DNA methylation…



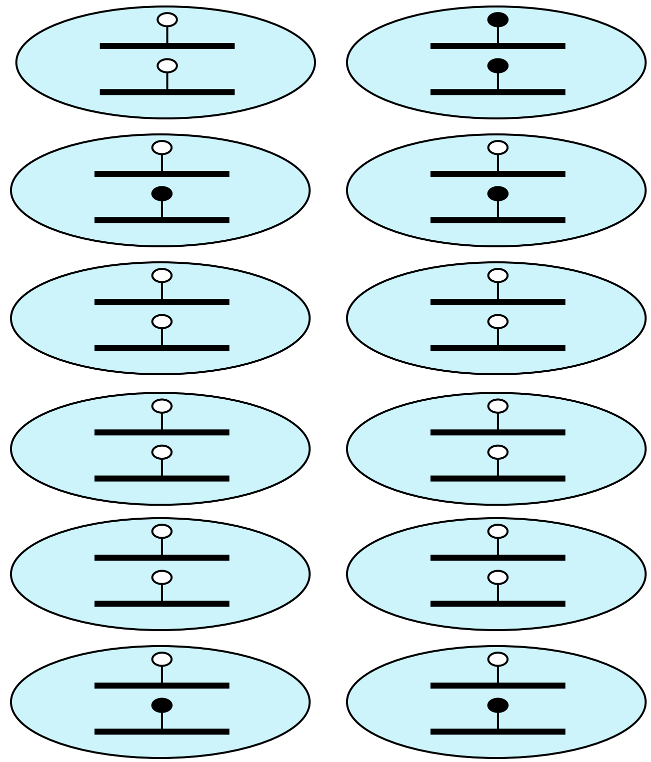
0% methylation



100% methylation



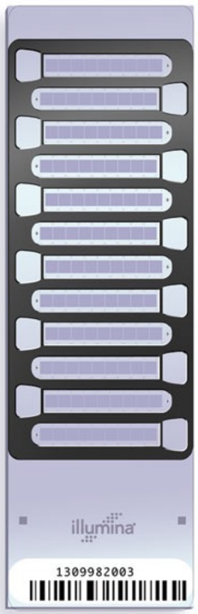
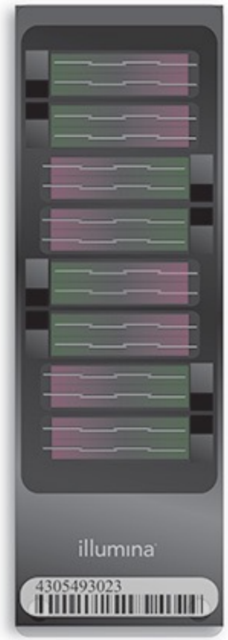
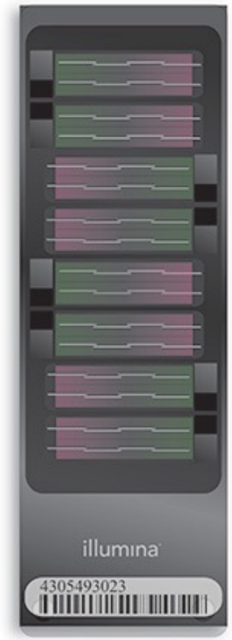
50% methylation



25% methylation

The value that I get is a mean of the methylation status of the population of DNA molecules

Infinium HumanMethylation family

	27K Bead Chip	450K Bead Chip	EPIC	EPIC v2
				
Release year	2009	2011	2016	2022
Probes	27578	485512	853307	937691

Infinium HumanMethylation450 BeadChip

From the *Illumina* website:

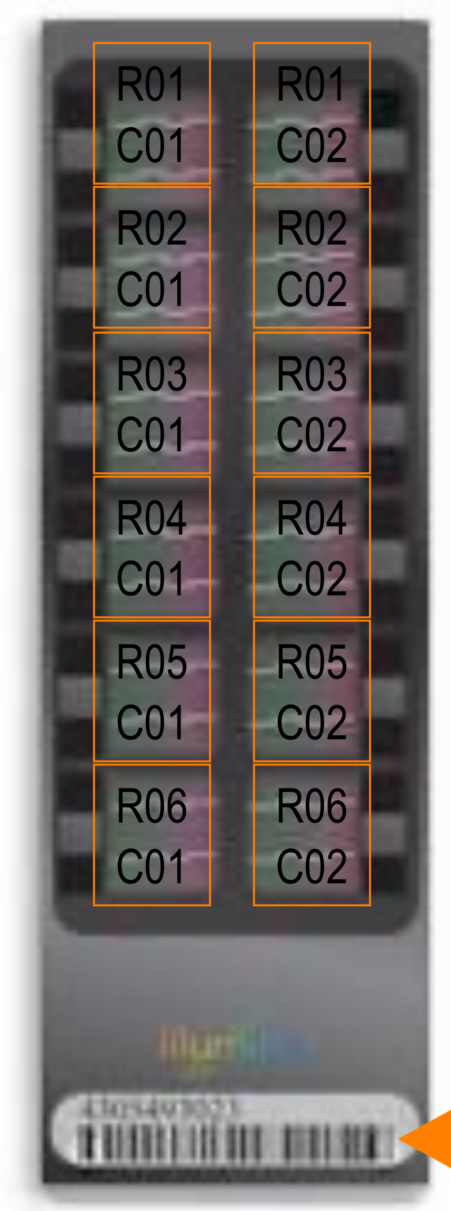
The Infinium HumanMethylation450 BeadChip's unique combination of comprehensive, expert-selected coverage, high sample throughput, and affordable price make it an ideal solution for epigenome-wide association studies (EWAS).

Powered by Illumina's revolutionary Infinium Methylation Assay, this BeadChip allows researchers to interrogate **> 485,000 methylation sites per sample at single-nucleotide resolution.**

It covers 99% of RefSeq genes, with an average of 17 CpG sites per gene region distributed across the promoter, 5'UTR, first exon, gene body, and 3'UTR. It covers **96% of CpG islands**, with additional coverage in **island shores and the regions flanking them**. Further content categories requested by the Consortium include:

- CpG sites outside of CpG islands
- Non-CpG methylated sites identified in human stem cells
- Differentially methylated sites identified in tumor versus normal (multiple forms of cancer) and across several tissue types
- CpG islands outside of coding regions
- miRNA promoter regions

Infinium HumanMethylation450 BeadChip



Arrays

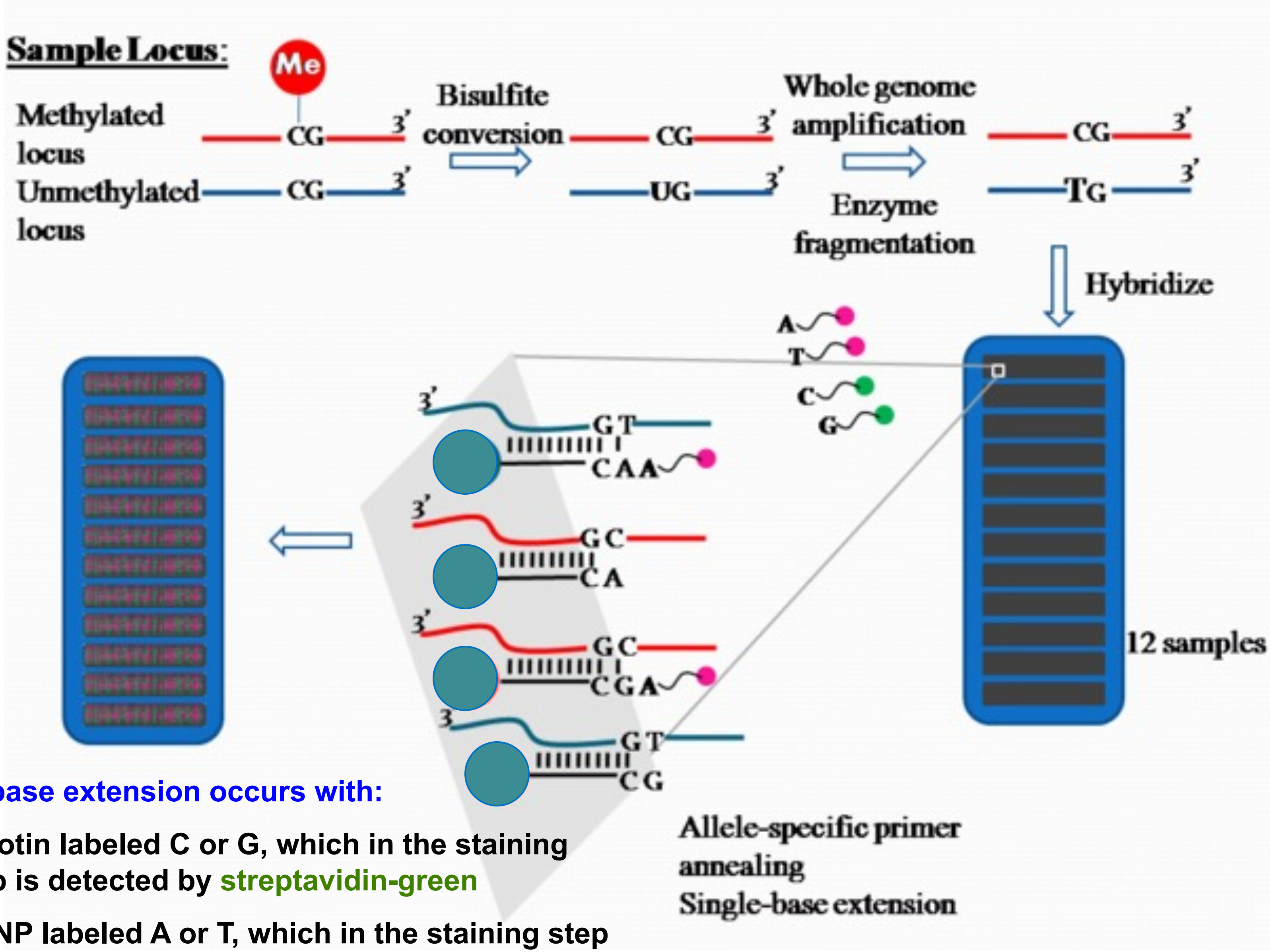
Each slide contains 12 arrays

1 array = 1 sample

For each probe sequence, **a median of 14 beads is randomly distributed on the array**. Each of these beads contains hundreds of thousands of oligonucleotides.

Slide

Samples preparation for Infinium microarrays



Single-base extension occurs with:

- a biotin labeled C or G, which in the staining step is detected by streptavidin-green
- a DNP labeled A or T, which in the staining step is detected by anti-DNP-red

Infinium I and Infinium II

Infinium I

1 IlmnID

2 bead types

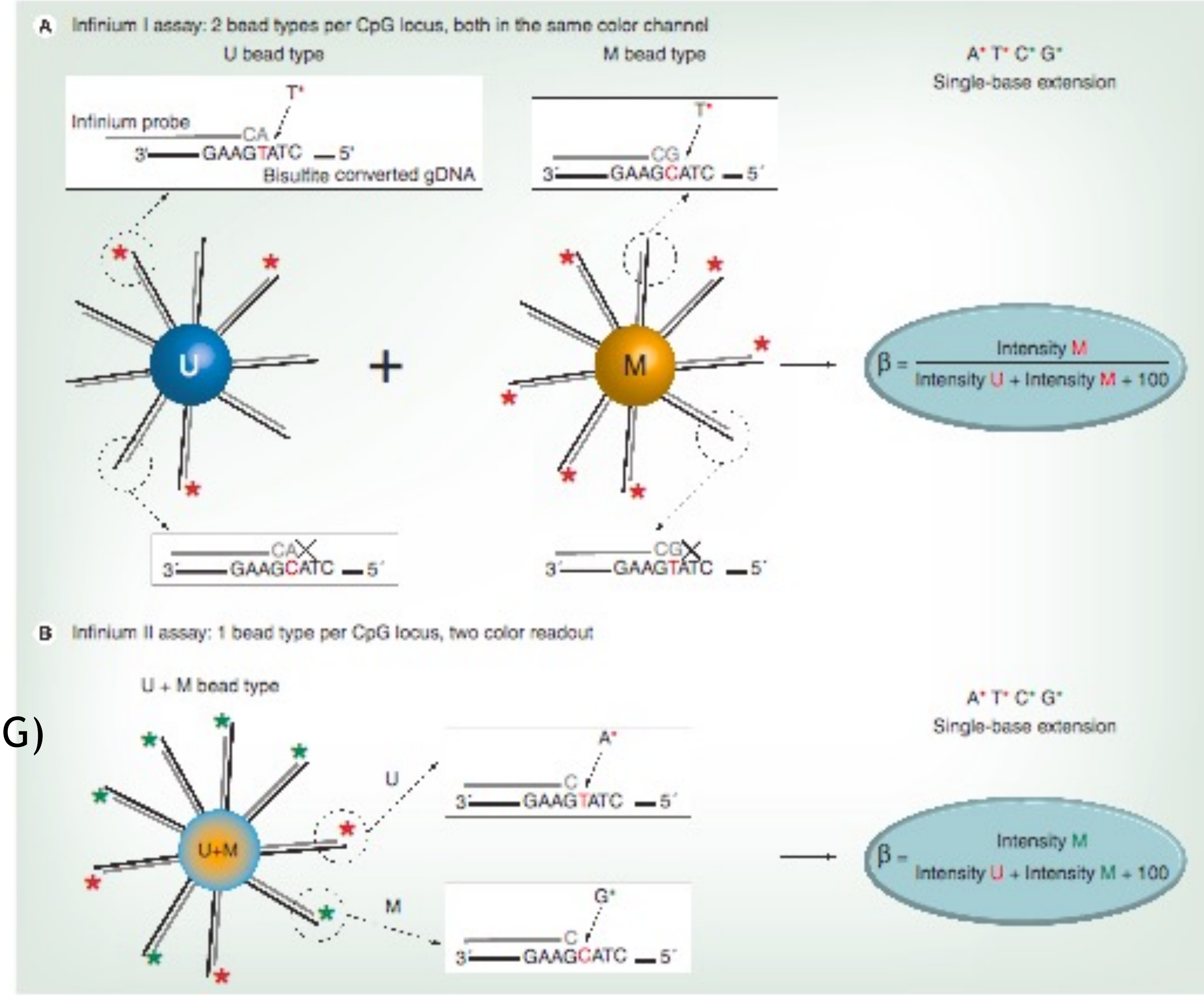
1 color (R or G)

Infinium II

1 IlmnID

1 bead type

2 colors (R and G)



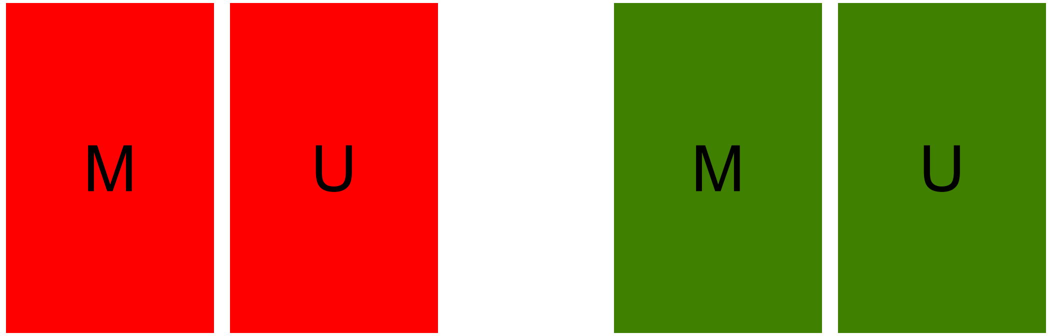
Infinium I and Infinium II

Infinium I

1 IlmnID

2 bead types

1 color (R or G)

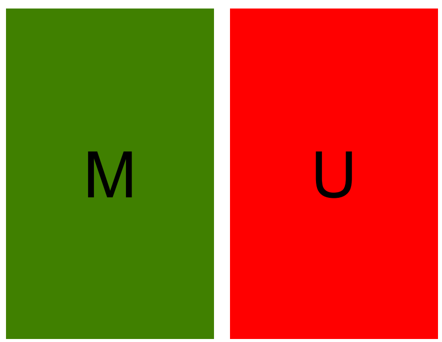


Infinium II

1 IlmnID

1 bead type

2 colors (R and G)



Infinium I and Infinium II

CpG islands are usually very dense in CpGs --> as the probes are 50 bp long, they can contain **underlying** CpG sites

Infinium I

2 bead types

2 probes

1 color (R or G)

Probe designs for Infinium I assays are based on the assumption that methylation is regionally correlated within a 50 bp span and, thus, underlying CpG sites are treated as in phase with the 'methylated' (C) or 'unmethylated' (T) query sites.

Infinium II

1 bead type

1 probe

2 colors (R and G)

A single, 50-mer probe is used to determine methylation state, making an all-or-none approach inapplicable. Underlying CpG sites are represented by degenerated R-bases. Illumina determined that Infinium II probes can have up to three underlying CpG sites within the 50-mer probe sequence without compromising data quality. This feature enables the methylation status at a query site to be assessed independently of assumptions on the status of neighboring CpG sites. The requirement for only a single bead type enables increased capacity for the number of CpG sites that can be queried. Infinium II designs are therefore applied whenever possible.

Example of Infinium I probe

5-TATCTCTGTCTGG**CG**AGGAGGCAA**CG**CACAACTGTGGTGGTTTTTGGAGTGGGTGGACCC**CG**GCCAA-3

3-ATAGAGACAGACC**GC**TCCTCCGTT**GC**GTGTTGACACCACCAAAAACCTCACCCACCTGGG**GC**CGGTT-5

After bisulfite conversion:

If the DNA is NOT methylated

3-ATAGAGATAGATT**GT**TTTTTTGTT**GT**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GT**TGGTT-5

If the DNA is methyalted

3-ATAGAGATAGATT**GC**TTTTTTGTT**GC**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GC**TGGTT-5

IlmnID cg00050873, AddressA 32735311 (against unmethylated DNA)

5-ACAAAAAAACAACACACAACTATAATAATTTTTAAAATAAATAAACCCCA

3-ATAGAGATAGATT**GT**TTTTTTGTT**GT**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GT**TGGTT-5

IlmnID cg00050873, AddressB 31717405 (against methylated DNA)

5-ACGAAAAAACAACGCACAACTATAATAATTTTTAAAATAAATAAACCCCG

3-ATAGAGATAGATT**GC**TTTTTTGTT**GC**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GC**TGGTT-5

Example of Infinium I probe

5-TATCTCTGTCTGG**CG**AGGAGGCAA**CG**CACAACTGTGGTGGTTTTTGGAGTGGGTGGACCC**CG**GCCAA-3

3-ATAGAGACAGACC**GC**TCCTCCGTT**GC**GTGTTGACACCACCAAAAACCTCACCCACCTGGG**GC**CGGTT-5

After bisulfite conversion:

If the DNA is NOT methylated

3-ATAGAGATAGATT**GT**TTTTTTGTT**GT**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GT**TGGTT-5

If the DNA is methyalted

3-ATAGAGATAGATT**GC**TTTTTTGTT**GC**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GC**TGGTT-5

IlmnID cg00050873, AddressA 32735311 (against unmethylated DNA)

5-ACAAAAAAAACAACACACAACTATAATAATTTTTAAAATAAATAAACCCCAA

3-ATAGAGATAGATT**GT**TTTTTTGTT**GT**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GT**TGGTT-5

IlmnID cg00050873, AddressB 31717405 (against methylated DNA)

5-ACGAAAAAAACAACGCACAACTATAATAATTTTTAAAATAAATAAACCCCGA

3-ATAGAGATAGATT**GC**TTTTTTGTT**GC**GTGTTGATATTATTAAAAATTTTATTTATTTGGG**GC**TGGTT-5

Single base extension: A T C G

Example of Infinium II probe

5-CAGCTAGATTGTGTTCTTCCACACAAAATACTGAA**CG**TATATTTACAAAAATGCTTCCAT**CG**AATCC-3

3-GTCGATCTAACACAAGAAGGTGTGTTTTATGACTT**GC**ATATAAATGTTTTTACGAAGGTA**GC**TTAGG-5

After bisulfite conversion:

If the DNA is NOT methylated

3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GT**TTAGG-5

If the DNA is methylated

3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GC**TTAGG-5

IlmnID cg00063477, Address 16712347 (independent of the methylation status of DNA)

5-TATTCTTCCACACAAAATACTAAA**CR**TATATTTACAAAAATACTTCCATC

3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GT**TTAGG-5

5-TATTCTTCCACACAAAATACTAAA**CR**TATATTTACAAAAATACTTCCATC

3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GC**TTAGG-5

Example of Infinium II probe

5-CAGCTAGATTGTGTTCTTCCACACAAAATACTGAA**CG**TATATTTACAAAAATGCTTCCAT**CG**AATCC-3
3-GTCGATCTAACACAAGAAGGTGTGTTTTATGACTT**GC**ATATAAATGTTTTTACGAAGGTA**GC**TTAGG-5

After bisulfite conversion:

If the DNA is NOT methylated

3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GT**TTAGG-5

If the DNA is methylated

3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GC**TTAGG-5

IlmnID cg00063477, Address 16712347 (independent of the methylation status of DNA)

5-TATTCTTCCACACAAAATACTAAA**CR**TATATTTACAAAAATACTTCCATCA
3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GT**TTAGG-5

5-TATTCTTCCACACAAAATACTAAA**CR**TATATTTACAAAAATACTTCCATCG
3-GTTGATTTAATATAAGAAGGTGTGTTTTATGATTT**G?**ATATAAATGTTTTTATGAAGGTA**GC**TTAGG-5

Single base extension: A T C G

Infinium 450k Annotation

Meaningful analysis of Infinium 450k (and, more in general, of genomic) data depends on annotating probes with genomic location and genomic features.

Probe sequences is fixed for a certain array **design**, but **annotation** can change!

(for example, check *WRN* locus in hg19 and hg38)

Design	$\neq$	Annotation
IlluminaHumanMethylation450kmanifest *(Illumina website)*		IlluminaHumanMethylation450kmanifest *(Illumina website)*
		IlluminaHumanMethylation450kanno.ilmn12.hg19 *(Bioconductor)*
		IlluminaHumanMethylation450k.db *(Bioconductor)*

Illumina Infinium450 Manifest

Illumina450Manifest.Rdata

486428 rows, 33 columns

- 485512 probes targeting CpG sites (probe name starts with *cg*) and not CpG methylation (probe name starts with *ch*)
- 65 probes targeting SNPs (probe name starts with *rs*)
- 850 control probes (probe name is numeric)
- 1 line with only the string [Controls]

Infinium 450k analysis pipeline

1. Import raw data files
2. Quality checks and probe filtering
3. Background correction and normalization
4. Batch effects
5. Differential methylation
6. Biological interpretation

Bioinformatic tools for Infinium 450k analysis

GenomeStudio: Illumina proprietary software package to visualize and analyze data generated on Illumina array platforms.

Genotyping Module
Gene Expression Module
Methylation Module
Protein Analysis Module
Polyploid Clustering Module

Bioinformatic tools for Infinium 450k analysis

GenomeStudio: Illumina proprietary software package to visualize and analyze data generated on Illumina array platforms.

Genotyping Module
Gene Expression Module
Methylation Module
Protein Analysis Module
Polyploid Clustering Module

R packages in Bioconductor:

ARRmNormalization, BEclear, bumphunter, charm, COHCAP, coMET, conumee, CopyNumber450k, DMRcate, DMRforPairs, ENmix, lumi, M3D, MEAL, methVisual, methyAnalysis, MethylAid, methylMnM, methylPipe, methylumi, minfi, missMethyl, Repitools, RnBeads, shinyMethyl, skewr, TurboNorm, watermelon
… and manu more!

R packages not in Bioconductor:

IMA

Infinium 450k analysis pipeline

1. Import raw data files
2. Quality checks and probe filtering
3. Background correction and normalization
4. Batch effects
5. Differential methylation
6. Biological interpretation

Import raw data

Starting point:

.idat files, a binary format containing the raw red and green channel intensities generated by the Illumina iScan

R packages to import data:

- **Minfi**
- Methylumi
- IMA
- lumi

Import raw data

Genome analysis Advance Access publication January 28, 2014

Minfi: a flexible and comprehensive Bioconductor package for the analysis of Infinium DNA methylation microarrays

Martin J. Aryee[1,†], Andrew E. Jaffe[2,3], Hector Corrada-Bravo[4], Christine Ladd-Acosta[5], Andrew P. Feinberg[2,6], Kasper D. Hansen[2,7,*] and Rafael A. Irizarry[2,*,‡]

Bioinformatics, 33(4), 2017, 558–560
doi: 10.1093/bioinformatics/btw691
Advance Access Publication Date: 29 November 2016
Applications Note

OXFORD

Genome analysis

Preprocessing, normalization and integration of the Illumina HumanMethylationEPIC array with minfi

Jean-Philippe Fortin[1], Timothy J. Triche Jr[2] and Kasper D. Hansen[1,3,*]

S4 classes

Representations of the array data through **S4 classes**

S4 stands for object oriented programming with S.

- Data inside an S4 class are organized into slots [‘@’ or slots()]. **As a user you should never have to access slots directly.**
- “Accessor” functions: for example *get* and the slot name. Not all slots have an accessor function. Accessor functions are usually documented on the same help page as the class itself.

https://kasperdanielhansen.github.io/genbioconductor/html/R_S4.html#s3-and-s4-classes

Minfi classes

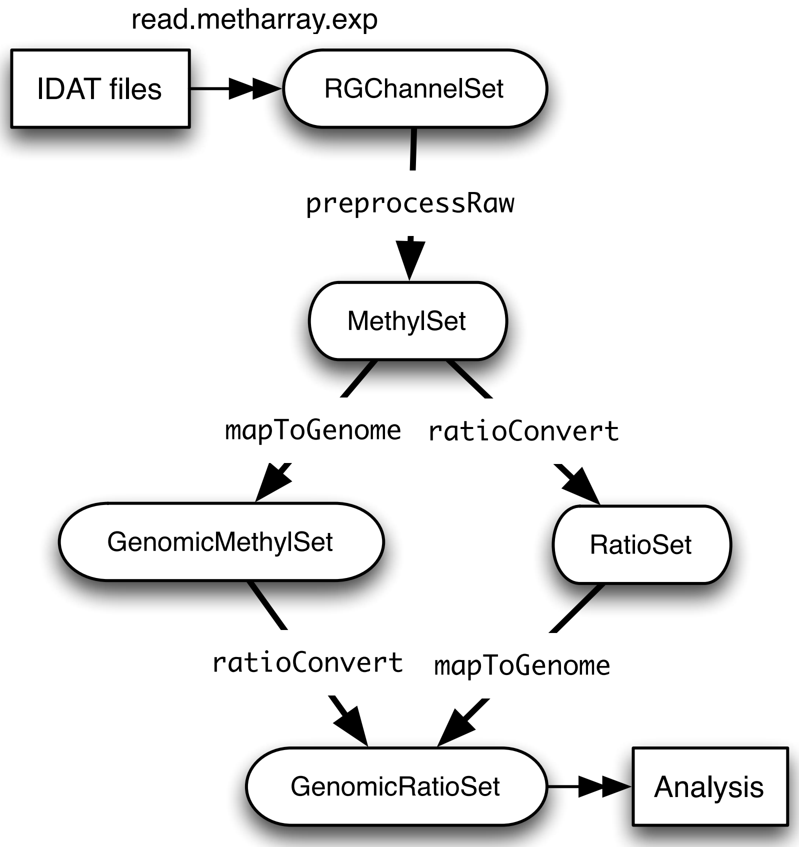


Figure 2: Flow chart of the `minfi`'s functions

- `RGChannelSet` : raw data from the IDAT files; this data is organized at the probe (not CpG locus) level. This data has two channels: Red and Green.
- `MethylSet` : data organized by the CpG locus level, but not mapped to a genome. This data has two channels: Meth (methylated) and Unmeth (unmethylated).
- `RatioSet` : data organized by the CpG locus level, but not mapped to a genome. The data has at least one of two channels: Beta and/or M (logratio of Beta). It may optionally include a CN channel (copy number).
- `GenomicMethylSet` : like a `MethylSet`, but mapped to a genome.
- `GenomicRatioSet` : like a `RatioSet`, but mapped to the genome.

Minfi

Import raw data (.idat) by minfi

RGChannelSet → stores **Red** and **Green** Intensities for each **Address**

Accessors Functions: getRed, getGreen

MethylSet → stores **Unmeth** and **Meth** values for each **IlmnID**

Accessors Functions: getUnmeth, getMeth

RGChannelSet and MethylSet

X9376538140_R04C01

cg00050873 (type I, Red)

AddressA	AddressB
32735311	31717405
1001	9478

cg02004872 (type I, Green)

AddressA	AddressB
25785404	58629399
7083	234

cg00035864

AddressA
31729416
5058
1351

RGSet (RGChannelSet class)

getRed(RGSet)

32735311	1001
31717405	9478
25785404	626 (Bkg)
58629399	451 (Bkg)
31729416	5058

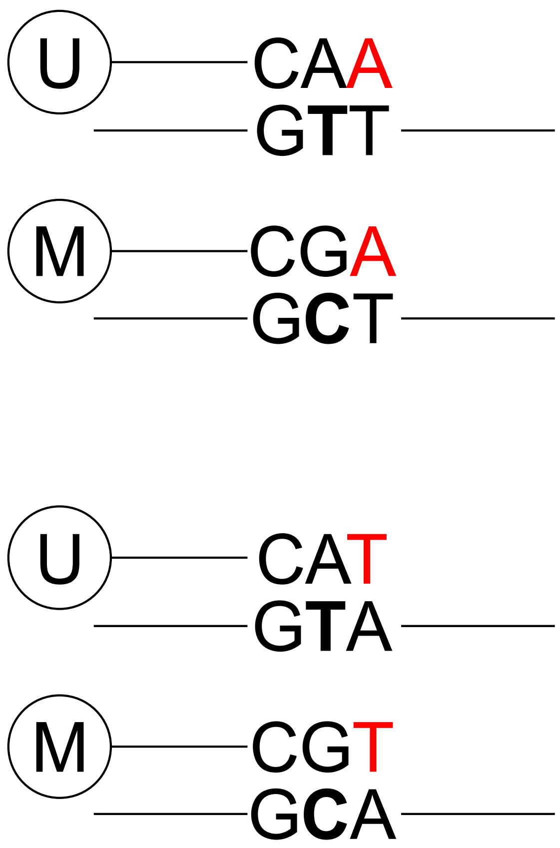
getGreen(RGSet)

32735311	332 (Bkg)
31717405	930 (Bkg)
25785404	7083
58629399	234
31729416	1351

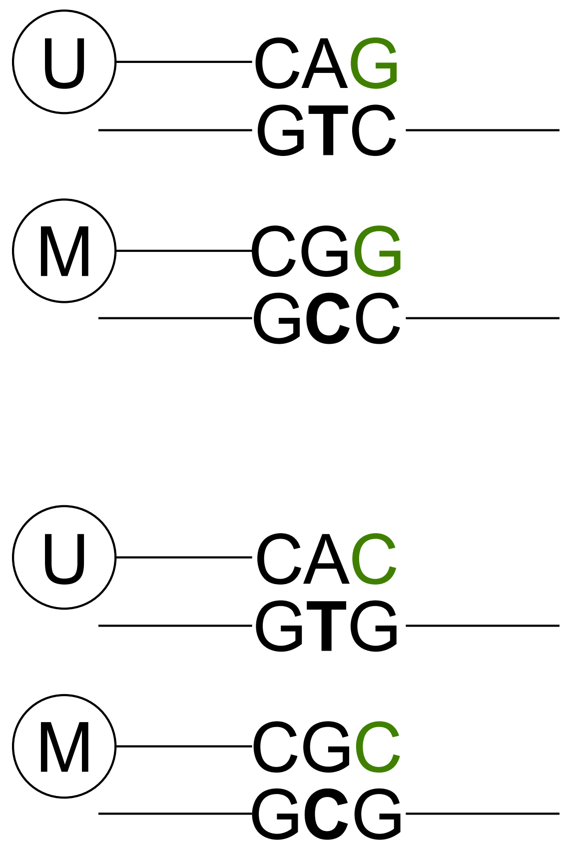
Out of band probes

These are the measurements of Type I probes in the wrong color channel → they can be used for estimating non-specific fluorescence in background correction

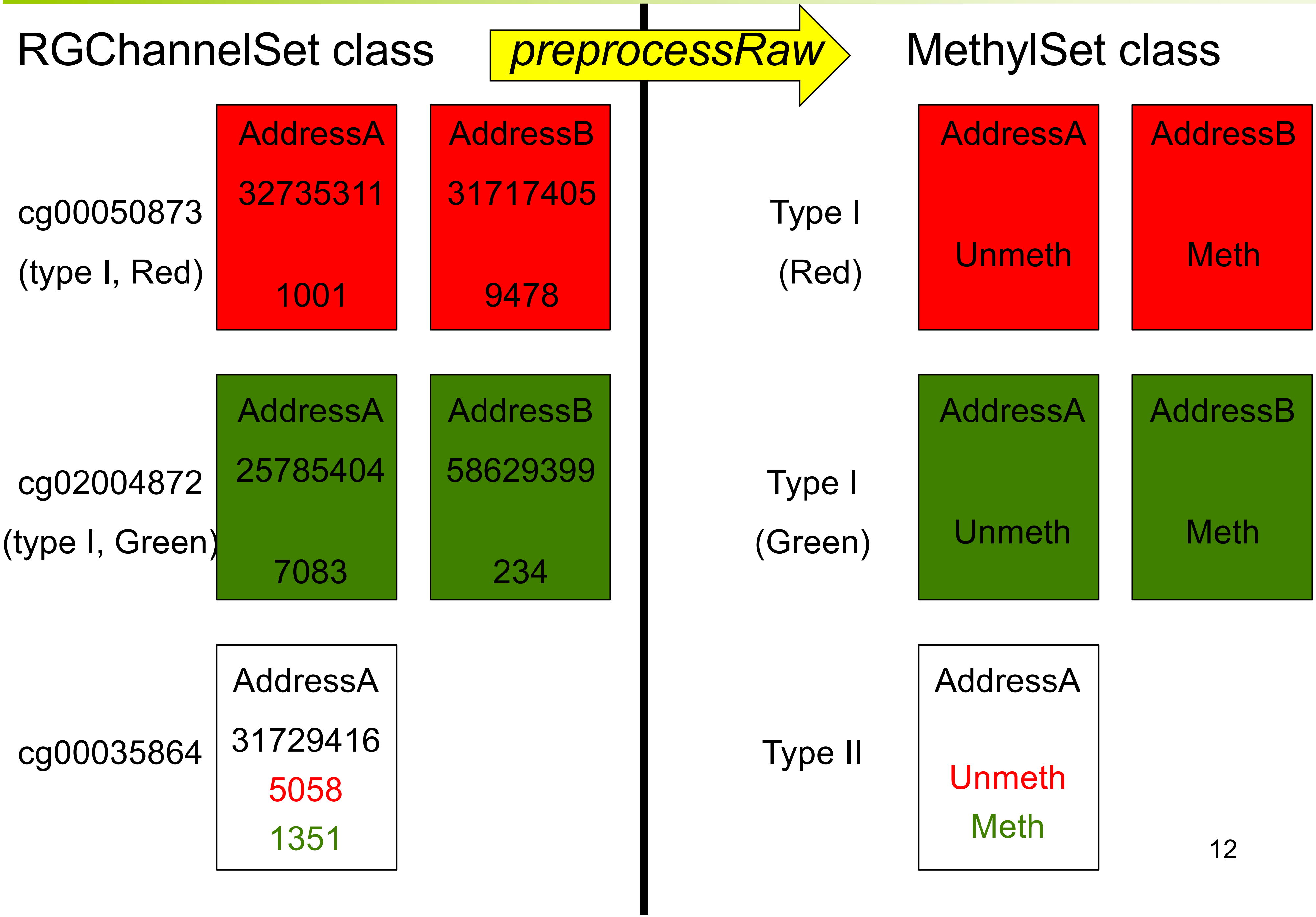
The methylation level is measured in the Red channel…but the scanner reads them also in the green channel
→ Green background signal



The methylation level is measured in the Green channel…but the scanner reads them also in the red channel
→ Red background signal



RGChannelSet and MethylSet



RGChannelSet and MethylSet

X9376538140_R04C01

RGSet (RGChannelSet class)

getRed(RGSet)

32735311	1001
31717405	9478
25785404	626 (Bkg)
58629399	451 (Bkg)
31729416	5058

getGreen(RGSet)

32735311	332 (Bkg)
31717405	930 (Bkg)
25785404	7083
58629399	234
31729416	1351

MSet.raw (MethylSet class)

getUnmeth(MSet.raw)

cg00050873	1001
cg02004872	7083
cg00035864	5058

getMeth(Mset.raw)

cg00050873	9478
cg02004872	234
cg00035864	1351

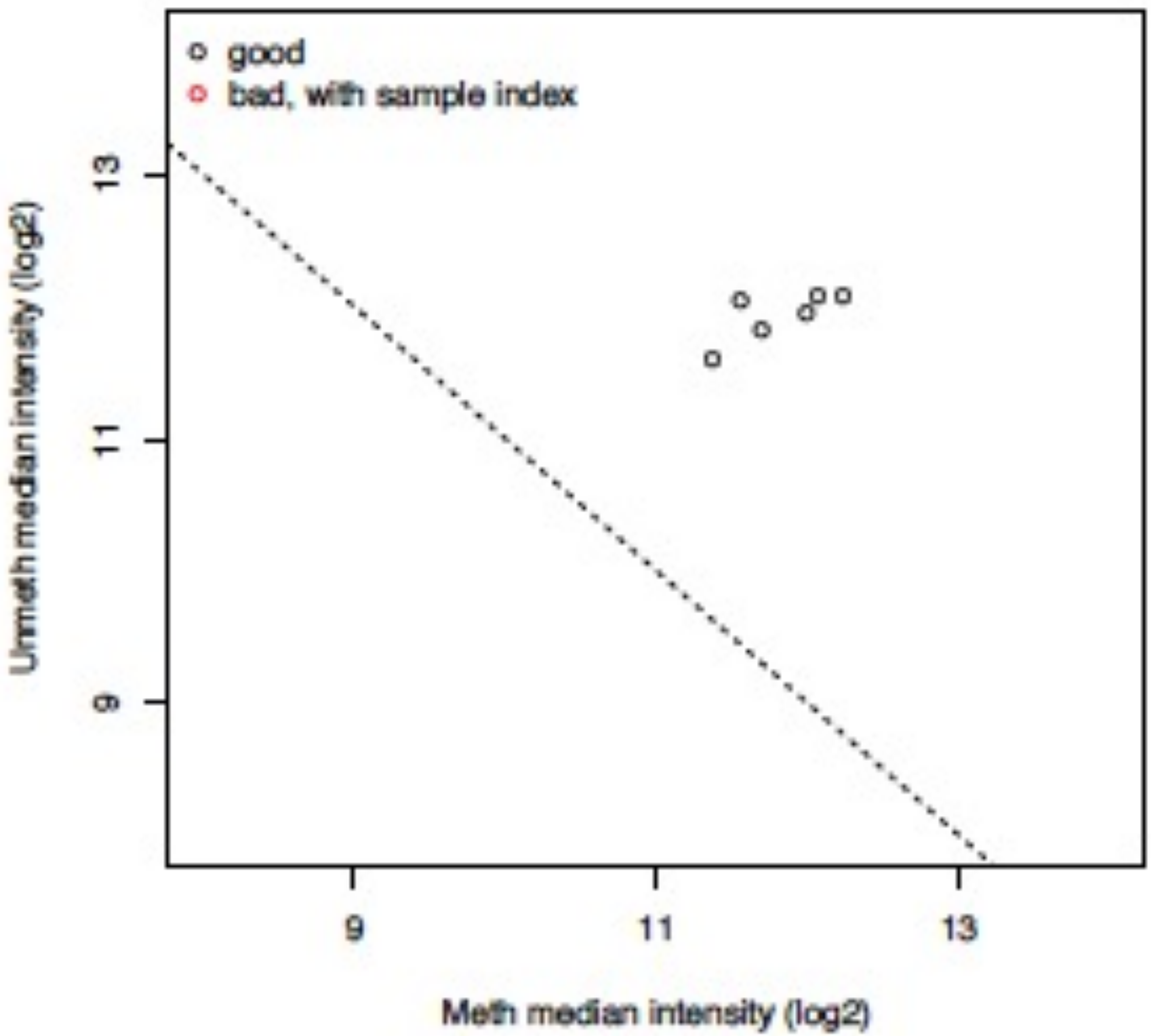
Infinium 450k analysis pipeline

1. Import raw data files
2. Quality checks and probe filtering
3. Background correction and normalization
4. Batch effects
5. Differential methylation
6. Biological interpretation

Quality checks

1. QC plot
2. Control probes
3. Detection pValues
4. SNPs
5. Batch effects (see lesson 5)

QC plot



good
bad, with sample index
Unmeth median intensity (log2)
13
11
9
9
11
13
Meth median intensity (log2)

Control Probes

Sample-Independent Controls

The sample-independent controls let you evaluate the quality of specific steps in the process flow, and include:

- Staining controls
- Extension controls
- Target removal controls
- Hybridization controls

They are **not** based on the hybridization between control probes and sample DNA

Sample-Dependent Controls

The sample-dependent controls let you evaluate performance across samples, and include:

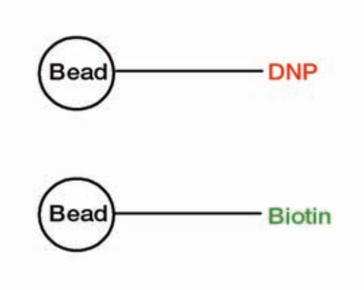
- Bisulfite conversion I controls
- Bisulfite conversion II controls
- Specificity I controls
- Specificity II controls
- Nonpolymorphic (NP) controls
- Negative controls

They are based on the hybridization between control probes and sample DNA

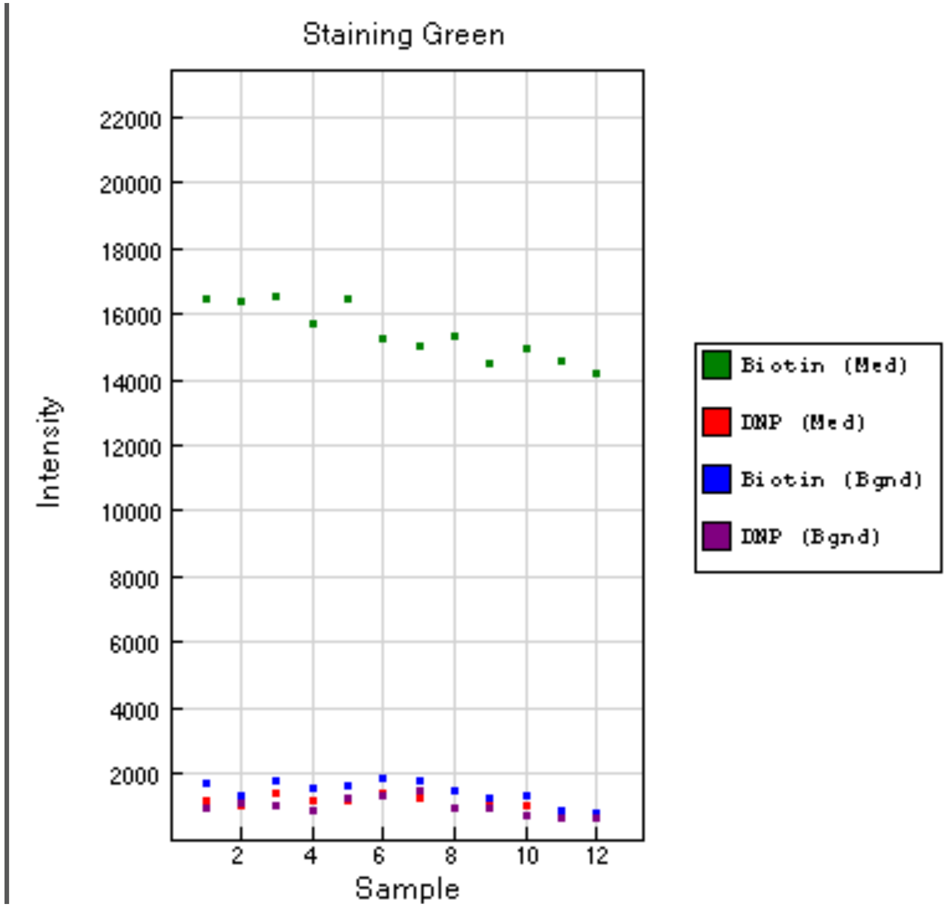
Several key steps in the Infinium HD Assay for Methylation require evaluation of both the red and green color channels. For these instances, both red and green channel controls are included.

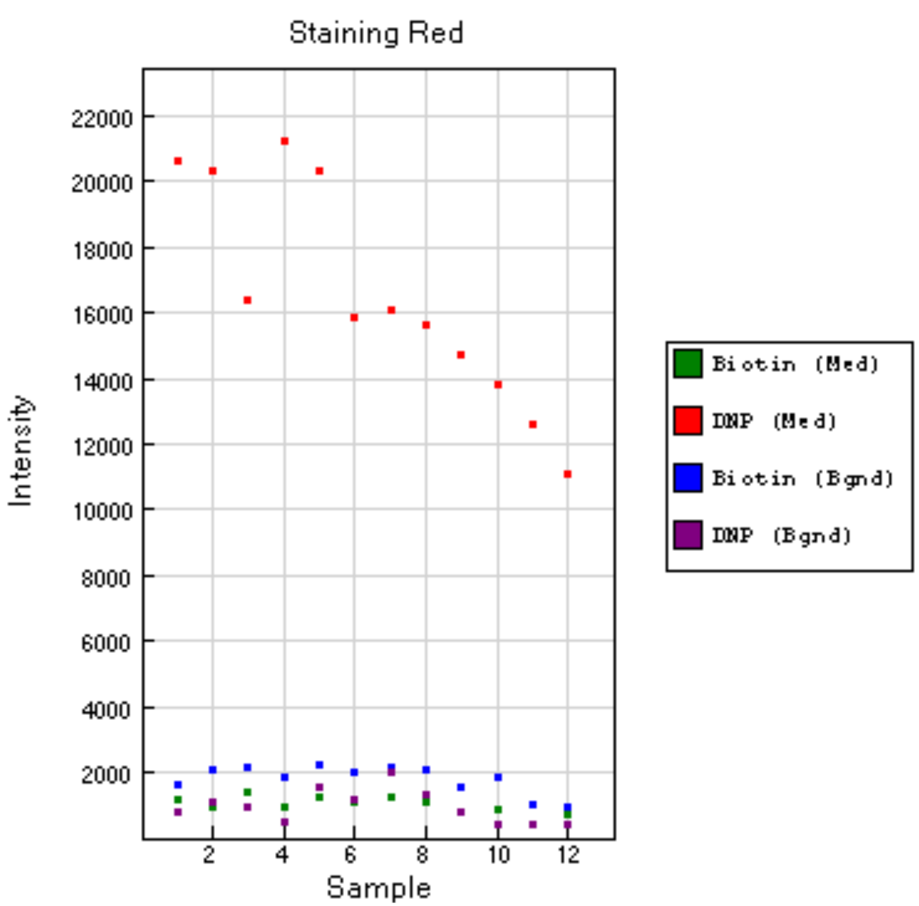
Staining controls

Measure the efficiency and sensitivity of staining step in both the red and the green channel (independent of hybridisation/extension step)



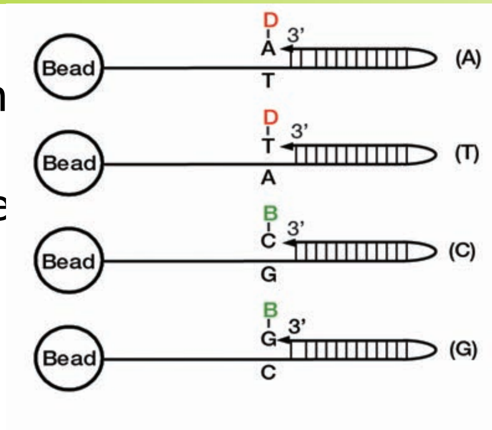
Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Staining	DNP (High)	1	-	+	High
Staining	DNP (Bgnd)	1	-	+	Background
Staining	Biotin (Med)	1	+	-	High
Staining	Biotin (Bgnd)	1	+	-	Background



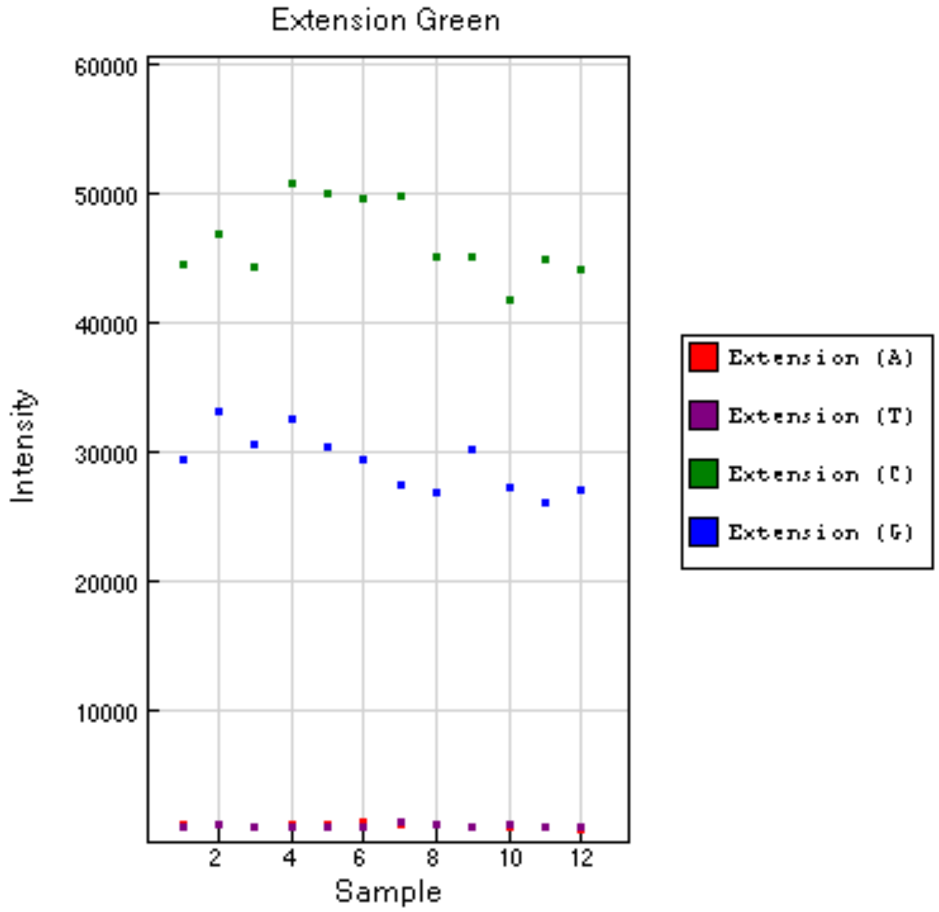


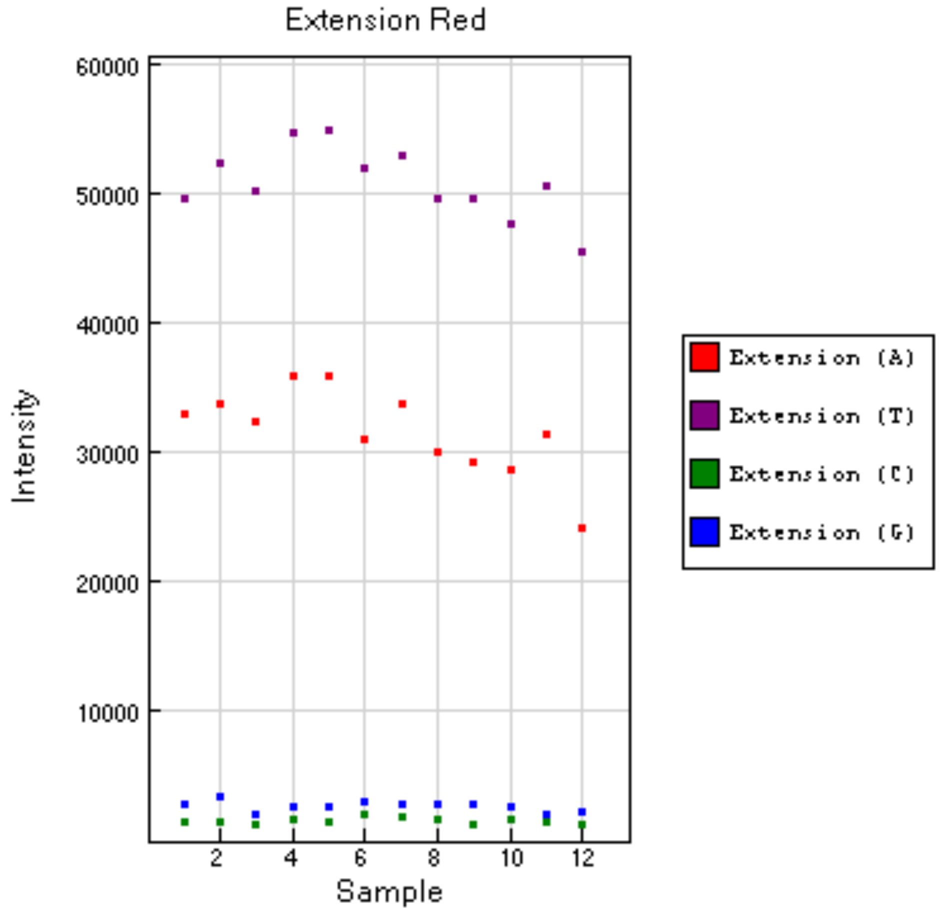
Extension controls

Test efficiency of extension of A, T, C an G nucleotides from a hairpin probe (sample independent).



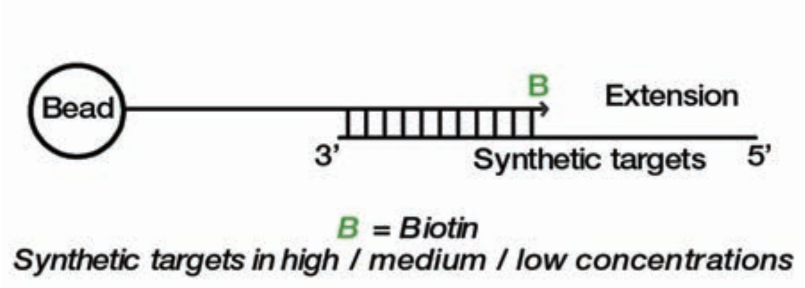
Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Extension	Extension (A), (T)	2	-	+	High
Extension	Extension (C), (G)	2	+	-	High



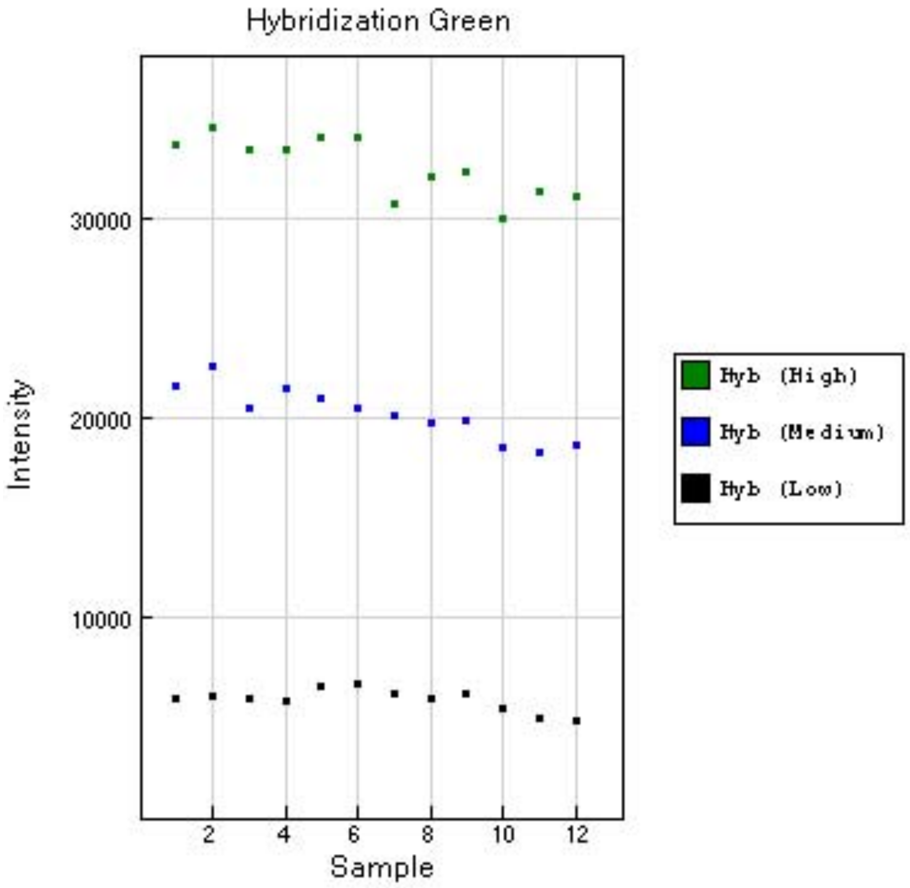


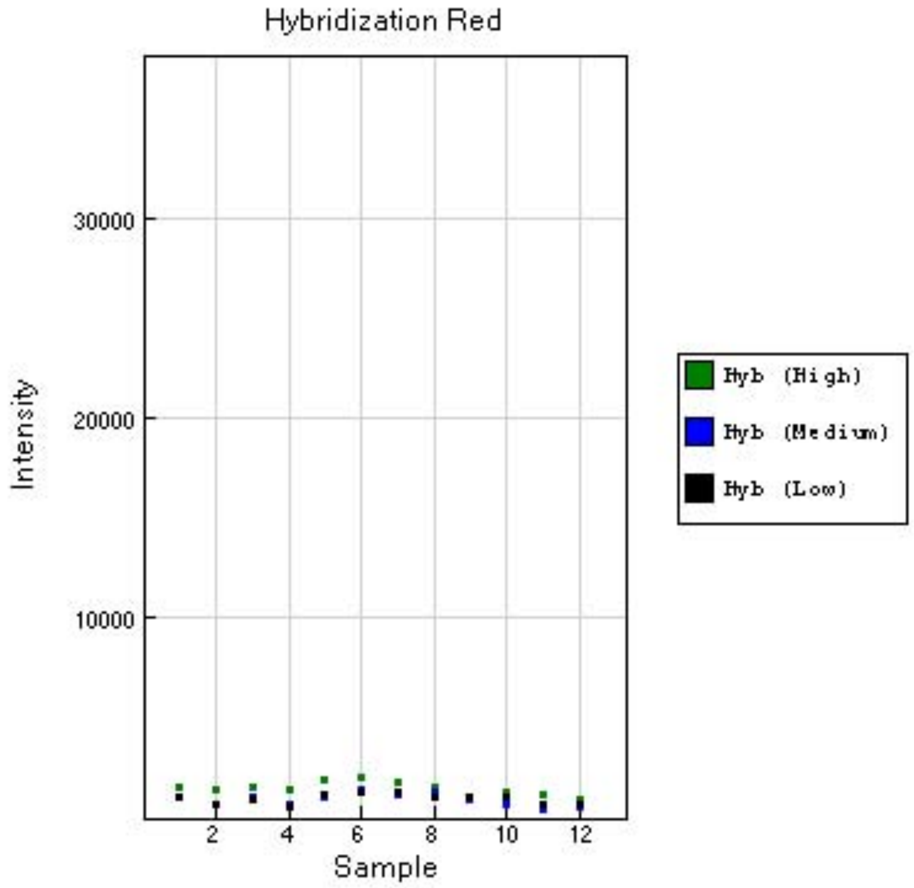
Hybridization controls

Test the overall performance of Infinium assay using synthetic targets (not DNA) at 3 concentrations



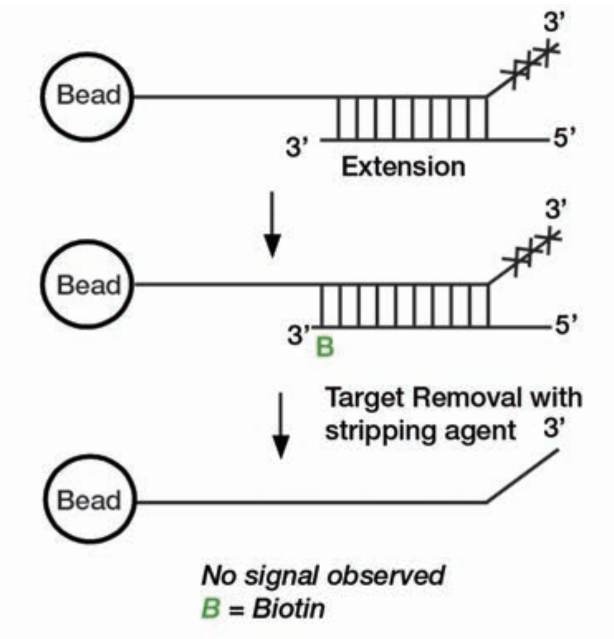
Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Hybridization	Hyb (Low)	1	+	-	Low
Hybridization	Hyb (Medium)	1	+	-	Medium
Hybridization	Hyb (High)	1	+	-	High



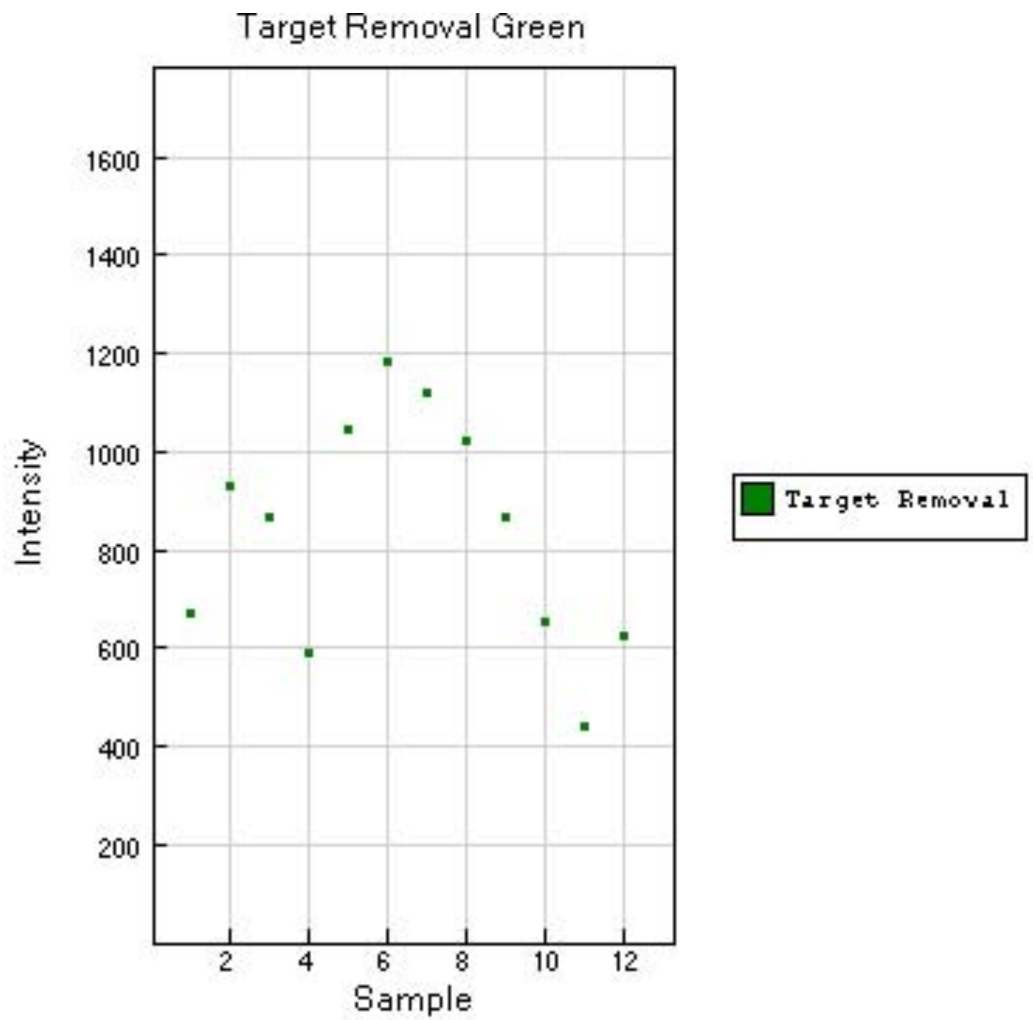


Target removal controls

Test the efficiency of stripping step after extension. The probe, which is non-extendable, hybridise with a extendable synthetic target



Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Target Removal	Target Removal 1, 2	2	+	-	Low

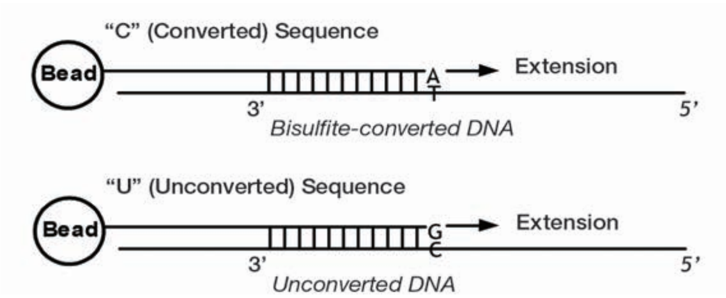


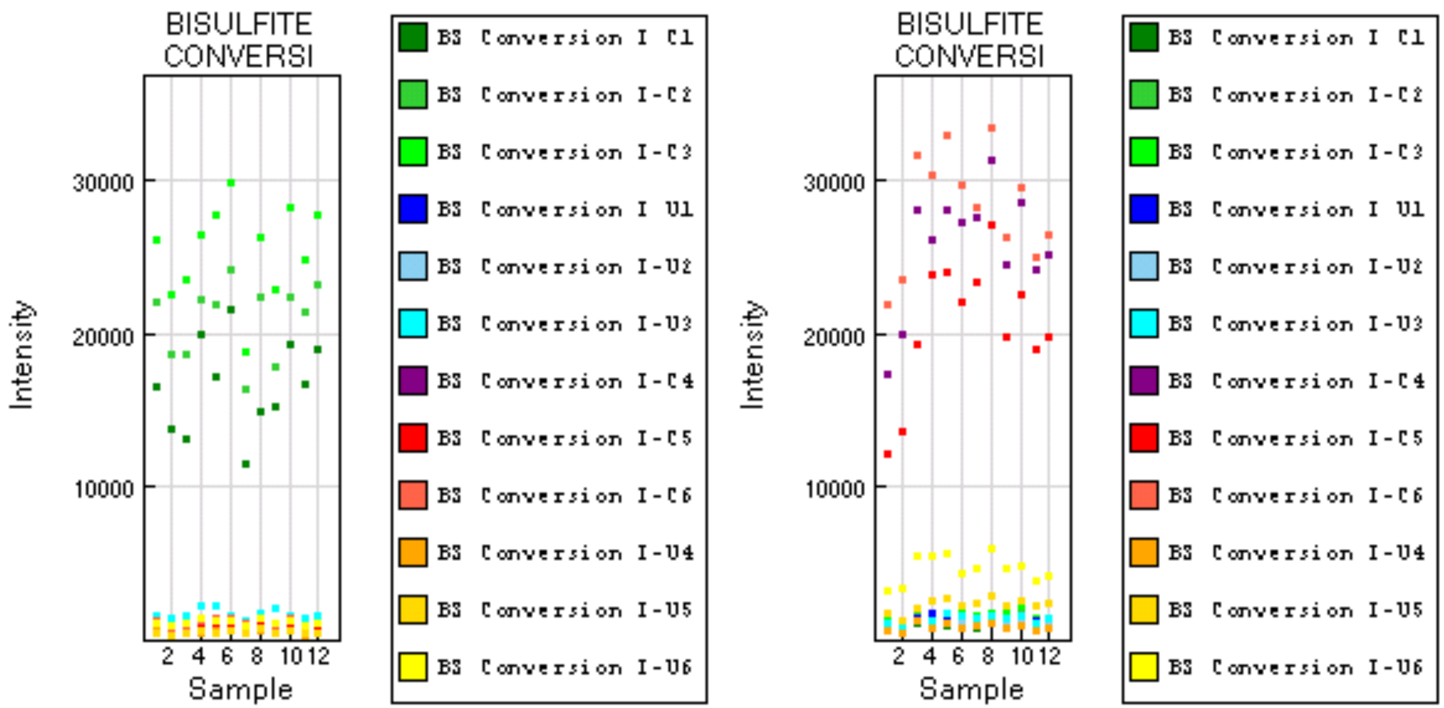
Bisulfite conversion controls

Test efficiency of bisulphite conversion of gDNA. They query a [C/T] polymorphism created by bisulfite conversion of non-CpG cytosines in the genome.

Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Bisulfite conversion I	BC conversion I C1, C2, C3	3	+	-	High
Bisulfite conversion I	BC conversion I U1, U2, U3	3	+	-	Background
Bisulfite conversion I	BC conversion I C4, C5, C6	3	-	+	High
Bisulfite conversion I	BC conversion I U4, U5, U6	3	-	+	Background

Type I





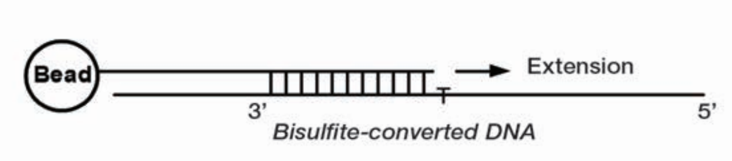
Performance of bisulfite conversion controls C1, C2 and C3 should be monitored in the Green channel, and controls C4, C5 and C6 should be monitored in Red channel.

Bisulfite conversion controls

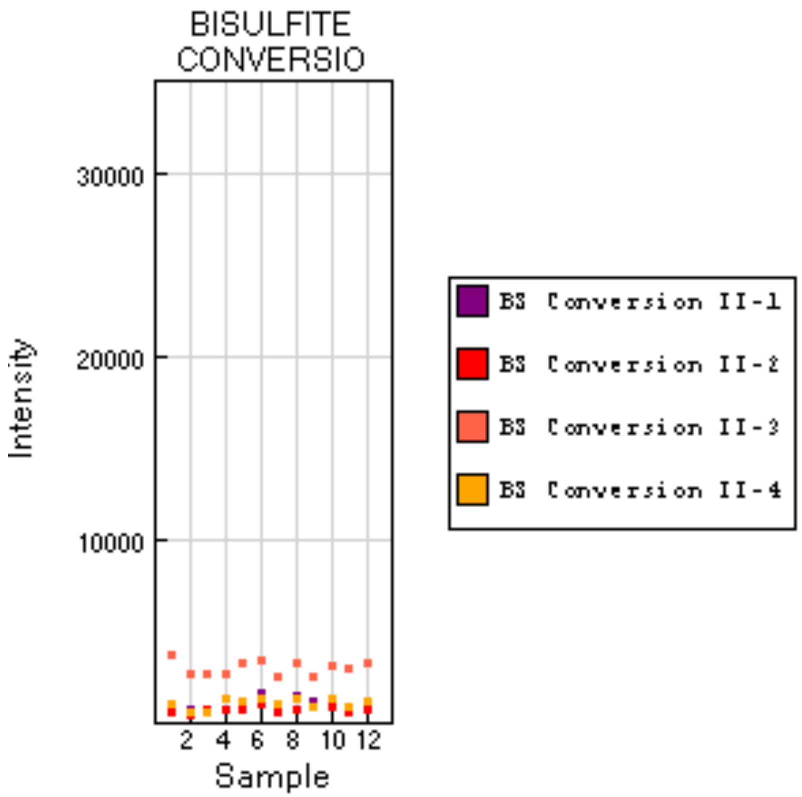
Test efficiency of bisulphite conversion of gDNA. They query a [C/T] polymorphism created by bisulfite conversion of non-CpG cytosines in the genome.

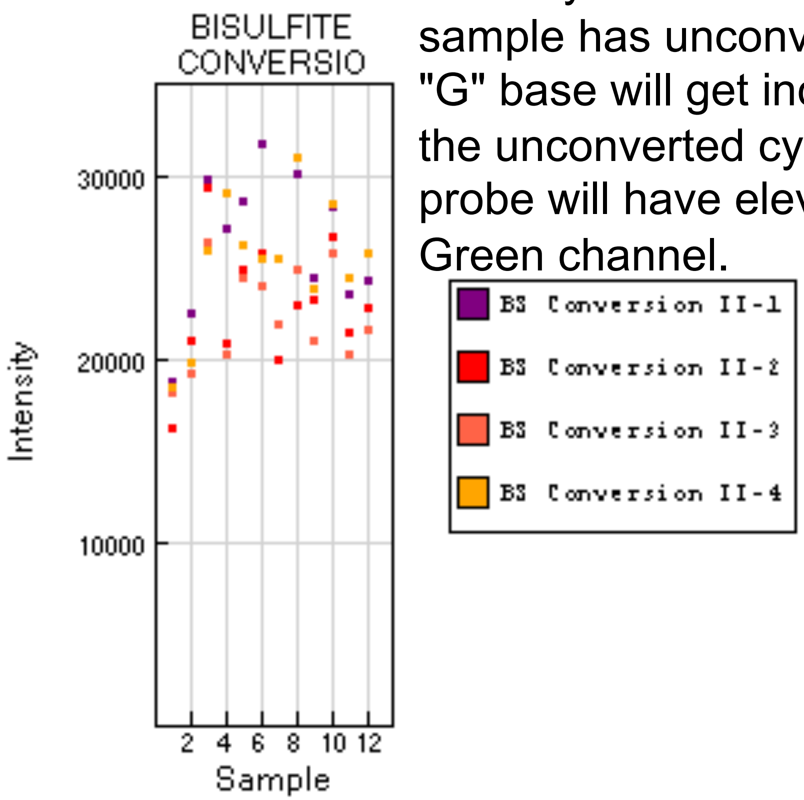
Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Bisulfite conversion II	BC conversion II 1, 2, 3, 4	4	-	+	High

Type II



If the bisulfite conversion reaction was successful, the "A" base will get incorporated and the probe will have intensity in the Red channel. If the sample has unconverted DNA, the "G" base will get incorporated across the unconverted cytosine, and the probe will have elevated signal in the Green channel.



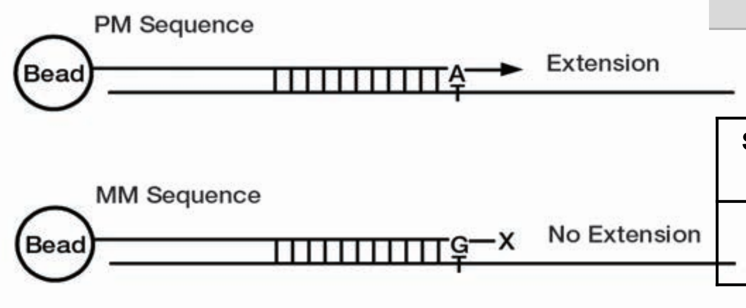


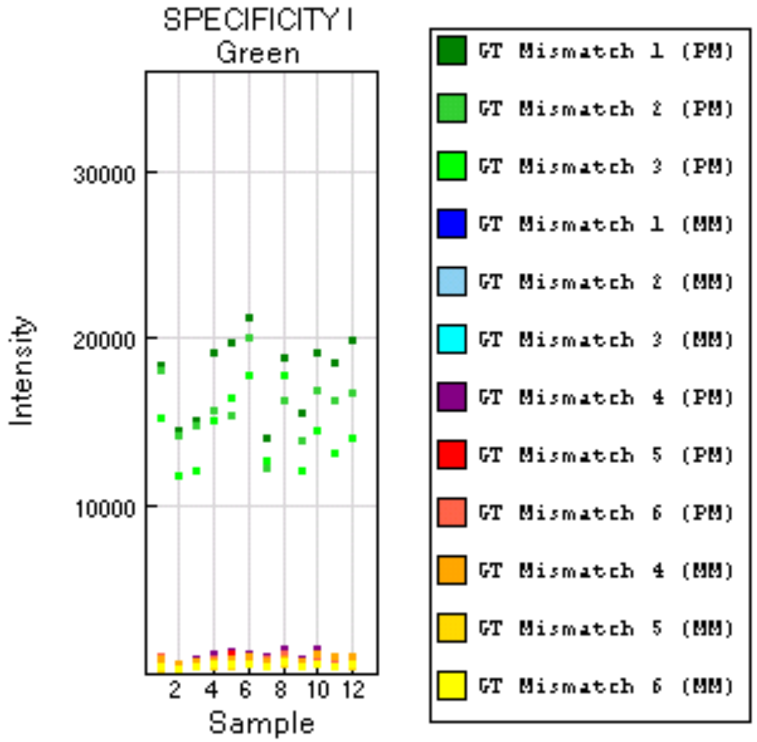
Specificity controls

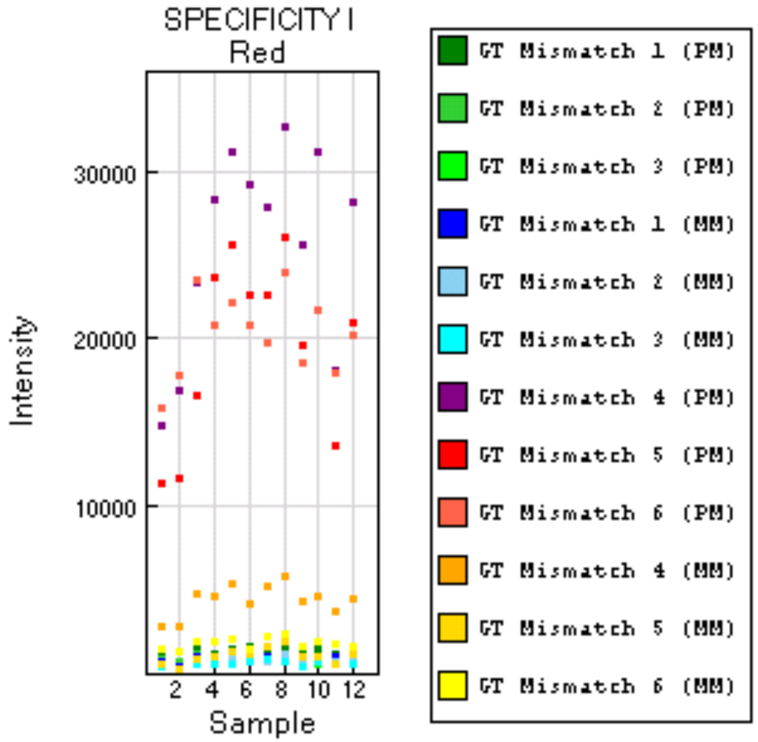
Check potential nonspecific primer extension for Infinium I and Infinium II assay probes → non-polymorphic T sites are assessed

Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Specificity I	GT mismatch 1, 2, 3 (PM)	3	+	-	High
Specificity I	GT mismatch 1, 2, 3 (MM)	3	+	-	Background
Specificity I	**GT mismatch 4,5,6 (PM)**	**3**	**-**	**+**	**High**
Specificity I	GT mismatch 4,5,6 (MM)	3	-	+	Background

Type I





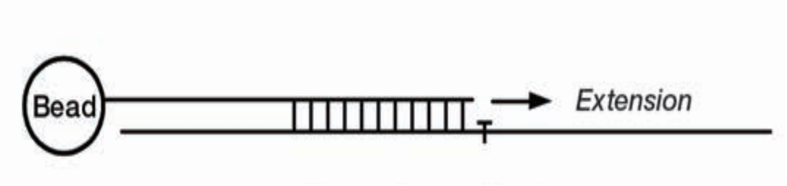


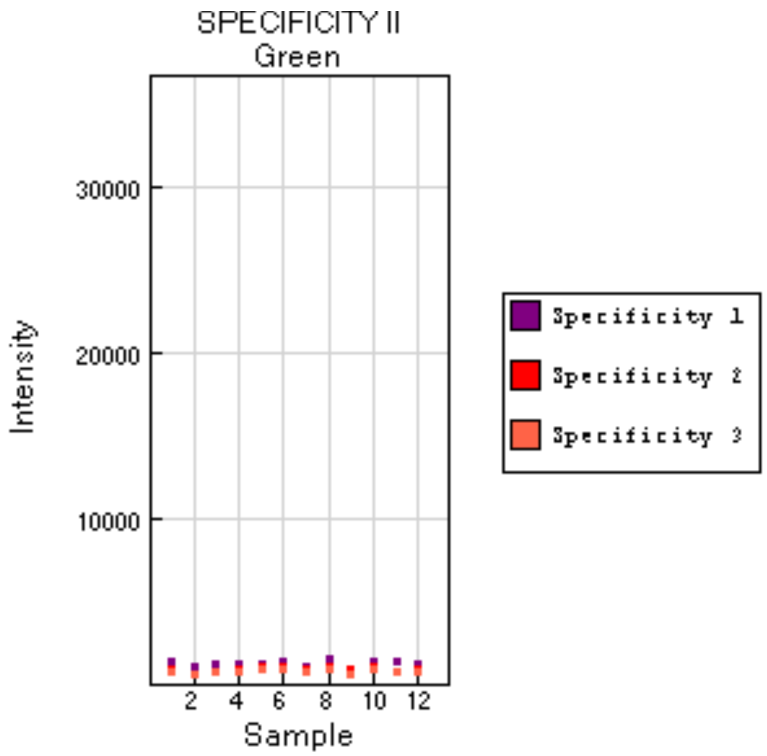
Specificity controls

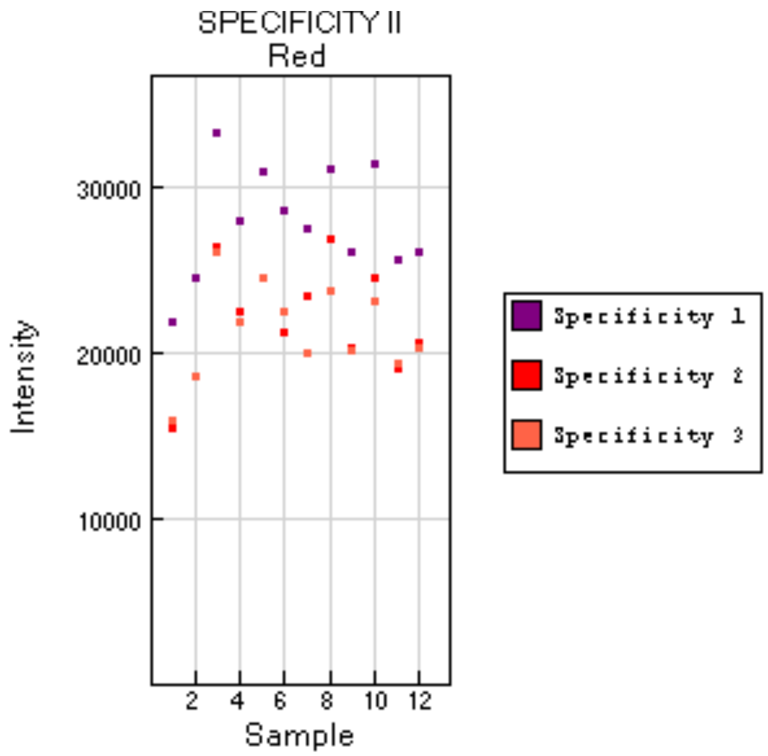
Check potential nonspecific primer extension for Infinium I and Infinium II assay probes → non-polymorphic T sites are assessed

Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Specificity II	Specificity 1, 2, 3	3	-	+	High

Type II







Non-polymorphic controls

Query a non polymorphic base A, T, C and G to test overall performance of the assay from amplification to detection

Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Non-Polymorphic	NP (A), (T)	2	-	+	High
Non-Polymorphic	NP (C), (G)	2	+	-	High

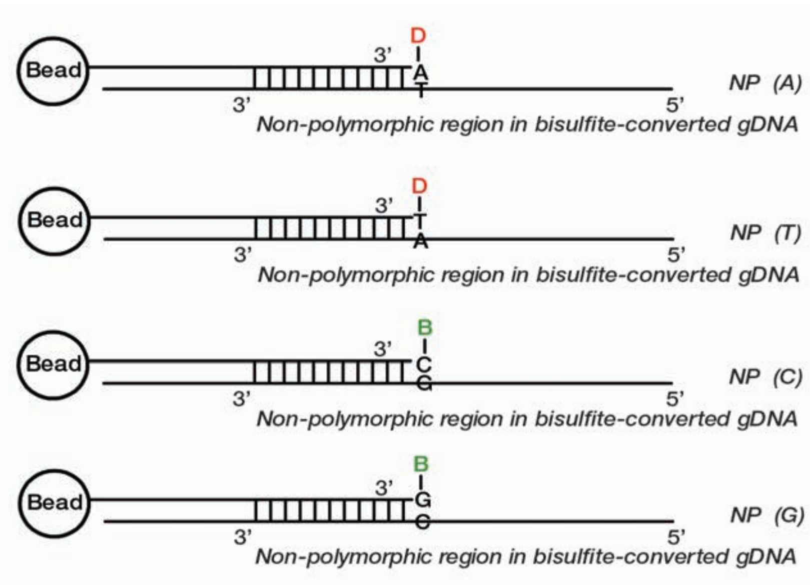
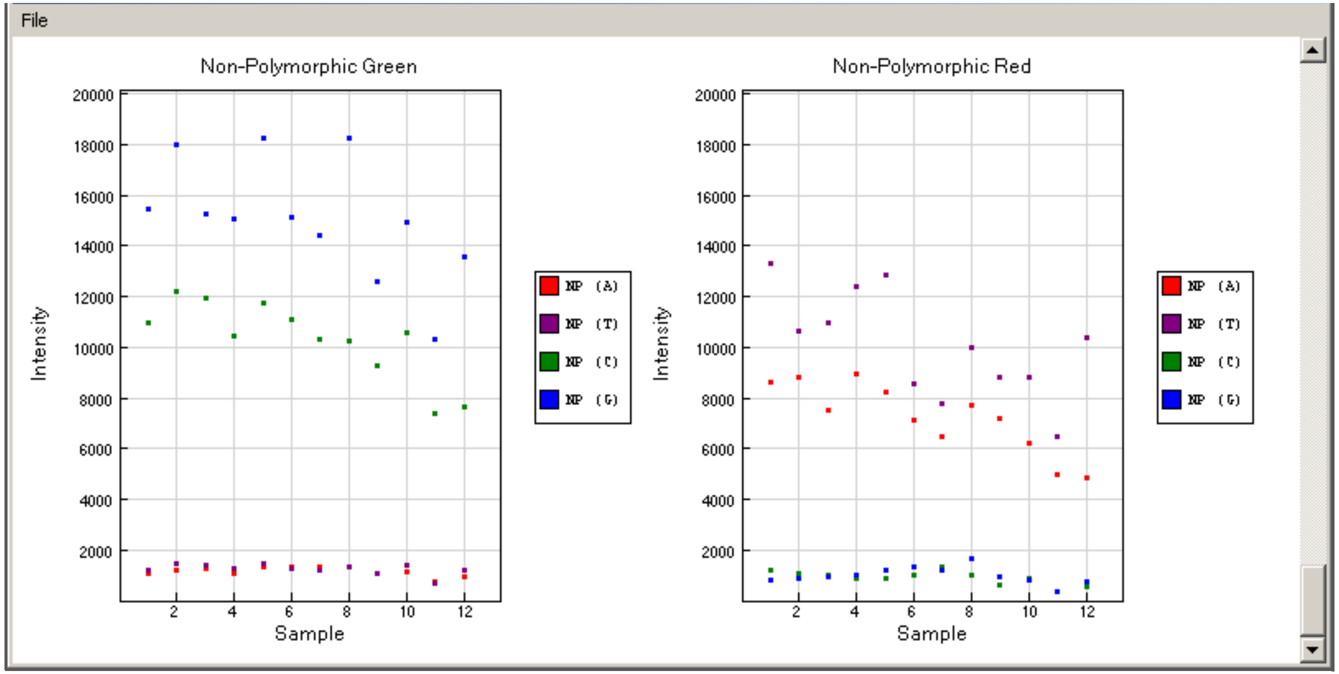


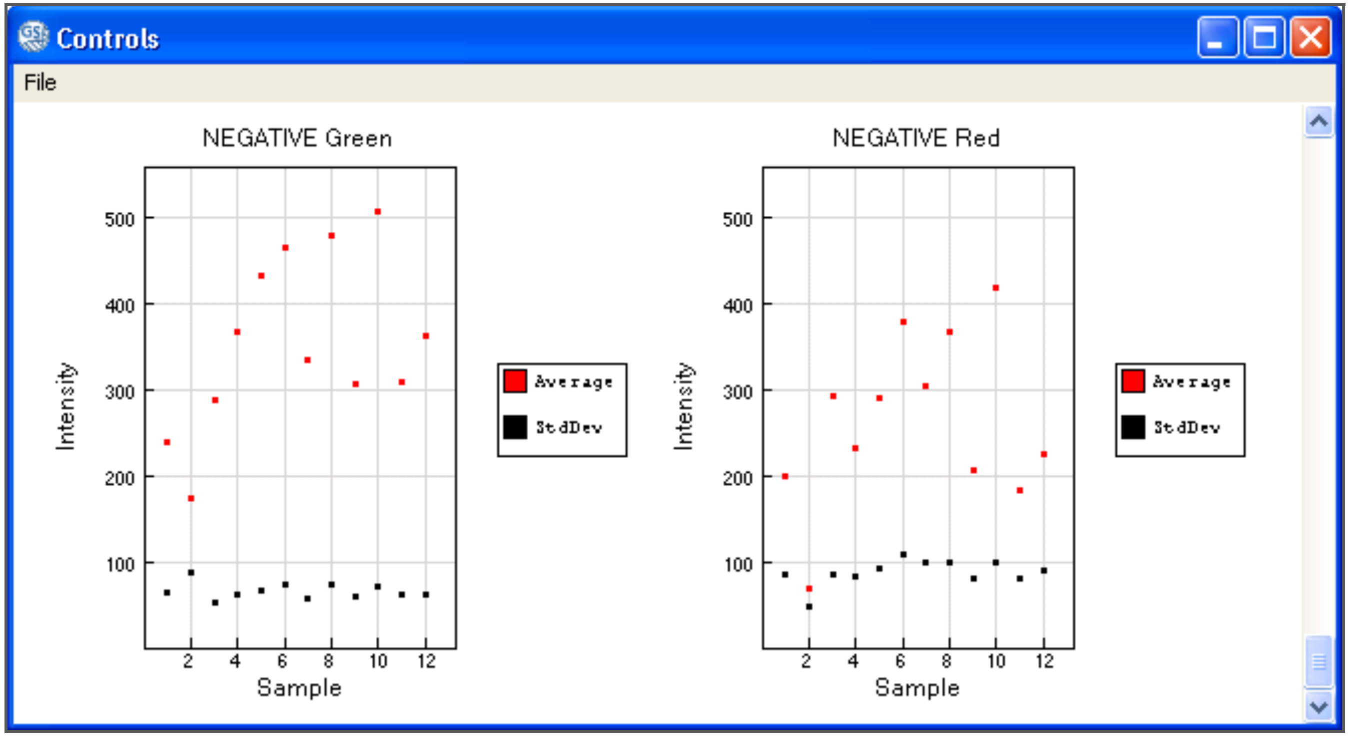
Figure 62 Non-polymorphic Controls



Negative controls

The probes are randomly permutated bisulphite-converted sequences that should not hybridise to DNA. The mean of these probes determines the system background

Purpose	Name	Number on the Array	Evaluate Green (GRN)	Evaluate Red (RED)	Expected Intensity
Negative	Average*	600	+	+	Background
Negative	StdDev**		+	+	Background



The graph displays average and standard deviation of 600 negative controls.

Under normal experimental conditions, the signal intensities from negative controls can vary from approximately 100 to 1000 units. Since these controls are sample-dependent, a dramatic increase in these counts may indicate poor DNA template quality prior to the bisulfite conversion step.

Detection p-value

It is computed from the background model and it indicates the chance that the target sequence signal is distinguishable from the negative controls.

It is as an objective measure of overall probe performance.

Small p-values indicate a good position. Positions with non-significant p-values (typically >0.01 or >0.05) should not be trusted and should be filtered out.

SNPs

Fo example: SNP in the target CpG site of a type II probe

```
Probe         5'----ACTTTTCTC

Target        3'----TGAAAAGAGCGA---5'  C ALLELE


Probe         5'----ACTTTTCTC

Target        3'----TGAAAAGAGTGA---5'  T ALLELE
```

SNPs

Fo example: SNP in the target CpG site of a yype II probe

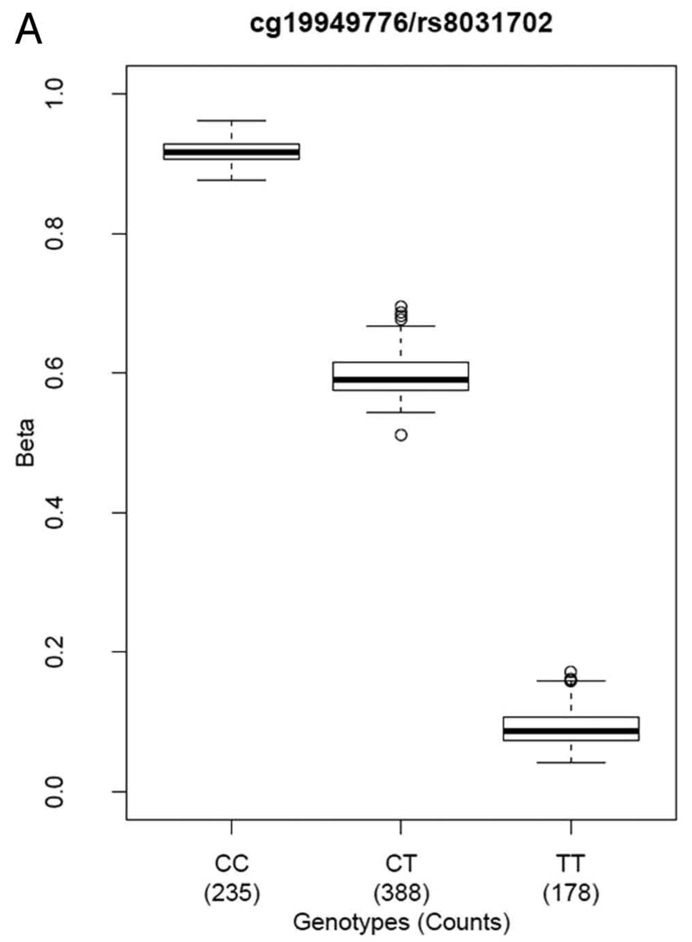
Probe	5'----ACTTTTCTCG
Target	3'----TGAAAAGAG<u>C</u>GA---5' **C ALLELE**
Probe	5'----ACTTTTCTCA
Target	3'----TGAAAAGAG<u>**T**</u>GA---5' **T ALLELE**

SNPs

SNPs can occur:

- In the target CpG site
- In the base flanking the target CpG site
- Within the probe sequence (the nearest to the 3’, the more likely it is that hybridization efficiency is affected)

- 59892 probes have a SNP >10bp from query site
- 36535 probes have a SNP ≤10bp from query site



SNPs

Comprehensive characterization, annotation and innovative use of Infinium DNA methylation BeadChip probes

Wanding Zhou, Peter W. Laird and Hui Shen*

Center for Epigenetics, Van Andel Research Institute, 333 Bostwick Ave NE, Grand Rapids, MI 49503, USA

Received July 08, 2016; Revised October 05, 2016; Editorial Decision October 10, 2016; Accepted October 12, 2016

Beta Values and M-Values

β-values and M-values

$$\beta = \frac{M}{M+U}$$

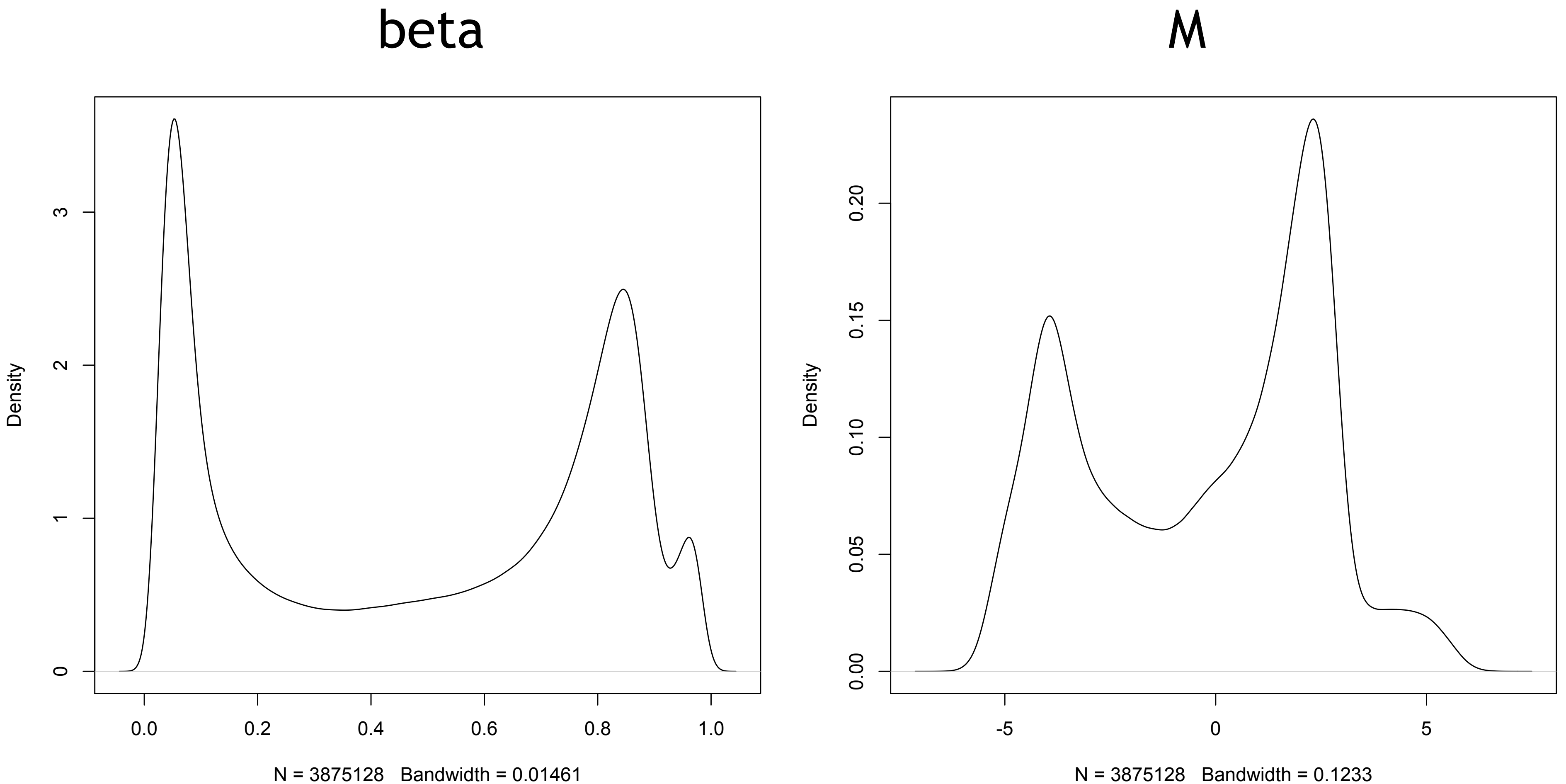
Percentage of methylation at the given CpG site; it is a continuous variable between 0 (absent methylation) and 1 (completely methylated)

$$M = \log_2 \frac{M}{U}$$

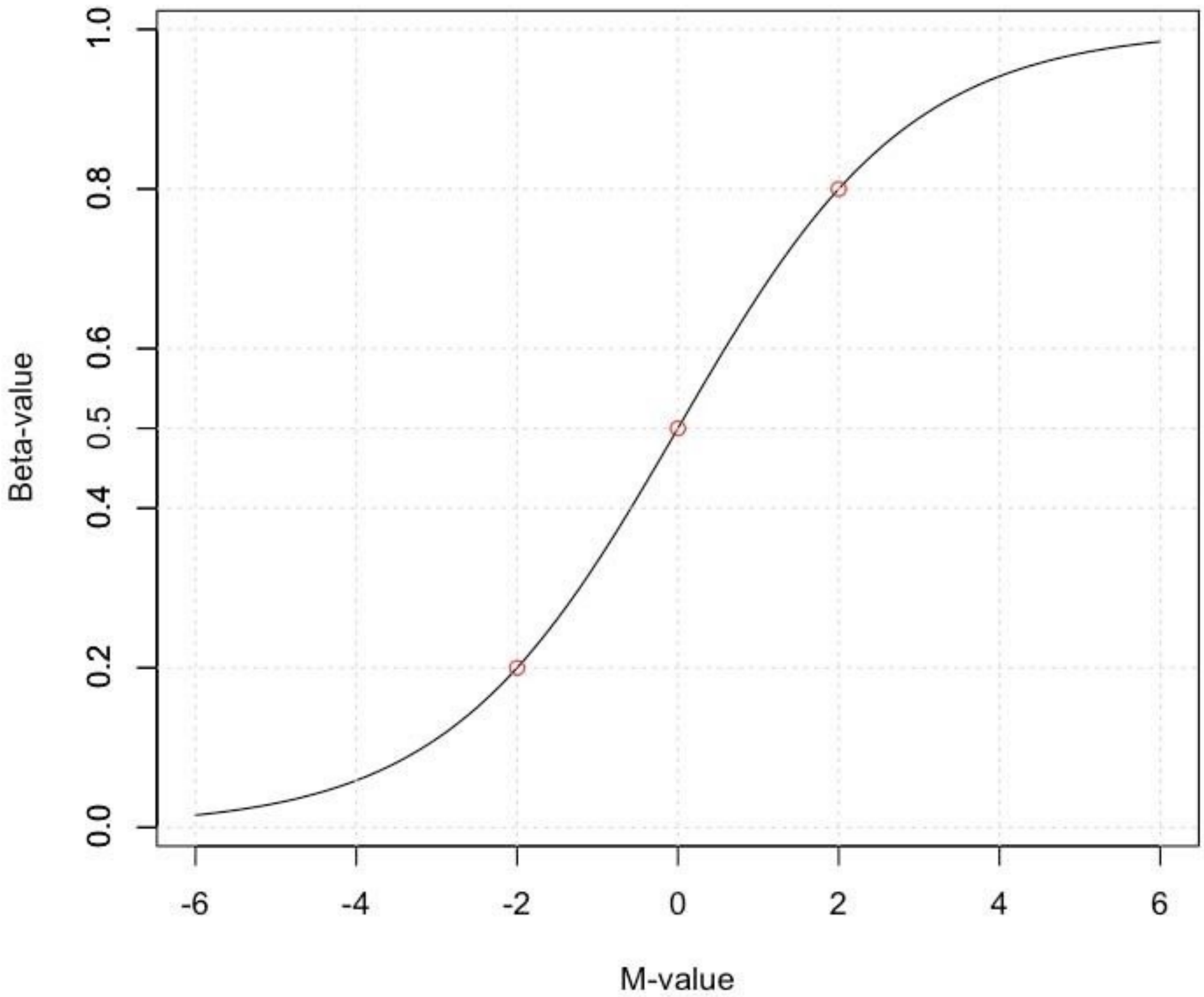
It is a continuous variable which can in principle take on any value on the real line

$$Beta_i = \frac{2^{M_i}}{2^{M_i} + 1} \qquad M_i = \log_2\left(\frac{Beta_i}{1 - Beta_i}\right)$$

β-values and M-values



β-values and M-values



Normalization

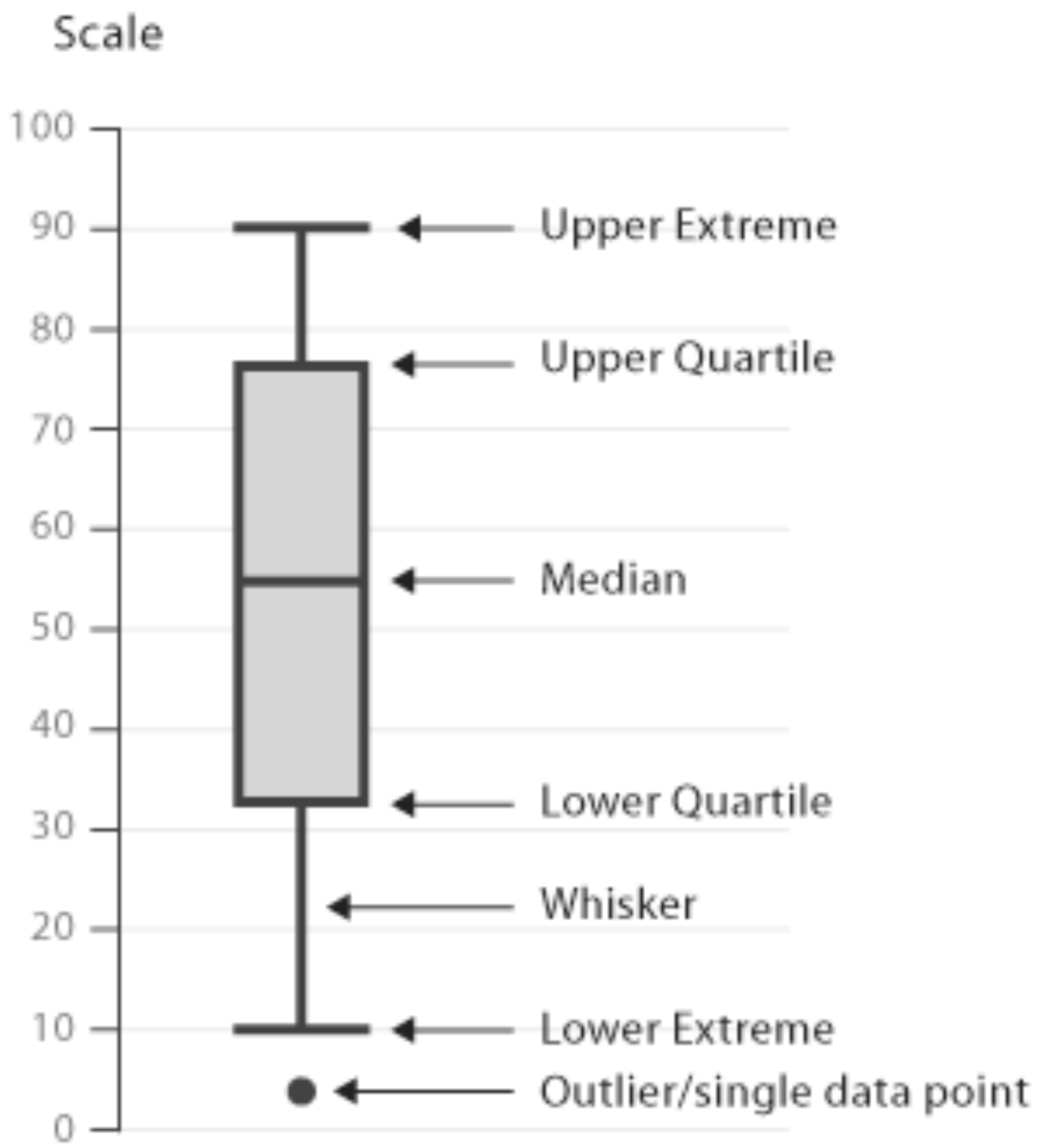
Why normalize the data?

Our experimental goal is to identify biological variation, but technical variation can hide the real data

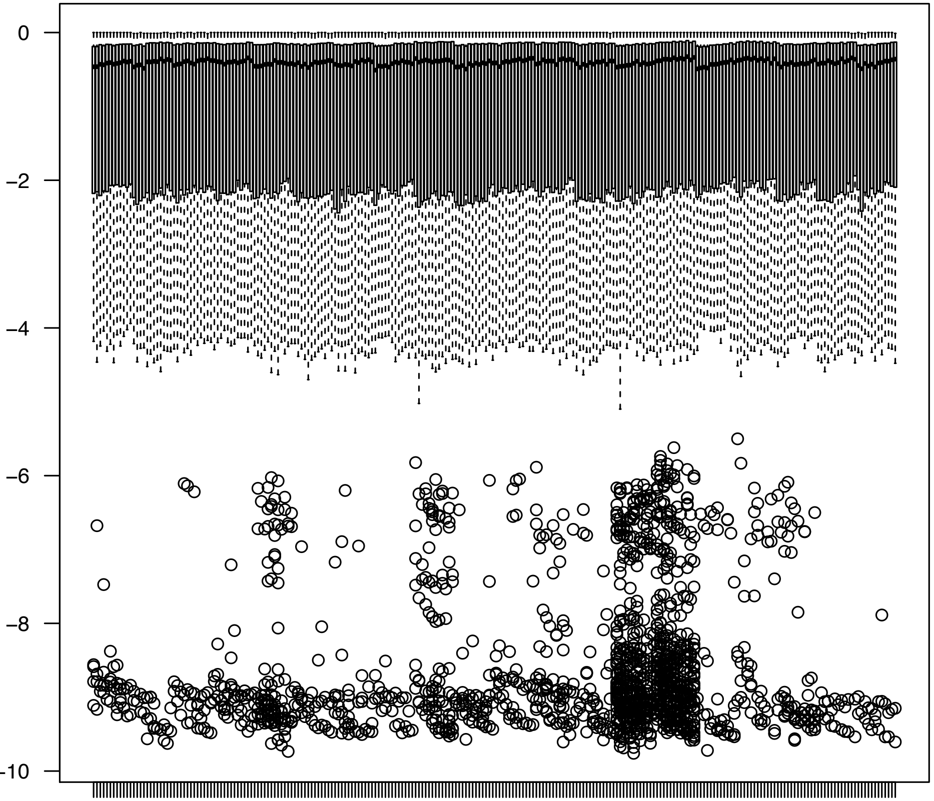
The aim of normalization methods is to remove unavoidable technical variation, in particular systematic bias.

Technical variation:	Normalization
- systematic (intrinsic to the experiment)	+
- batch effetcs	+/-
- random	-

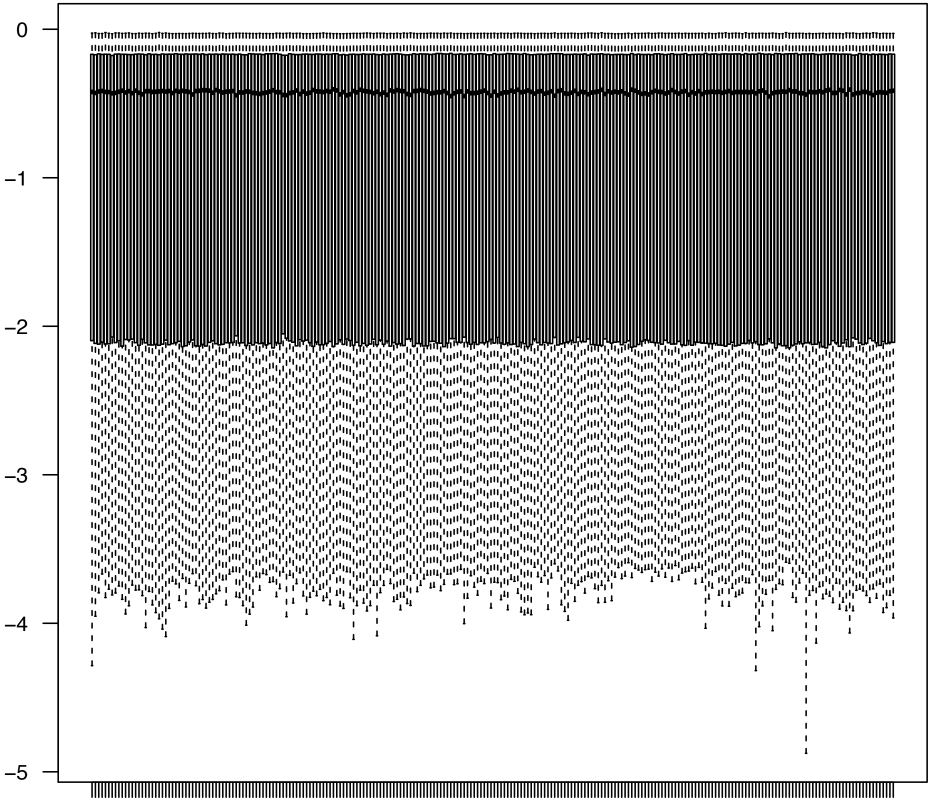
Boxplots


Scale
100
90
80
70
60
50
40
30
20
10
0
Upper Extreme
Upper Quartile
Median
Lower Quartile
Whisker
Lower Extreme
Outlier/single data point

Why normalize the data?



Before normalization



After normalization

Within- and between-array normalization

Whithin-array normalization

Technical biases linked to the array design

1. Background correction
2. Color bias adjustement
3. Infinium I/II-type bias correction

Between-array normalization

Array-to-array variations

Background correction - GenomeStudio

- Background is calculated in the two channels separately.
- Channel background is 5% percentile of the negative controls in the given channel. Negative probes outliers are not removed.
- **Background is subtracted from probe intensities in the same channel. If intensity becomes negative, it is set to 0.**

BUT… truncation of the data at low intensity signals

Background correction - noob

Low-level processing of Illumina Infinium DNA Methylation BeadArrays

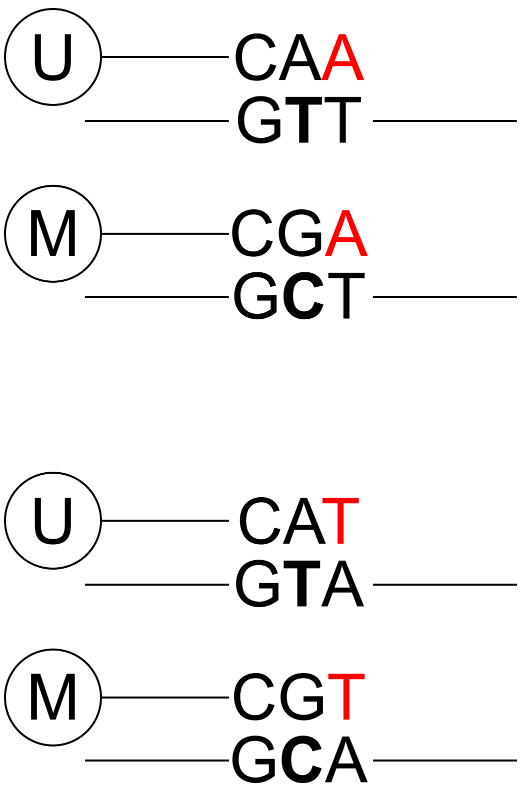
Timothy J. Triche Jr[1,2,*], Daniel J. Weisenberger[2,3], David Van Den Berg[1,2], Peter W. Laird[2,3,4] and Kimberly D. Siegmund[1,*]

Out-of-band probes (OOB)

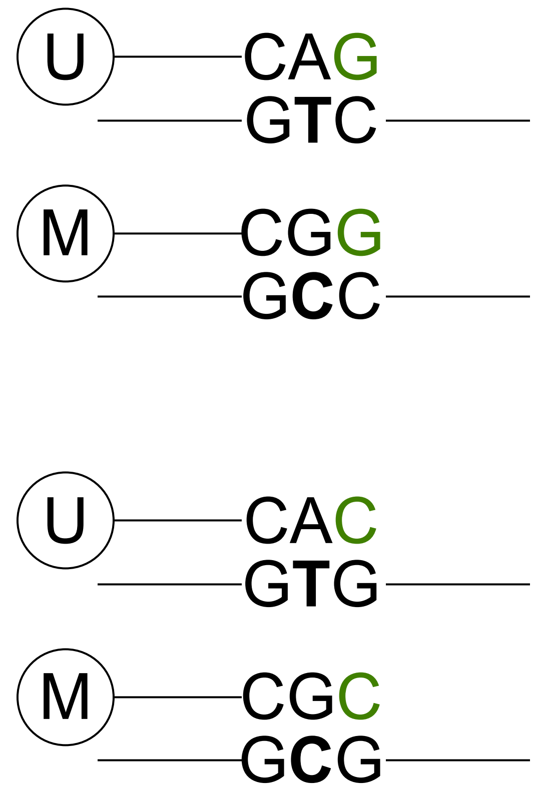
Out of band probes

These are the measurements of Type I probes in the wrong color channel → they can be used for estimating non-specific fluorescence in background correction

The methylation level is measured in the Red channel…but the scanner reads them also in the Green channel → Green background signal



The methylation level is measured in the Green channel…but the scanner reads them also in the Red channel → Red background signal



Low-level processing of Illumina Infinium DNA Methylation BeadArrays

Timothy J. Triche Jr[1,2,*], Daniel J. Weisenberger[2,3], David Van Den Berg[1,2], Peter W. Laird[2,3,4] and Kimberly D. Siegmund[1,*]

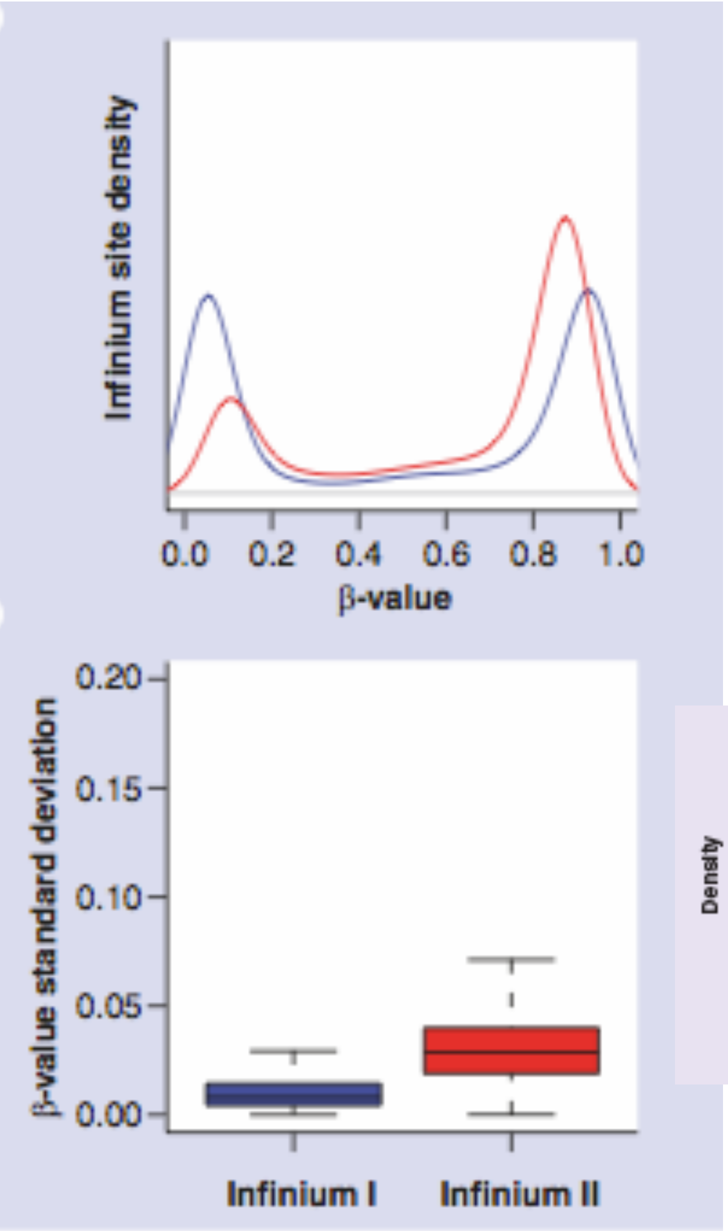
Out-of-band probes (OOB)

provide tens of thousands of features for measuring background instead of the much smaller number of negative control probes on the platforms

normal-exponential convolution and gamma convolution model

Within- and between-platform artefacts are substantially reduced, with concomitant improvement in sensitivity

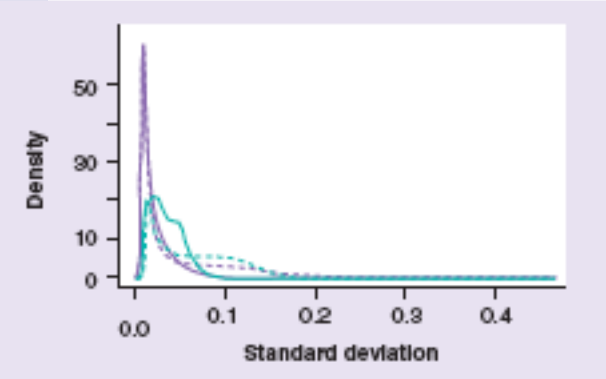
Infinium I/II-type bias



Desity of ß-values from Infinium II probes is shifted towards the centre: Infinium II assay is less sensitive for the detection of extreme methylation values (i.e., 0 and 1) than the Infinium I assay

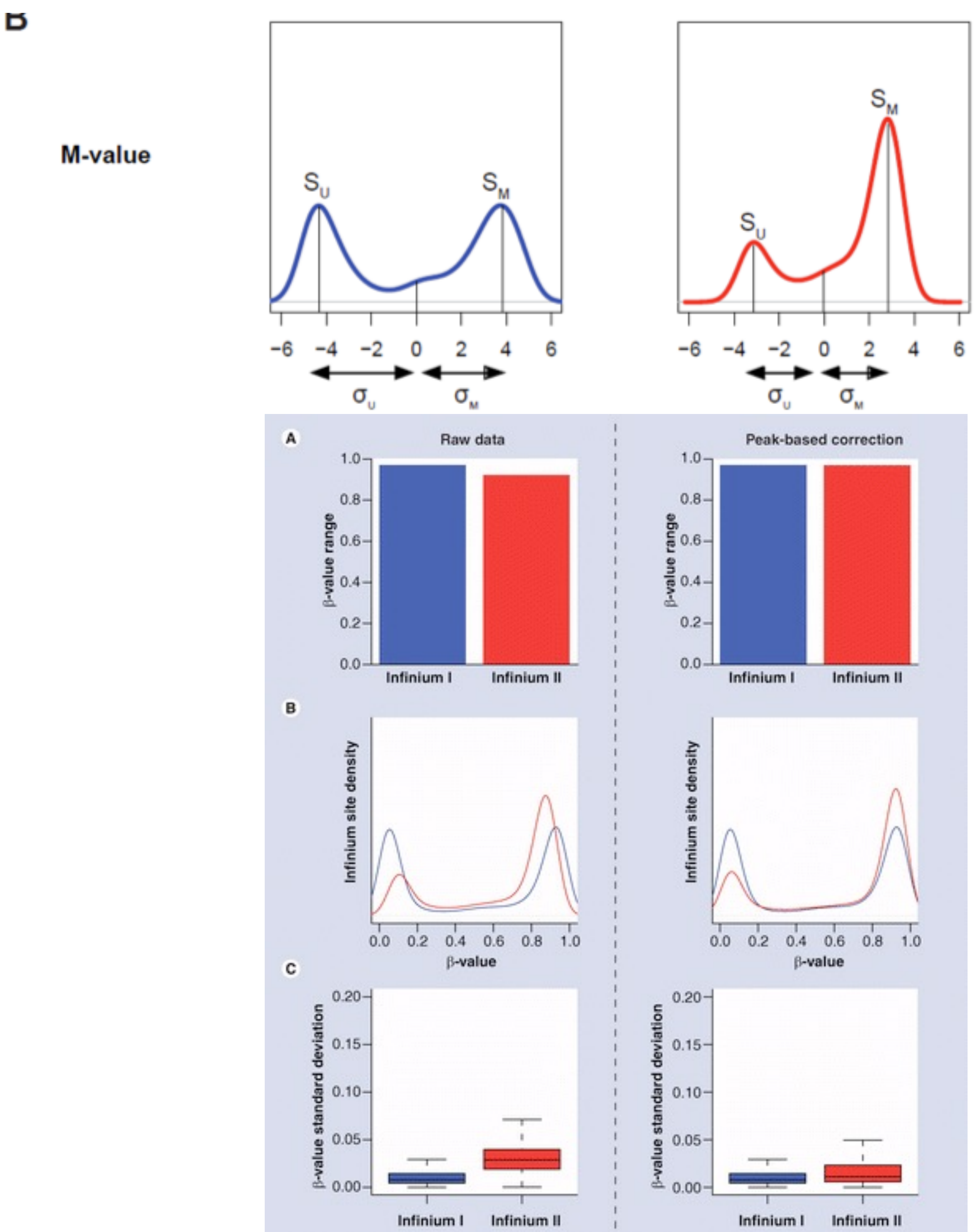
Dedeurwaeder et al, 2011

Touleimat, 2012



Infinium II probes have higher standard deviation between replicates: ß-values obtained from Infinium II probes are less accurate and reproducible than those obtained from Infinium I probes.

Infinium I/II-type bias: peak-based correction (PBC)



1. The raw β-values are converted to M-values;
2. Using the kernel density estimation, the unmethylated and methylated peaks are determined independently for both Infinium I and II
3. Positive and negative M-values are first corrected using the positive and negative peak summits as references, for both Infinium I and II
4. Corrected positive and negative M-values are re-corrected to match Infinium I positive and negative peaks
5. Corrected M-values are converted back to β-values

Reduction of the variance between replicates!

Infinium I/II-type bias: peak-based correction (PBC)

PBC efficiently corrects for InfI/Inf2 shift and improves results.

PBC implemented in R packages Illumina Methylation Analyzer (IMA) and wateRmelon (function **fuks**)

However, PBC depends on bimodal shape of methylation density profiles and it breaks down when the methylation density distribution does not exhibit well-defined peaks/modes.

Quantile normalization

Quantile normalization → non-parametric normalization procedure in which each feature value (row) is replaced with the mean of the features across all the samples with the same rank or quantile.

Raw data

2	4	4	5
5	14	4	7
4	8	6	9
3	8	5	8
3	9	3	5

***Order* values within each sample (or column)**

2	4	3	5
3	8	4	5
3	8	4	7
4	9	5	8
5	14	6	9

***Average* across rows and substitute value with average**

3.5	3.5	3.5	3.5
5.0	5.0	5.0	5.0
5.5	5.5	5.5	5.5
6.5	6.5	6.5	6.5
8.5	8.5	8.5	8.5

***Re-order* averaged values in original order**

3.5	3.5	5.0	5.0
8.5	8.5	5.5	5.5
6.5	5.0	8.5	8.5
5.0	5.5	6.5	6.5
5.5	6.5	3.5	3.5

Quantile normalization

Quantile normalization can be used:

To normalize Type I and Type II probes→ within array normalization

Probes

Type I	Type II
2	4
5	14
4	8
3	8
3	9

To normalize distribution between the arrays → between array normalization

Samples

Probes

2	4	4	5
5	14	4	7
4	8	6	9
3	8	5	8
3	9	3	5

Quantile normalization

Quantile normalization can be used:

To normalize Type I and Type II probes→ within array normalization

Probes	*Type I*	*Type II*
	2	4
	5	14
	4	8
	3	8
	3	9

To normalize distribution between the arrays → between array normalization

Samples

Probes				
	2	4	4	5
	5	14	4	7
	4	8	6	9
	3	8	5	8
	3	9	3	5

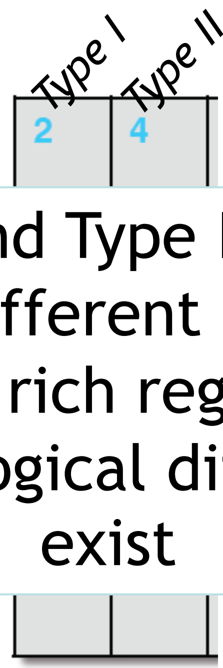
BUT…

Quantile normalization is a global-adjustment method which assumes the **statistical** distribution of each sample is the same.

Quantile normalization

Quantile normalization can be used:

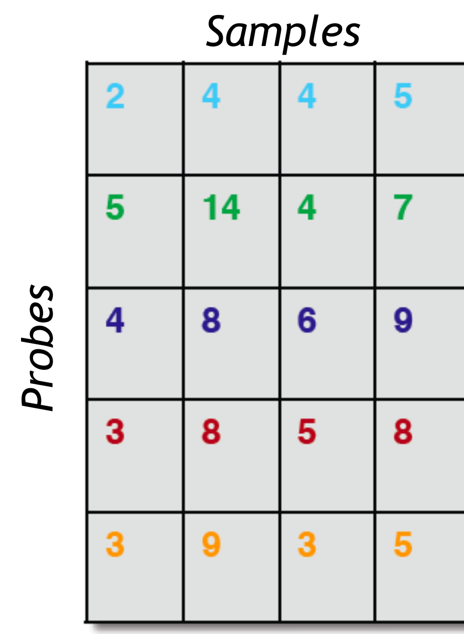
To normalize Type I and Type II probes→ within array normalization



Type I and Type II probes have a different coverage of CpG rich regions → True biological differences exist

BUT…

To normalize distribution between the arrays → between array normalization



Quantile normalization is a global-adjustment method which assumes the **statistical** distribution of each sample is the same.

Subset quantile normalization

Reference quantiles are estimated on a small set on probes that are considered more reliable and biologically similar and that are used as anchors. Then the intensities of the remaining probes are adjusted by interpolation onto the distribution of the subset probes.
Touleimat

SWAN (Subset-quantile Within Array Normalization) (Maksimovic et al., 2012) → probes are defined as biologically similar on the basis of the CpG content.

SQN (preprocessQuantile) (Touleimat and Tost, 2012) → probes are defined as biologically similar on the basis of the region where they map (based on the 'relation to CpG island' and based on the 'relation to gene sequence'). SQN is applied to all samples simultaneously → at the same time within and between array normalization

Quantile normalization

Quantile normalization can be used:

To normalize Type I and Type II probes→ within array normalization

Type I	Type II
2	4

Type I and Type II probes have a different coverage of CpG rich regions → True biological differences exist

To normalize distribution between the arrays → between array normalization

Samples

Probes				
	2	4	4	5
	5	14	4	7
	4	8	6	9
	3	8	5	8
	3	9	3	5

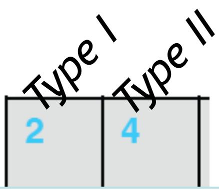
BUT...

Quantile normalization is a global-adjustment method which assumes the **statistical** distribution of each sample is the same.

Quantile normalization

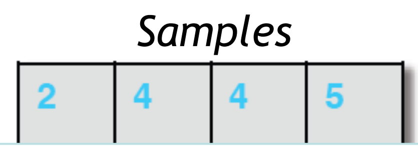
Quantile normalization can be used:

To normalize Type I and Type II probes→ within array normalization



Type I and Type II probes have a different coverage of CpG rich regions → True biological differences exist

To normalize distribution between the arrays → between array normalization



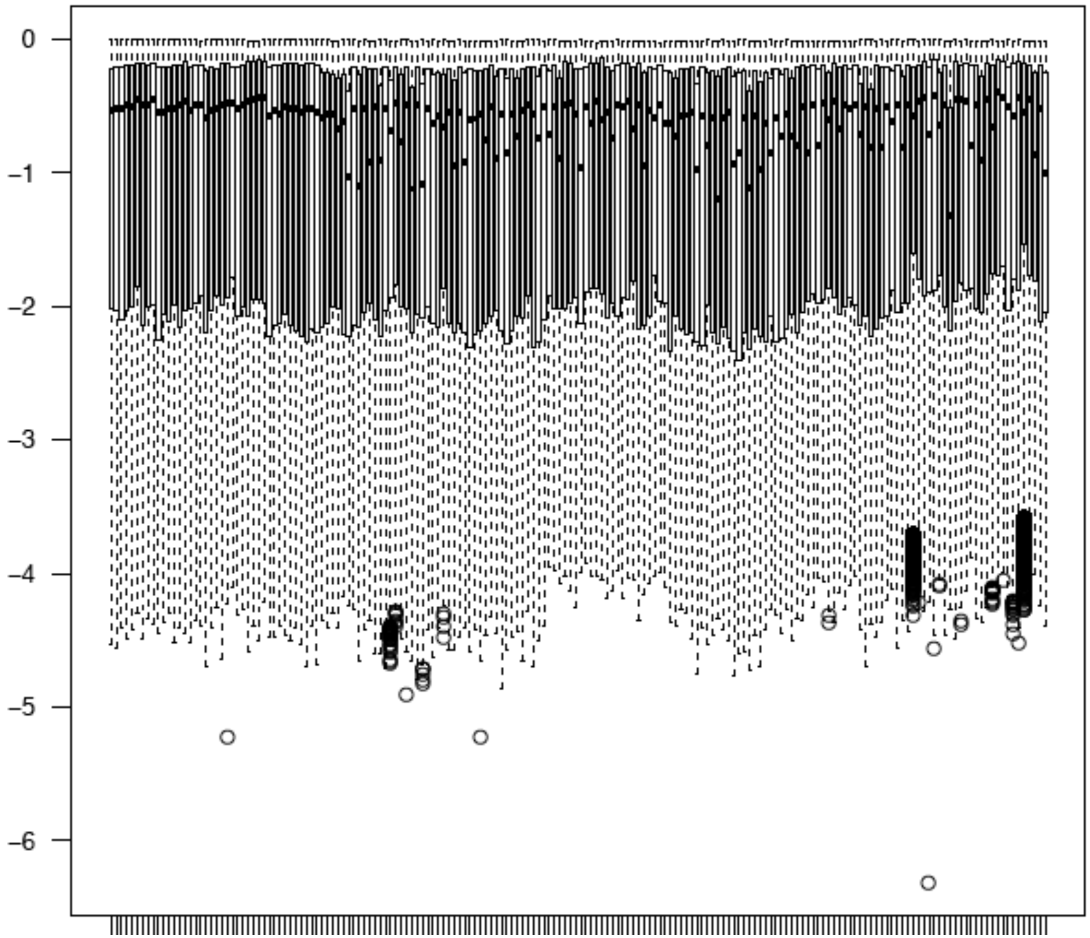
Large differences in DNA methylation can exist between samples (for example, normal vs tumor)

BUT…

Quantile normalization is a global-adjustment method which assumes the **statistical** distribution of each sample is the same.

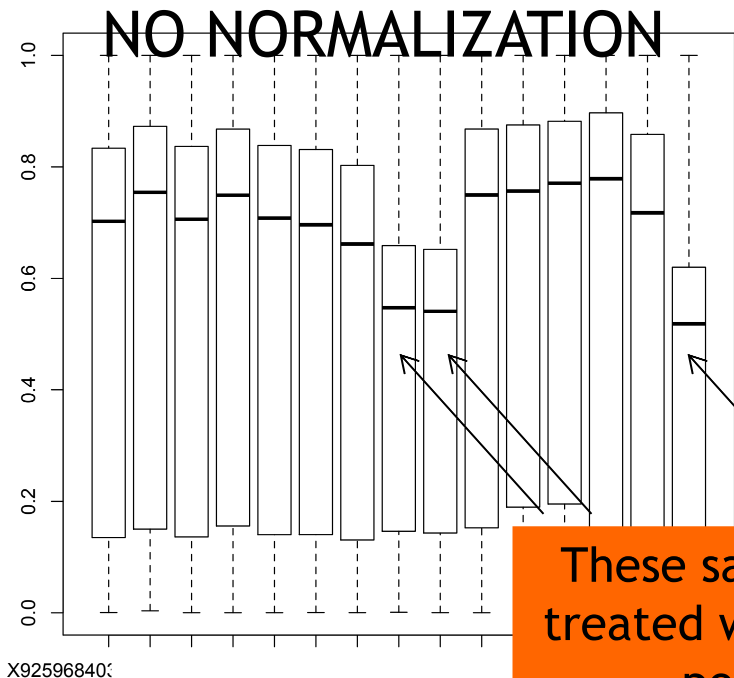
(Subset) quantile normalization

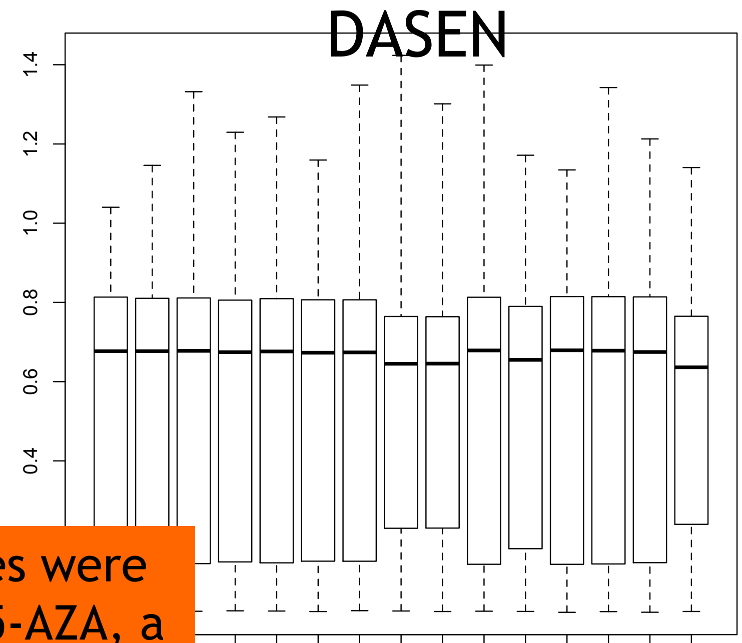
In general, approaches that are based on (subset) quantile normalization are not recommended for cases where global changes in DNA methylation are expected



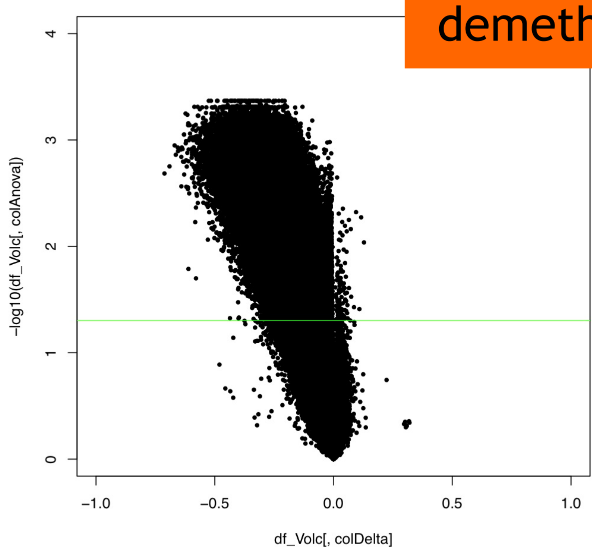
They can lead to misleading results!

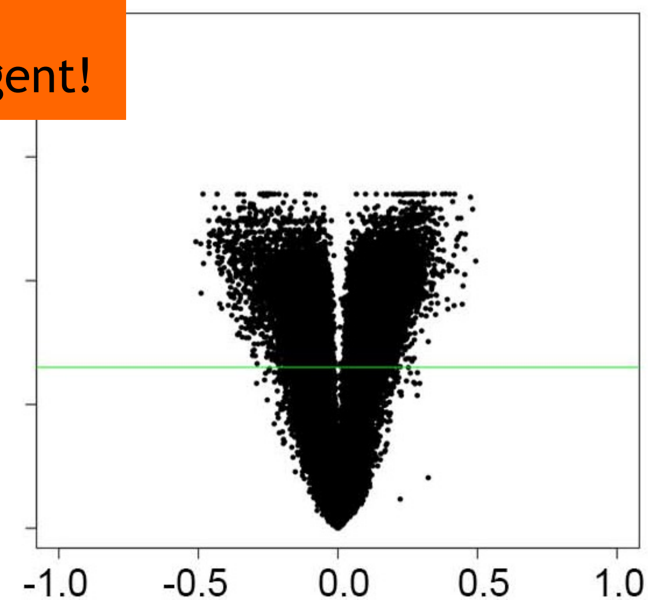
Improper use of dasen normalization





These samples were treated with 5-AZA, a powerful demethylating agent!





Functional normalization of 450k methylation array data improves replication in large cancer studies

Jean-Philippe Fortin[1], Aurélie Labbe[2,3,4], Mathieu Lemire[5], Brent W Zanke[6], Thomas J Hudson[5,7], Elana J Fertig[8], Celia MT Greenwood[2,9,10] and Kasper D Hansen[1,11*]

Abstract

We propose an extension to quantile normalization that removes unwanted technical variation using control probes. We adapt our algorithm, functional normalization, to the Illumina 450k methylation array and address the open problem of normalizing methylation data with global epigenetic changes, such as human cancers. Using data sets from The Cancer Genome Atlas and a large case–control study, we show that our algorithm outperforms all existing normalization methods with respect to replication of results between experiments, and yields robust results even in the presence of batch effects. Functional normalization can be applied to any microarray platform, provided suitable control probes are available.

Quantile normalization is applied to the control probes (Infinium control probes + out of band probes)

A practical guide to 450k normalization

If large differences exist between the conditions under study:

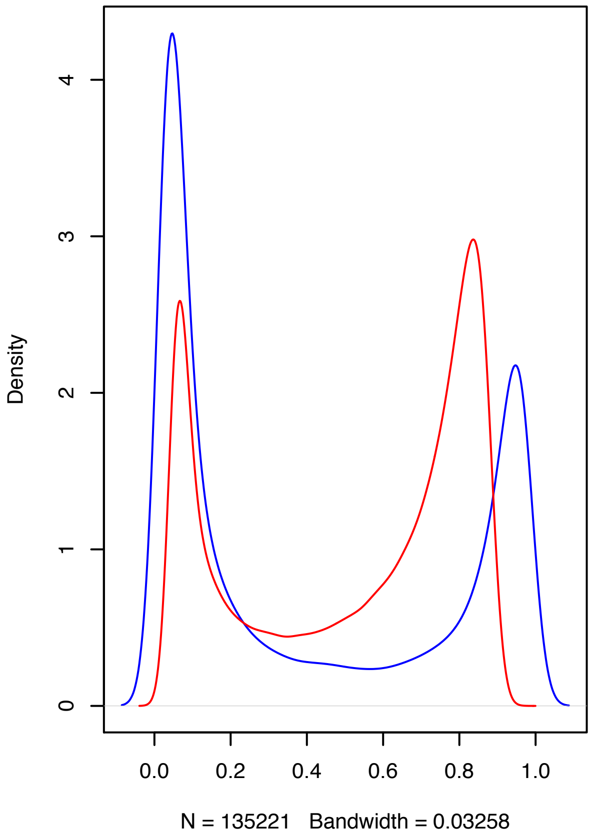
- Do not normalize
- If you really want to normalize, use Functional Normalization.

On the contrary, for the investigation of subtle differences, there is a need to ensure maximum sensitivity to detect differential DNA methylation → the use of the correct normalization approach can help!

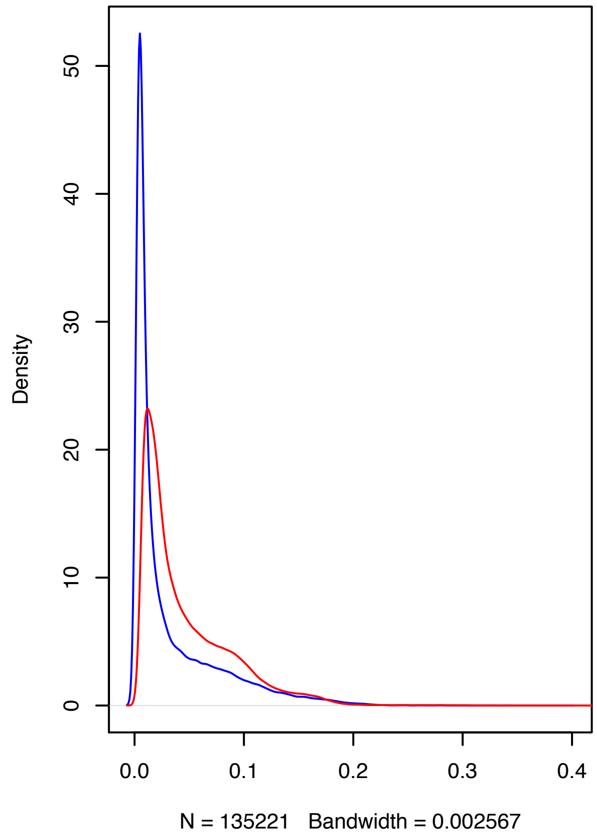
- Usually **preprocessQuantile** performs very well!

To check the effects of normalization approaches

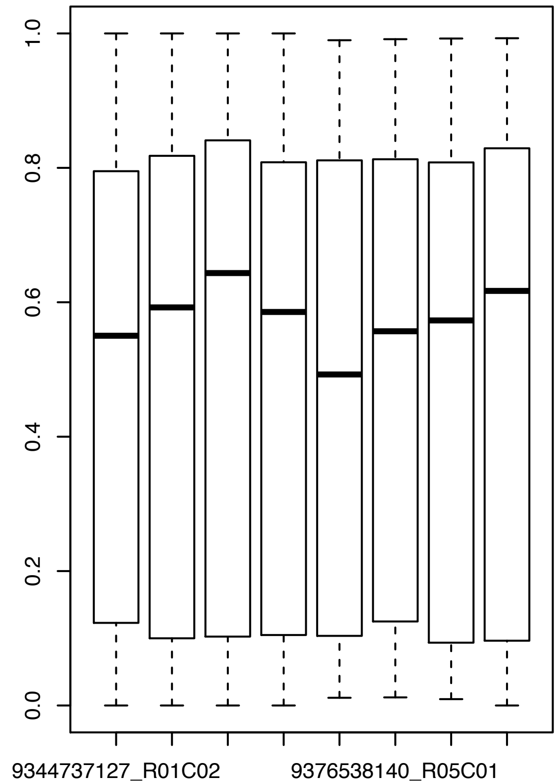
Density of means of beta values



Density of standard deviations of beta values

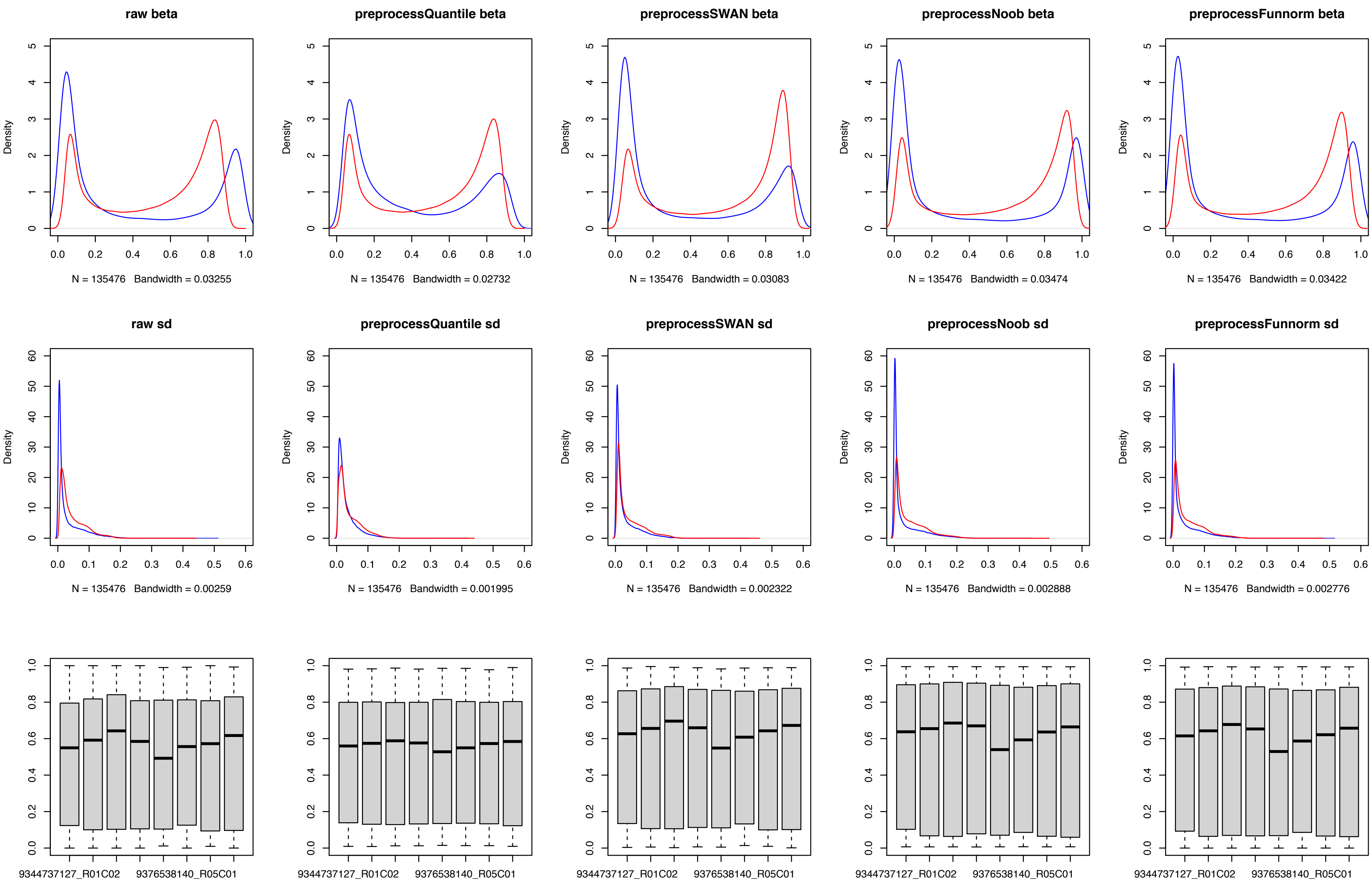


Boxplots of beta values



Normalization functions in minfi:

Function name	Package	Input	Output	Ref
preprocessNoob	minfi	RGset	Mset	Triche 2013
preprocessSWAN	minfi	RGset	Mset	Maksimovic 2012
preprocessQuantile	minfi	RGset or Mset	Genomic Ratio Set	Toulemait 2012
preprocessFunnorm	minfi	RGset	Mset	Fortin 2014


raw beta
preprocessQuantile beta
preprocessSWAN beta
preprocessNoob beta
preprocessFunnorm beta
Density
N = 135476 Bandwidth = 0.03255
N = 135476 Bandwidth = 0.02732
N = 135476 Bandwidth = 0.03083
N = 135476 Bandwidth = 0.03474
N = 135476 Bandwidth = 0.03422
raw sd
preprocessQuantile sd
preprocessSWAN sd
preprocessNoob sd
preprocessFunnorm sd
N = 135476 Bandwidth = 0.00259
N = 135476 Bandwidth = 0.001995
N = 135476 Bandwidth = 0.002322
N = 135476 Bandwidth = 0.002888
N = 135476 Bandwidth = 0.002776
9344737127_R01C02
9376538140_R05C01

Why normalize the data?

Our experimental goal is to identify biological variation, but technical variation can hide the real data

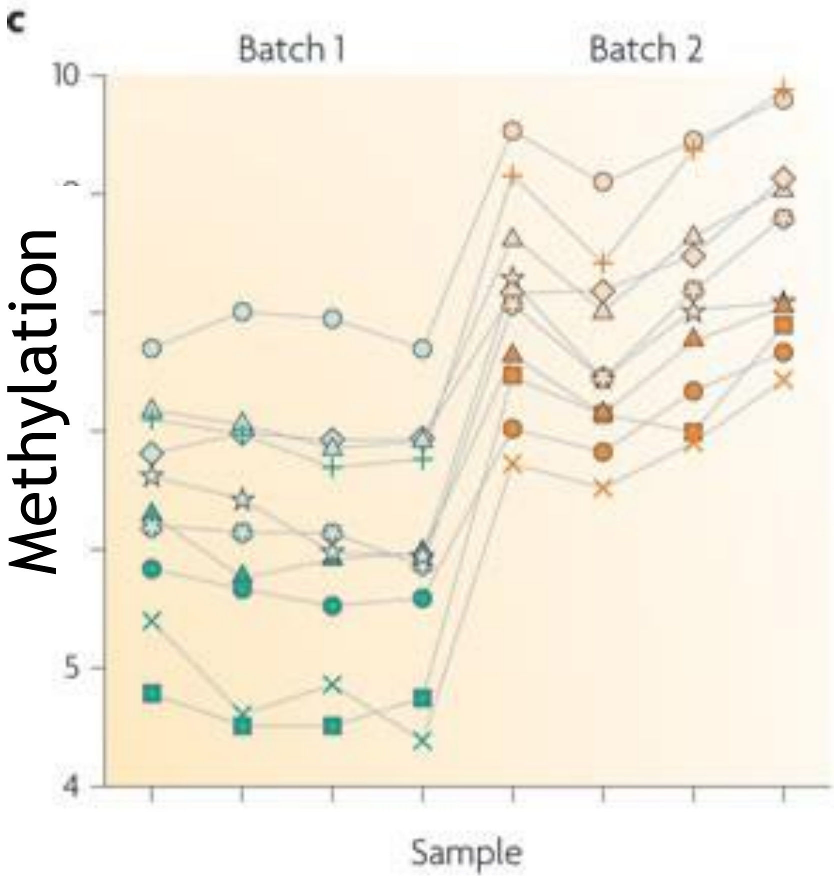
The aim of normalization methods is to remove unavoidable technical variation, in particular systematic bias.

Technical variation:	Normalization
- systematic (intrinsic to the experiment)	**+**
- batch effects	**+/-**
- random	**-**

Batch effects

Batch effects are one of the worst researchers' nightmares

Batch effects --> sub-groups of measurements that have qualitatively different behaviour across conditions and are unrelated to the biological or scientific variables in a study.

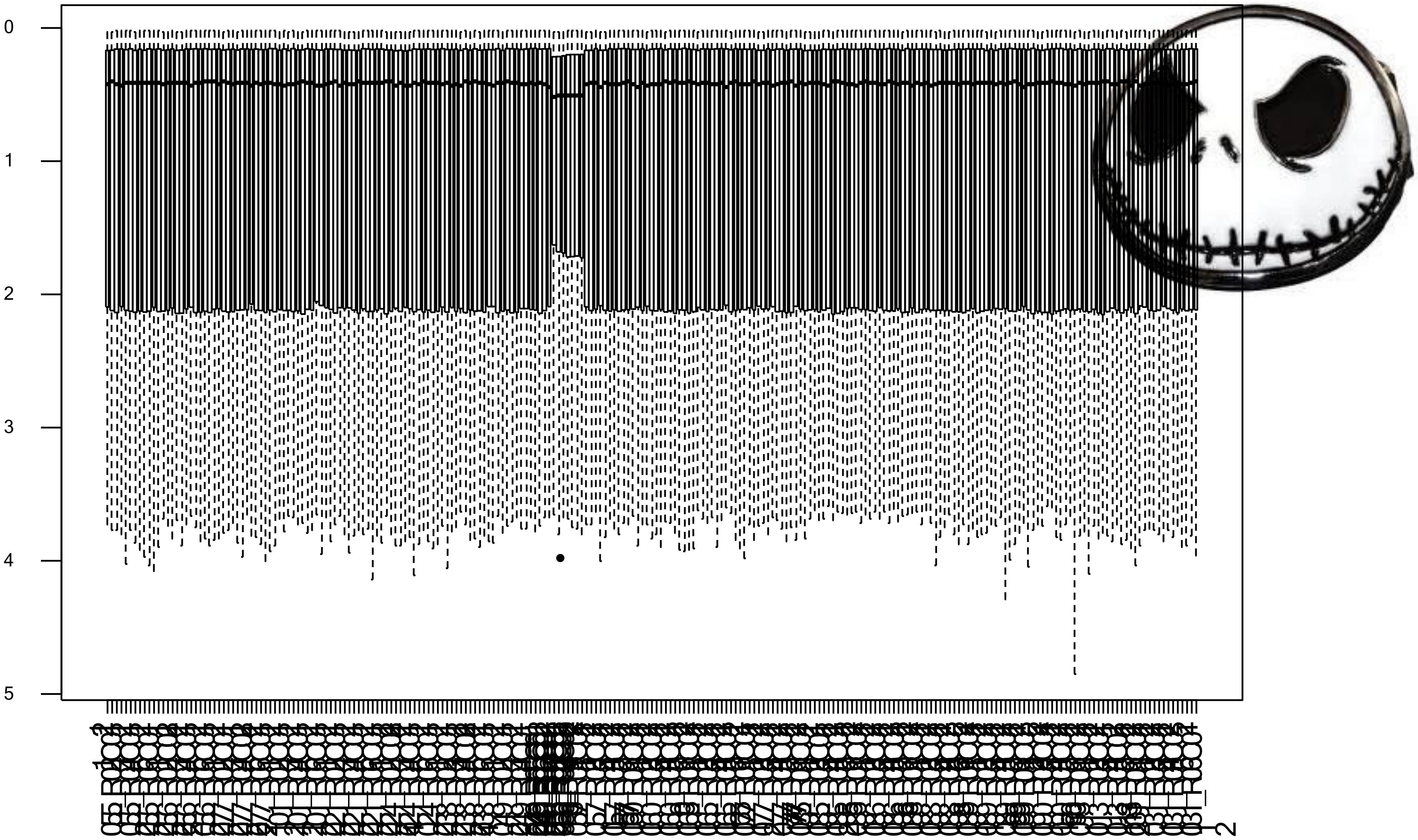


Batch effects can affect both low-dimensional and high-throughput experiments.

If not properly dealt with, batch effects can have a particularly strong and pervasive impact.

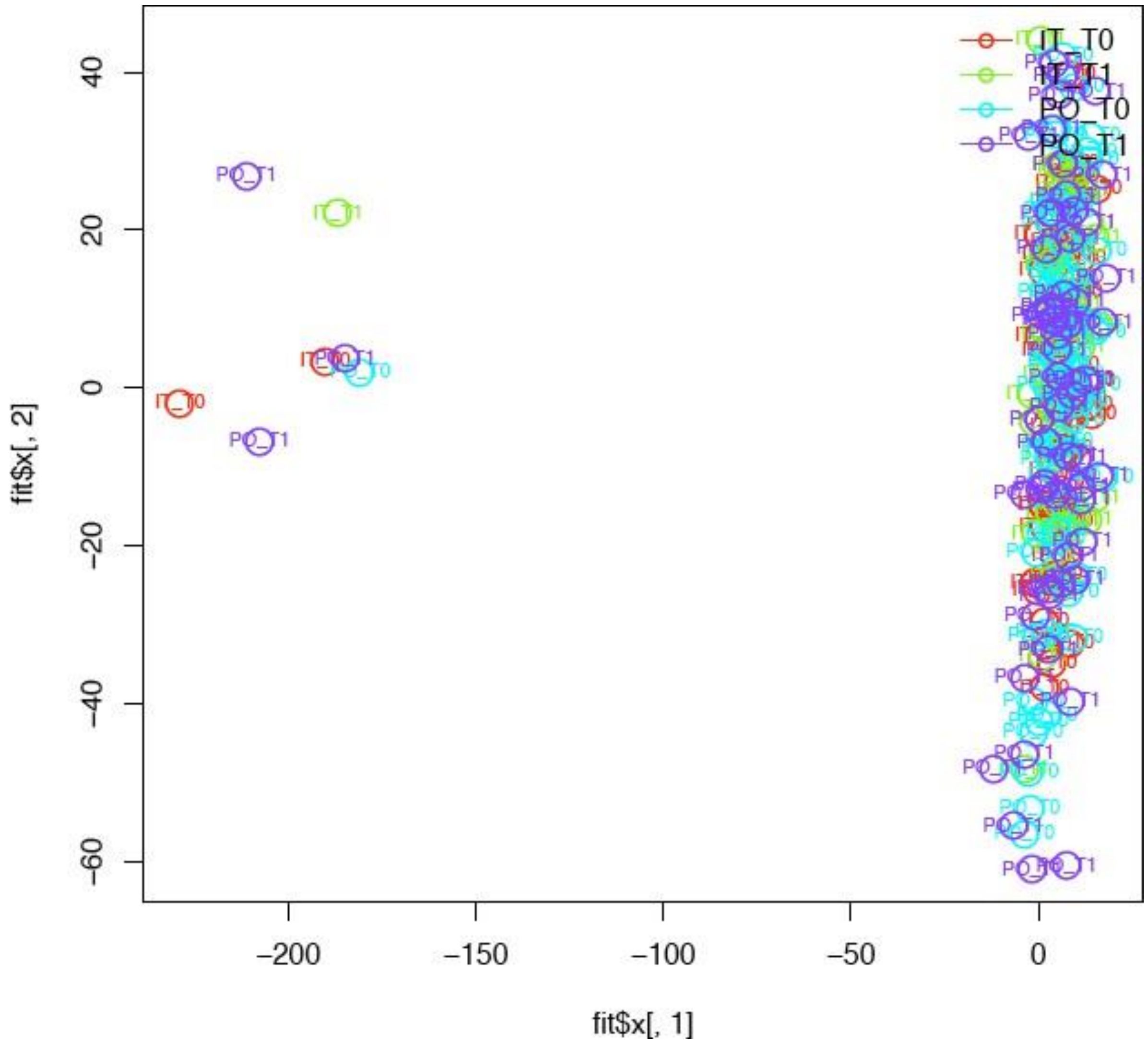
Leek et al., 2010

Batch effects



After normalization

PCA – Technical Variations – Batch Effect



After normalization

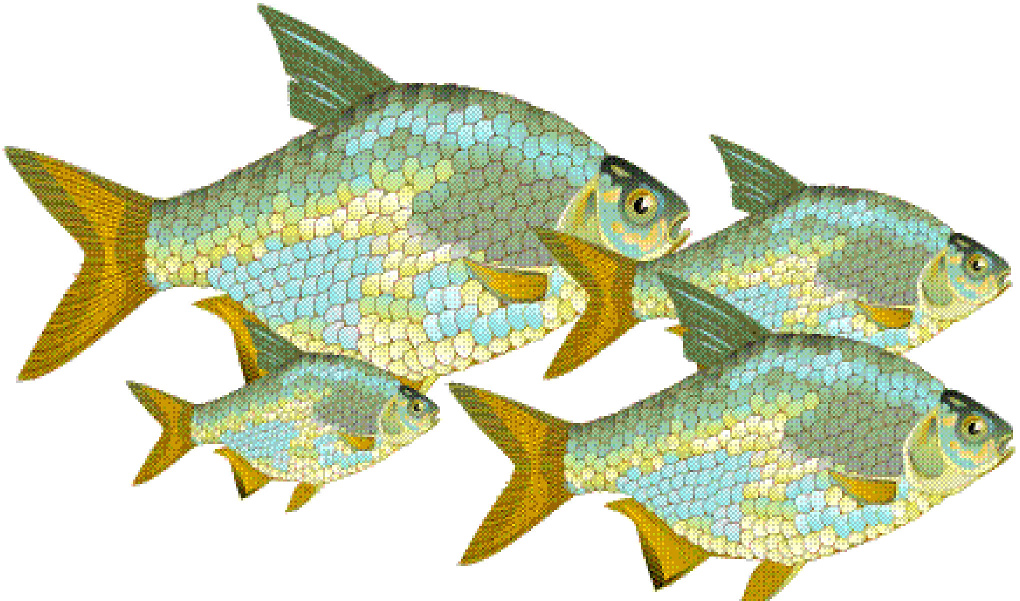
Dimension-reduction techniques

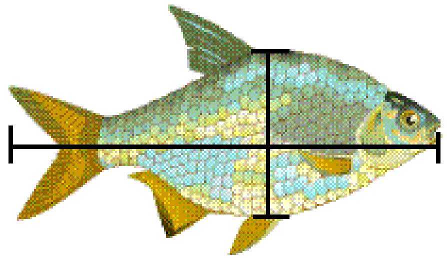
How can we detect low dimensional structure in high dimensional data?

The objective of dimension-reduction techniques is to project points in multidimensional space onto a space of fewer dimension

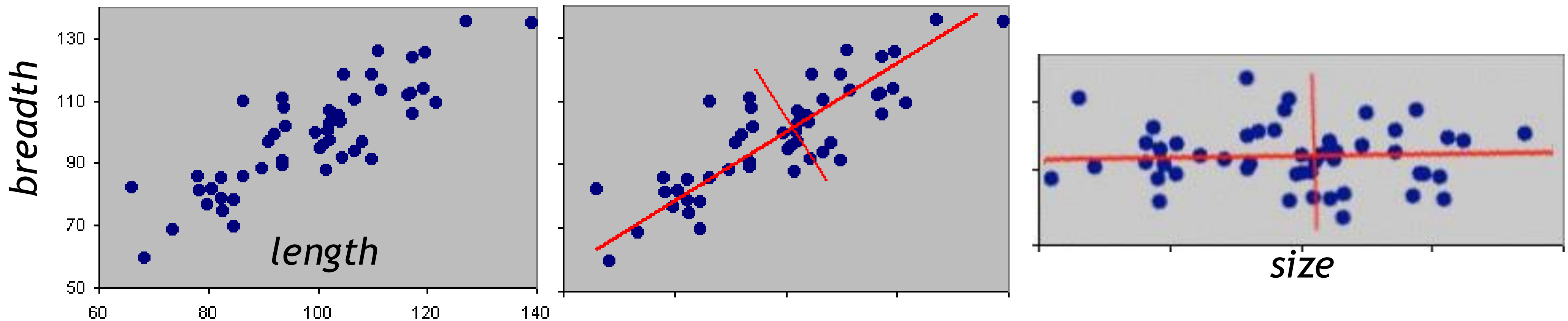
We can use PCA to check if there are large differences in DNA methyaltion profiles in our sample, due to biological causes or batch effects

PCA



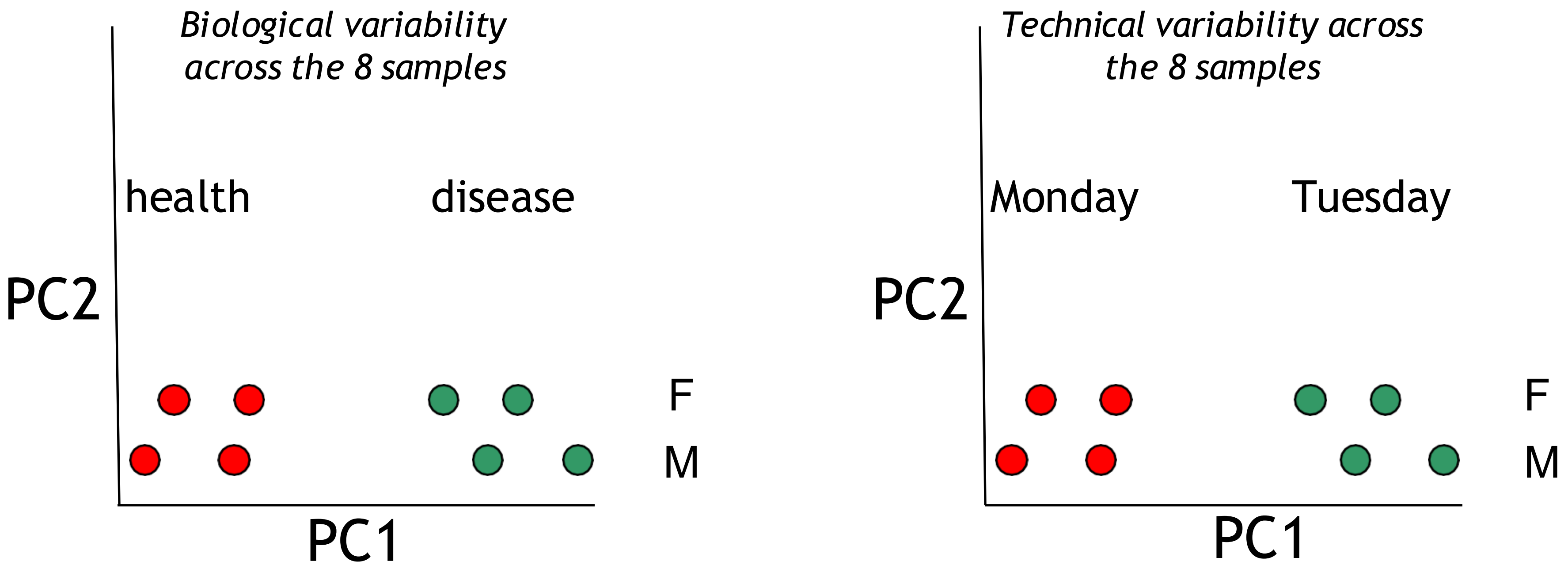


There is a relationship between lenght and breadth, longer fishes tend to be broader.



We can reduce the data's dimensionality from two (length and breadth) to one (size)

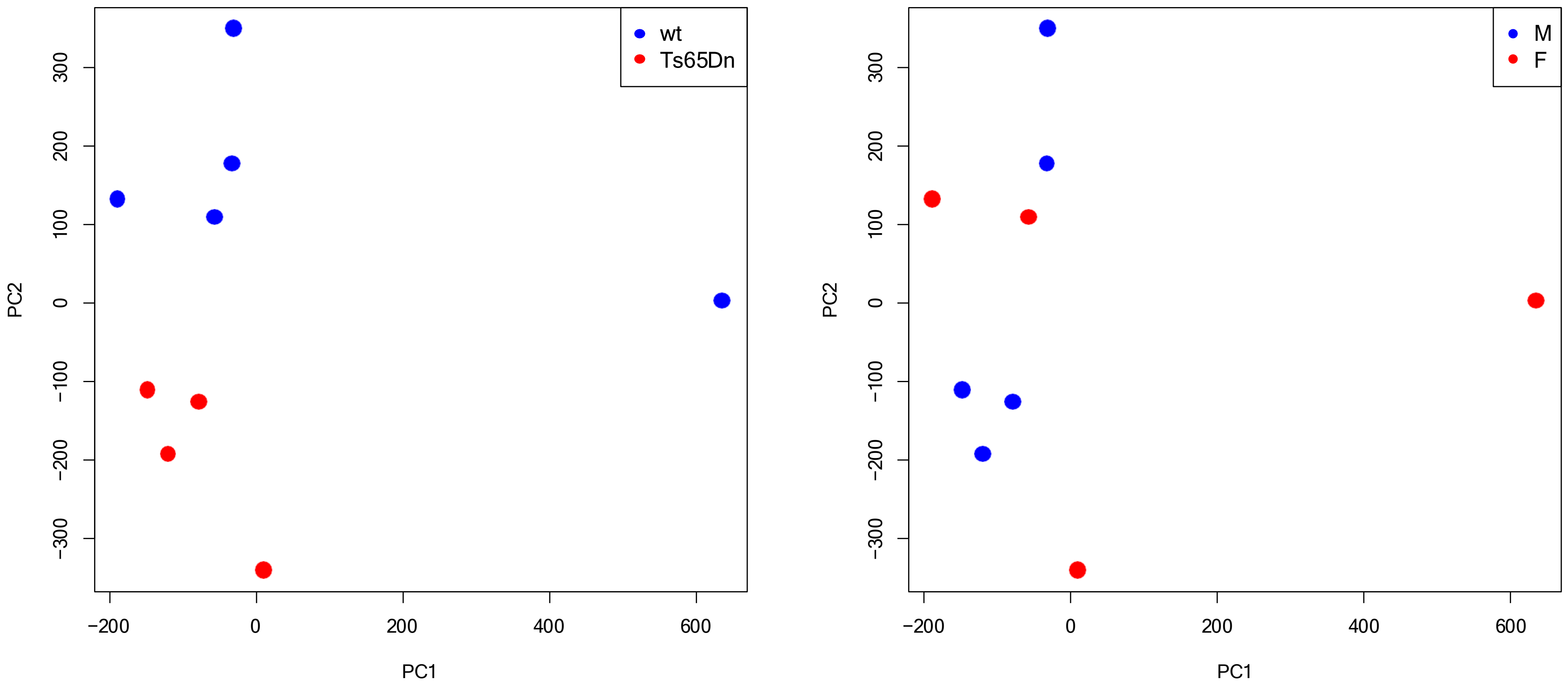
Dimension-reduction techniques



Principal components are patterns that exist across estimates of the most common features.

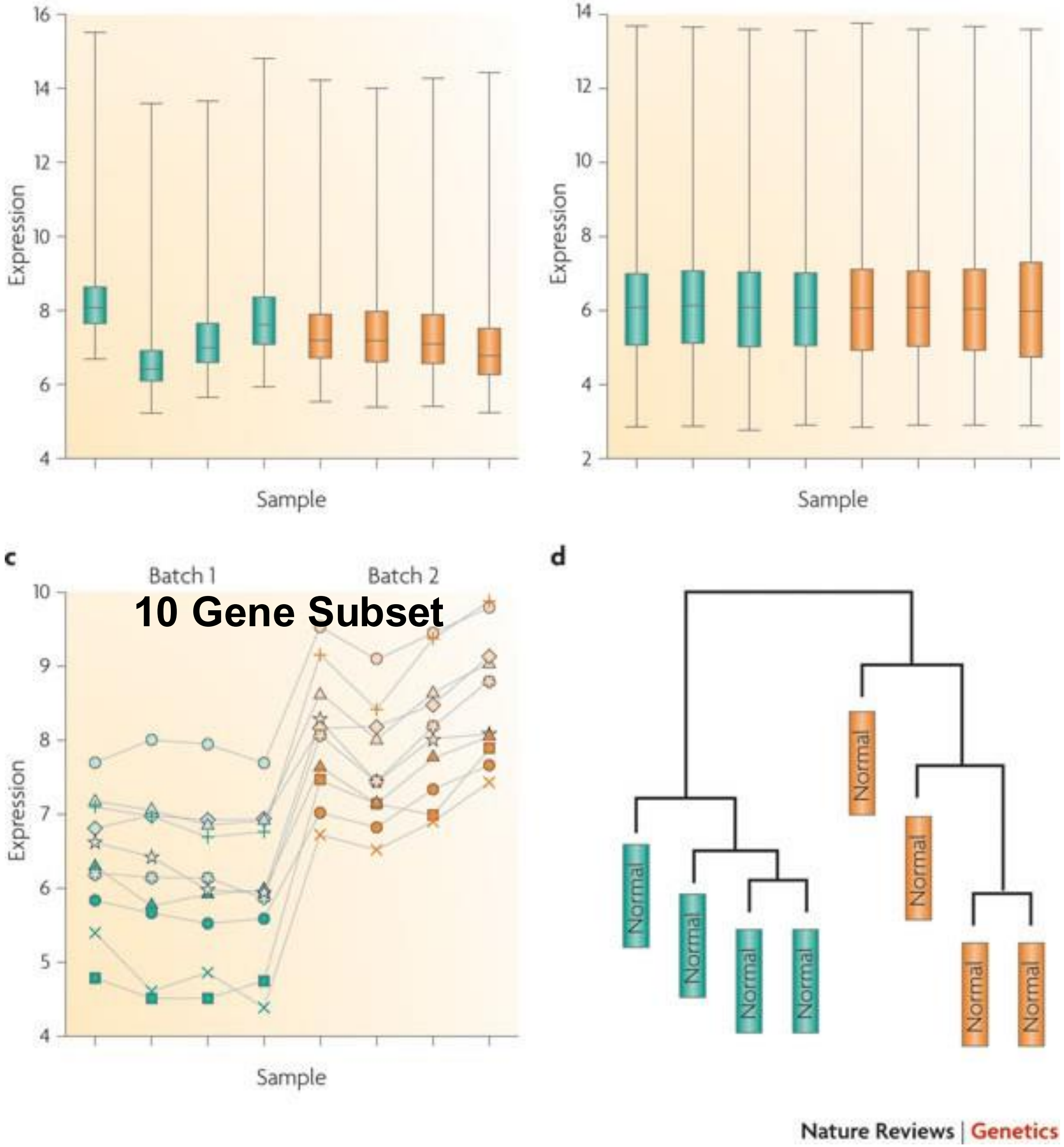
Principal components capture both biological and technical variability

PCA - Biological Variations



Batch effects

Normalization does not always remove batch effects, which can affect specific subsets of genes and may affect different genes in different ways



Leek et al., 2010

Batch effects

How to get rid of batch effects?

- Experimental design: **ALWAYS RANDOMIZE the samples!!!**
- Statistical approaches:

  ComBat package

  Surrogate variable analysis (SVA): estimates the source of unknown variation

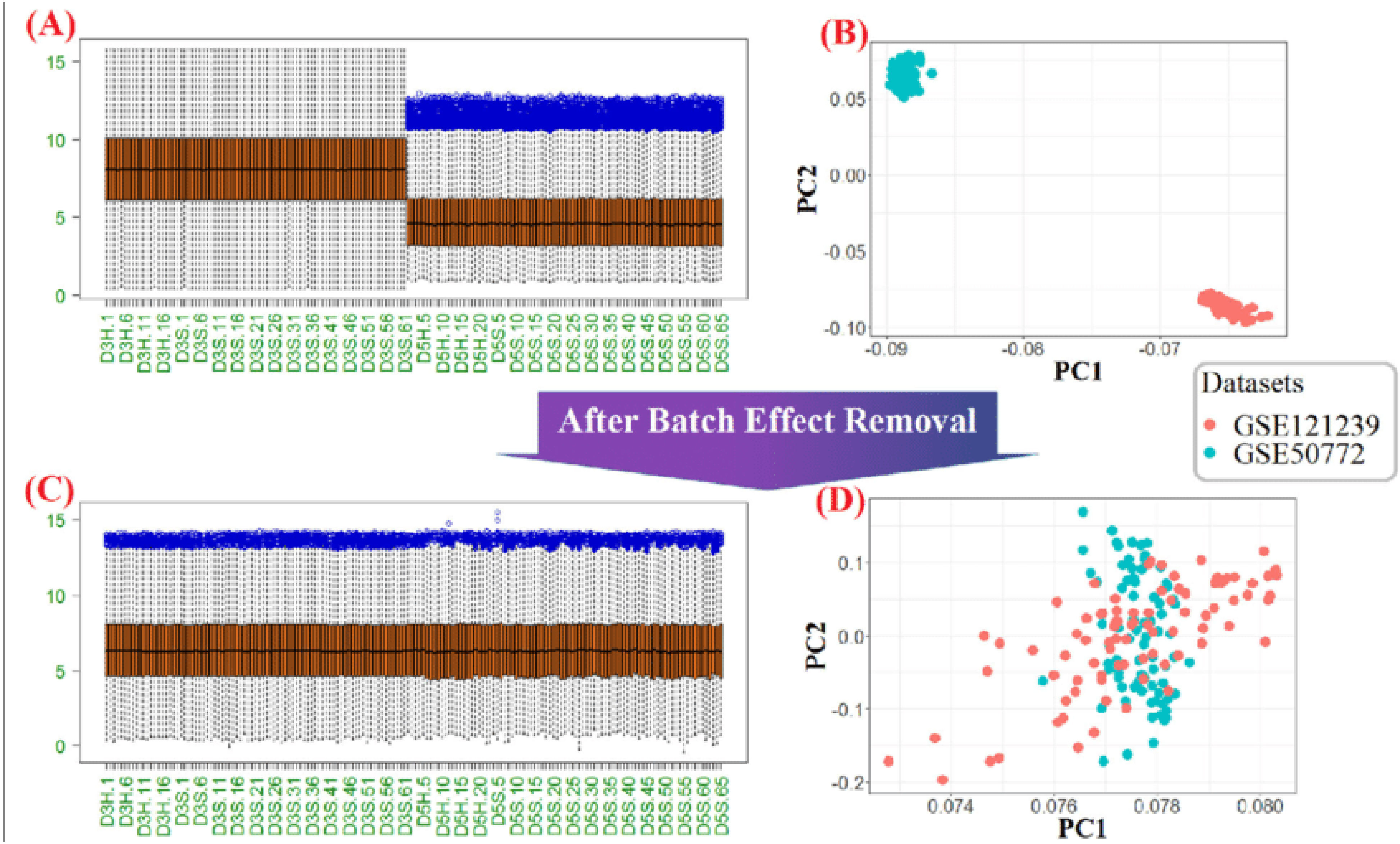
- Add batches as covariates in the statistical tests

An ounce of prevention is worth a pound of cure

Batch effects

ComBat package

Adjust data for known batches using and empirical Bayesian framework



Identification of differentially methylated probes and regions

β-values and M-values

What to Use?

β-values and M-values

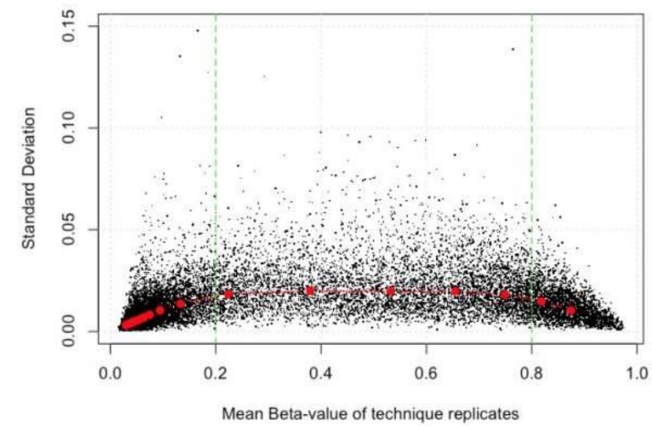
2 studies on 27k:

Du et al., 2010 --> better use M-values than β-values

Zhuang et al., 2012 --> M-values significantly outperform β-values but only if the sample size is small (3 samples)

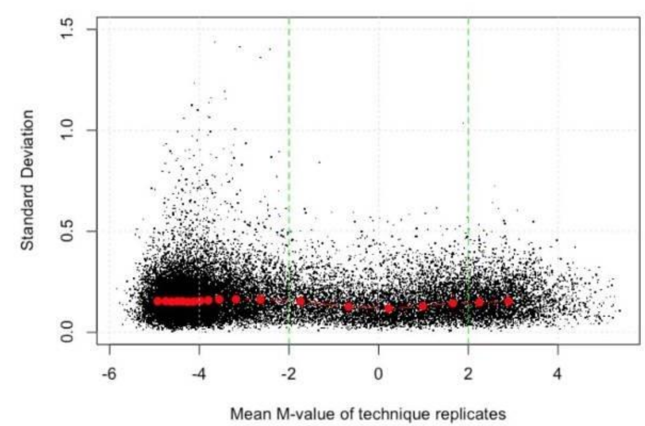
$$\beta = \frac{M}{M+U}$$

Beta values tend to be HETEROSCEDASTIC!



$$M = \log_2 \frac{M}{U}$$

M values tend to be HOMOSCEDASTIC!



β-values and M-values

What to Use?

- Beta-value has a more intuitive biological interpretation
- M-value is more statistically valid for the differential analysis of methylation levels
- M-value may be capturing significant but very small absolute changes in methylation
- One possibility is to use M-value method for conducting differential methylation analysis and including the Beta-value statistics when reporting the results

Our suggestion:

For small datasets (up to 4-5 samples) → M values

For large datasets → beta value or M values

Identification of differential methylation

Parametric or non parametric tests?

Parametric tests → data are sampled from a Gaussian distribution.

Non parametric tests → do not make assumptions about the population distribution. All commonly used nonparametric tests rank the outcome variable from low to high and then analyze the ranks.

Kolmogorov-Smirnoff test → checks whether the distribution of the data significantly differs from a Gaussian distribution.

Identification of differential methylation

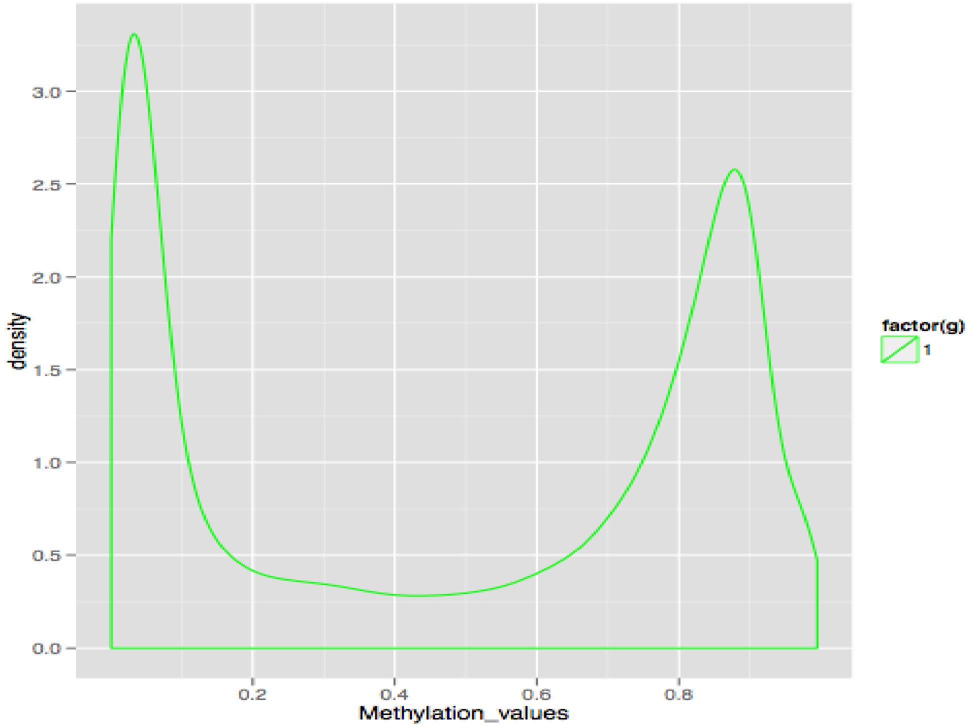
Parametric or non parametric tests?

	Type of Data			
Goal	**Measurement (from Gaussian Population)**	**Rank, Score, or Measurement (from Non- Gaussian Population)**	**Binomial (Two Possible Outcomes)**	**Survival Time**
Describe one group	Mean, SD	Median, interquartile range	Proportion	Kaplan Meier survival curve
Compare one group to a hypothetical value	One-sample *t*test	Wilcoxon test	Chi-square or Binomial test **	
Compare two unpaired groups	Unpaired *t* test	Mann-Whitney test	Fisher's test (chi-square for large samples)	Log-rank test or Mantel-Haenszel*
Compare two paired groups	Paired *t* test	Wilcoxon test	McNemar's test	Conditional proportional hazards regression*
Compare three or more unmatched groups	One-way ANOVA	Kruskal-Wallis test	Chi-square test	Cox proportional hazard regression**
Compare three or more matched groups	Repeated-measures ANOVA	Friedman test	Cochrane Q**	Conditional proportional hazards regression**
Quantify association between two variables	Pearson correlation	Spearman correlation	Contingency coefficients**	
Predict value from another measured variable	Simple linear regression or Nonlinear regression	Nonparametric regression**	Simple logistic regression*	Cox proportional hazard regression*
Predict value from several measured or binomial variables	Multiple linear regression* or Multiple nonlinear regression**		Multiple logistic regression*	Cox proportional hazard regression*

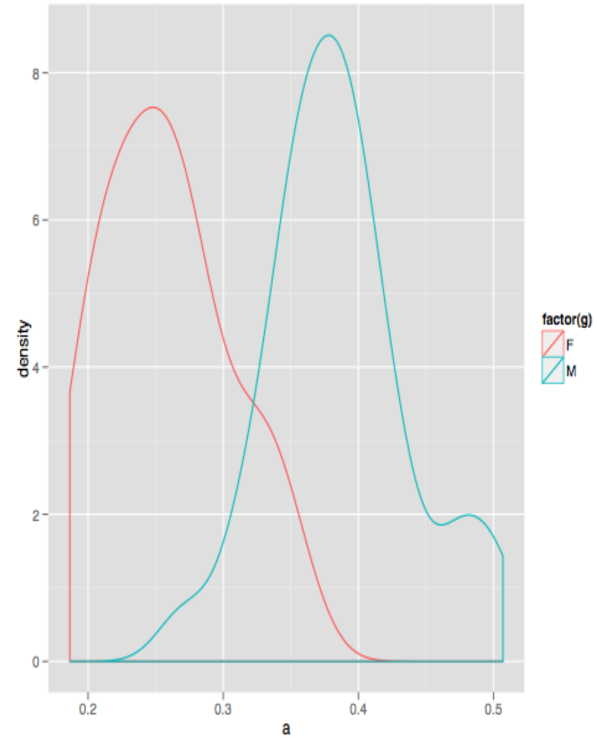
Identification of differential methylation

Parametric or non parametric tests?

Distribution of Beta within
a sample across CpG sites



Distribution of Beta within
a CpG site across samples



Identification of differential methylation

Does it matter whether you choose a parametric or nonparametric test? The answer depends on sample size.

Large sample

Parametric tests are robust to deviations from Gaussian distributions

Nonparametric tests are only slightly less powerful than parametric tests with large samples

Small samples

Parametric tests can lead to inaccurate p-values

Nonparametric tests lack statistical power with small samples

Identification of differential methylation

Limma Package Introduction

Gordon Smyth

23 October 2004, Revised 21 October 2013

Limma is an R package for the analysis of gene expression microarray data, especially the use of linear models for analysing designed experiments and the assessment of differential expression. Limma provides the ability to analyse comparisons between many RNA targets simultaneously in arbitrary complicated designed experiments. Empirical Bayesian methods are used to provide stable results even when the number of arrays is small. The normalization and data analysis functions are for two-color spotted microarrays. The linear model and differential expression functions apply to all microarray technologies including Affymetrix and other single-channel oligonucleotide platforms.

Identification of differentially methylated regions

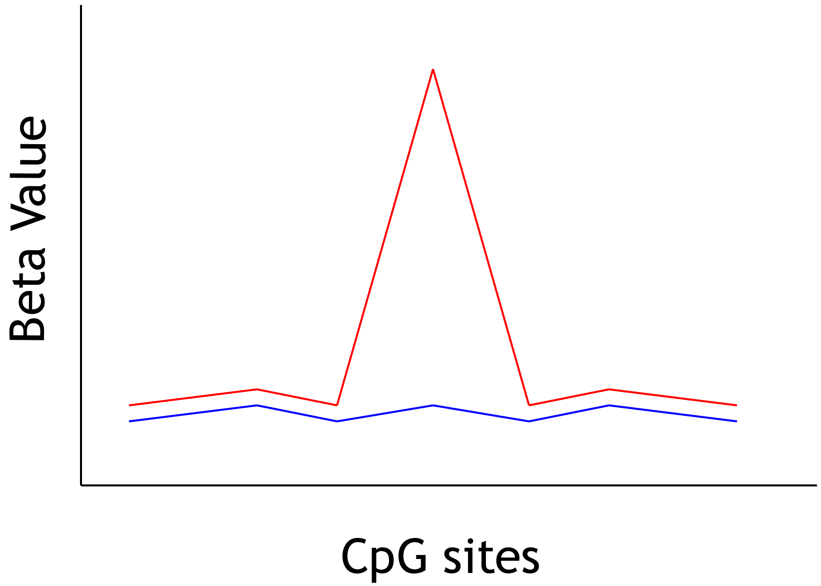
The real challenge is to identify biologically relevant DNA methylation changes and their relative contribution to the phenotype under study.

Changes in DNA methylation, especially in the CpG islands, usually involve **groups of adjacent CpG sites** whose methylation levels are **correlated**, thus potentially affecting chromatin structure. On the contrary the biological relevance of alterations at individual CpGs, although potentially interesting at specific genomic regions, is less characterized.

Differential DNA methylation

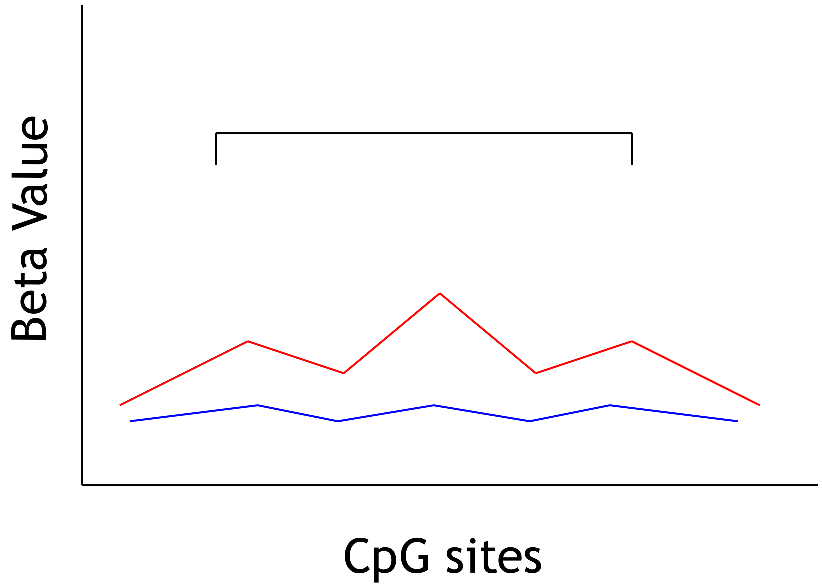
DMP

Differentially methylated **probes**



DMR

Differentially methylated **regions**



Single-probe *versus* region-centric approach

Single-probe approach: compare each CpG probe between the two gropus using parametric or not parametric tests

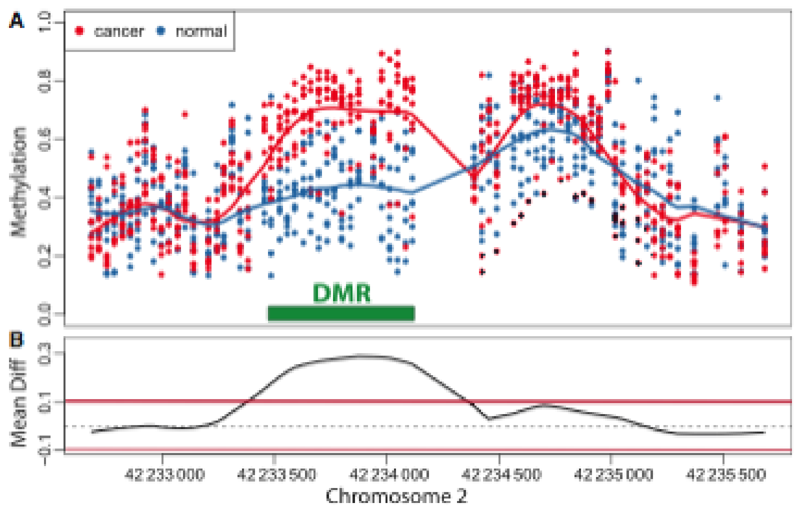
Region-centric approach: compare a group of adjacent CpG probes between the two groups

- minfi → “bump hunting” method
- DMRcate → dmrcate
- comb-p https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3496335/

Bump hunting

Identify **the spatial intervals (bumps)** in which an estimated function (for example, the average difference between two groups) is different from 0

Red and blue curves represent the smooth estimates of the population-level methylation for cancer and normal samples



The black line represent the estimate of the population-level differences between normal and cancer

Candidate DMRs are defined as the regions for which the black curve is outside the threshold

Jaffe A. et al., 2012

combp

> Bioinformatics. 2012 Nov 15;28(22):2986-8. doi: 10.1093/bioinformatics/bts545.
Epub 2012 Sep 5.

Comb-p: software for combining, analyzing, grouping and correcting spatially correlated P-values

Brent S Pedersen [1], David A Schwartz, Ivana V Yang, Katerina J Kechris

Affiliations + expand

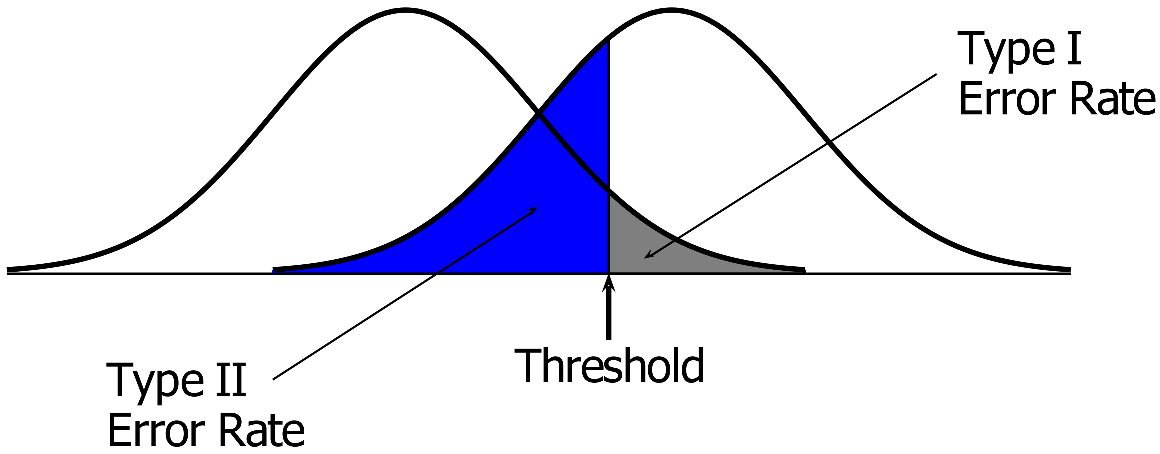
PMID: 22954632 PMCID: PMC3496335 DOI: 10.1093/bioinformatics/bts545

'moving averages' method of *P*-value correction that does not depend on the test used to generate the *P*-values

- calculates correlation among *p-values* of CpGs at varying distance
- uses the calculated correlation to correct CpG pvalues
- assigns FDR-corrected p-values to each region

«A given P-value will be pulled lower if its neighbors also have low P-values (and little auto-correlation) and likely remain insignificant if the neighboring P-values are also high. »

Multiple testing correction



Actual Situation "Truth"

Decision	H_0 True	H_0 False
Do Not Reject H_0	Correct Decision 1 - α	Incorrect Decision Type II Error β
Rejct H_0	Incorrect Decision Type I Error α	Correct Decision 1 - β

$\alpha = P(Type\ I\ Error) \quad \beta = P(Type\ II\ Error)$

Multiple testing correction

p-value cutoff=0.05

Number of comparisons: 485512

We expect to have 5% of 485512 = 24275 probes identified as differentially methylated by chance alone (false positives)!!!

We need to correct for multiple testing!

Adjust p-values in order to maintain the desired level of significance of the identified probes

Multiple testing correction

Cg1 pValue=0.020

Cg2 pValue=0.122

Cg3 pValue=0.004

Cg4 pValue=0.330

Cg5 pvalue=0.001

Multiple testing correction

Family Wise Error Rate (FWER) (for ex, Bonferroni):

It is the probability of incorrectly rejecting even one null hypothesis. In other words, it is the probability of observing at least 1 false positive probe in the outputted list

P of making at least 1 mistake:

Without correction: $1-(1-0.05)^5 = 22.6\%$

With Bonferroni correction: $1-(1\mathbf{-0.05/5})^5 = 5\%$

Bonferroni considers as differential only those probes with p-value < (α/ m), where: m=number of tests; α=0.01, 0.05, etc

Corrected α threshold : 0.05/5 = 0.01

Unadjusted p-value < 0.016

Cg1 pValue=0.020	Reject
Cg2 pValue=0.122	Reject
Cg3 pValue=0.004	Accept
Cg4 pValue=0.330	Reject
Cg5 pValue=0.001	Accept

Multiple testing correction

Family Wise Error Rate (FWER) (for ex, Bonferroni):

It is the probability of incorrectly rejecting even one null hypothesis. In other words, it is the probability of observing at least 1 false positive probe in the outputted list

P of making at least 1 mistake:

Without correction: $1-(1-0.05)^5 = 22.6\%$

With Bonferroni correction: $1-(1-\mathbf{0.05/3})^5 = 8\%$

Bonferroni considers as differential only those probes with
p-value < (α/ m), where: m=number of tests; α=0.01, 0.05, etc

Corrected α threshold : 0.05/5 = 0.01

Unadjusted p-value < 0.016

Cg1 pValue=0.020	Reject
Cg2 pValue=0.122	Reject
Cg3 pValue=0.004	Accept
Cg4 pValue=0.330	Reject
Cg5 pValue=0.001	Accept

Adjusted p-value <0.05

0.020*5 = 0.1	Reject
0.122*5 = 0.56	Reject
0.004*5 = 0.02	Accept
0.330*5 = 1	Reject
0.001*5 = 0.005	Accept

Multiple testing correction

False Discovery Rate (FDR, for example Benjamini & Hochberg):

Rank probes according to their p-values: p(1)<p(2)…<p(N)
Consider as differential probes the highest k that satisfy:

$p(k) < \alpha^*(k/N)$; where p(k) is BH critical value

Cg1 pValue=0.020	k=1, Cg5 pValue=0.001	0.05*(1/5)=0.010 Accept
Cg2 pValue=0.122	k=2, Cg3 pValue=0.004	0.05*(2/5)=0.020 Accept
Cg3 pValue=0.004	k=3, Cg1 pValue=0.020	0.05*(3/5)=0.030 Accept
Cg4 pValue=0.330	k=4, Cg2 pValue=0.122	0.05*(4/5)=0.04 Reject
Cg5 pvalue=0.001	k=5, Cg4 pValue=0.330	0.05*(5/5)=0.05 Reject

Multiple testing correction

False Discovery Rate (FDR, for example Benjamini & Hochberg):

Rank probes according to their p-values: p(1)<p(2)…<p(N)
Consider as differential probes the highest k that satisfy:

$$p(k) < \alpha^*(k/N)$$; where p(k) is BH critical value

		BH-corrected pValue	BH critical value
Cg1 pValue=0.020	k=1, Cg5 pValue=0.001	0.05*(1/5)=0.010 Accept	0.001*(5/1)=0.001 Accept
Cg2 pValue=0.122	k=2, Cg3 pValue=0.004	0.05*(2/5)=0.020 Accept	0.004*(5/2)=0.01 Accept
Cg3 pValue=0.004	k=3, Cg1 pValue=0.020	0.05*(3/5)=0.030 Accept	0.02*(5/3)=0.03 Accept
Cg4 pValue=0.330	k=4, Cg2 pValue=0.122	0.05*(4/5)=0.04 Reject	0.122*(5/4)=0.15 Reject
Cg5 pvalue=0.001	k=5, Cg4 pValue=0.330	0.05*(5/5)=0.05 Reject	0.330*(5/5)=0.330 Reject

Multiple testing correction

FWER control:

Ensure very low probability for having any false positive (less than α) → very clean list of differential probes

Limit: the list usually contains very few features… unacceptable high rate of false negatives

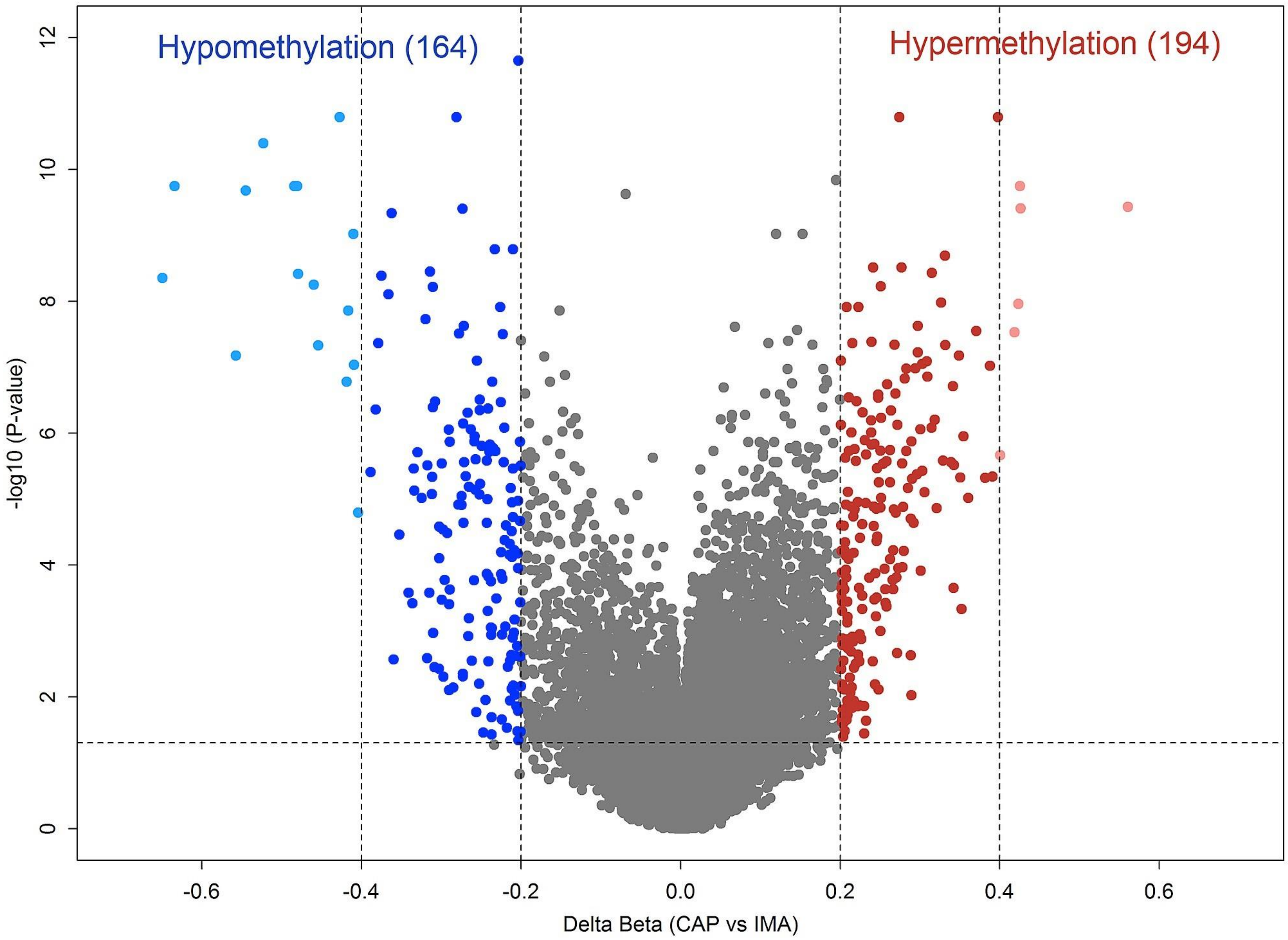
Choose control of the FWER if high confidence in all selected genes is desired.

FDR control:

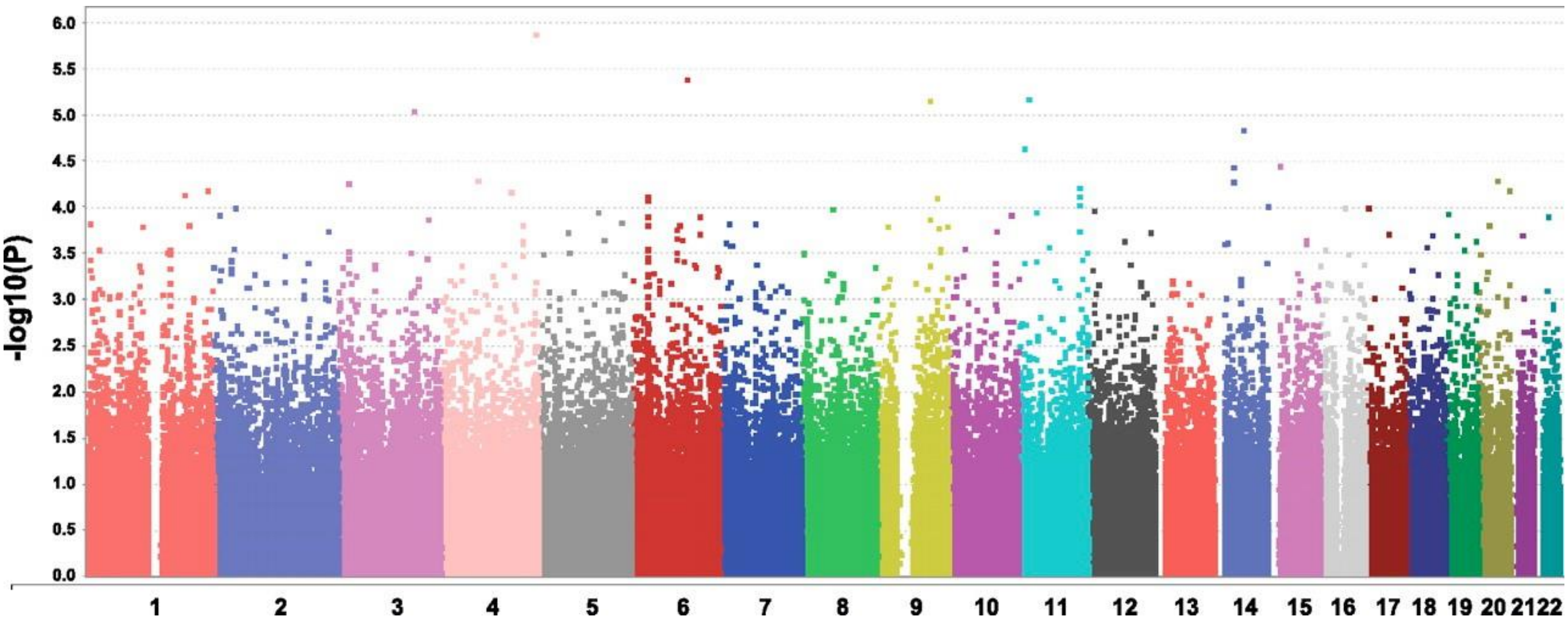
Choose control of the FDR if a certain proportion of false positives is tolerable.

Visualization of the results of DNA methylation

Visualize differential methylation – Volcano plots

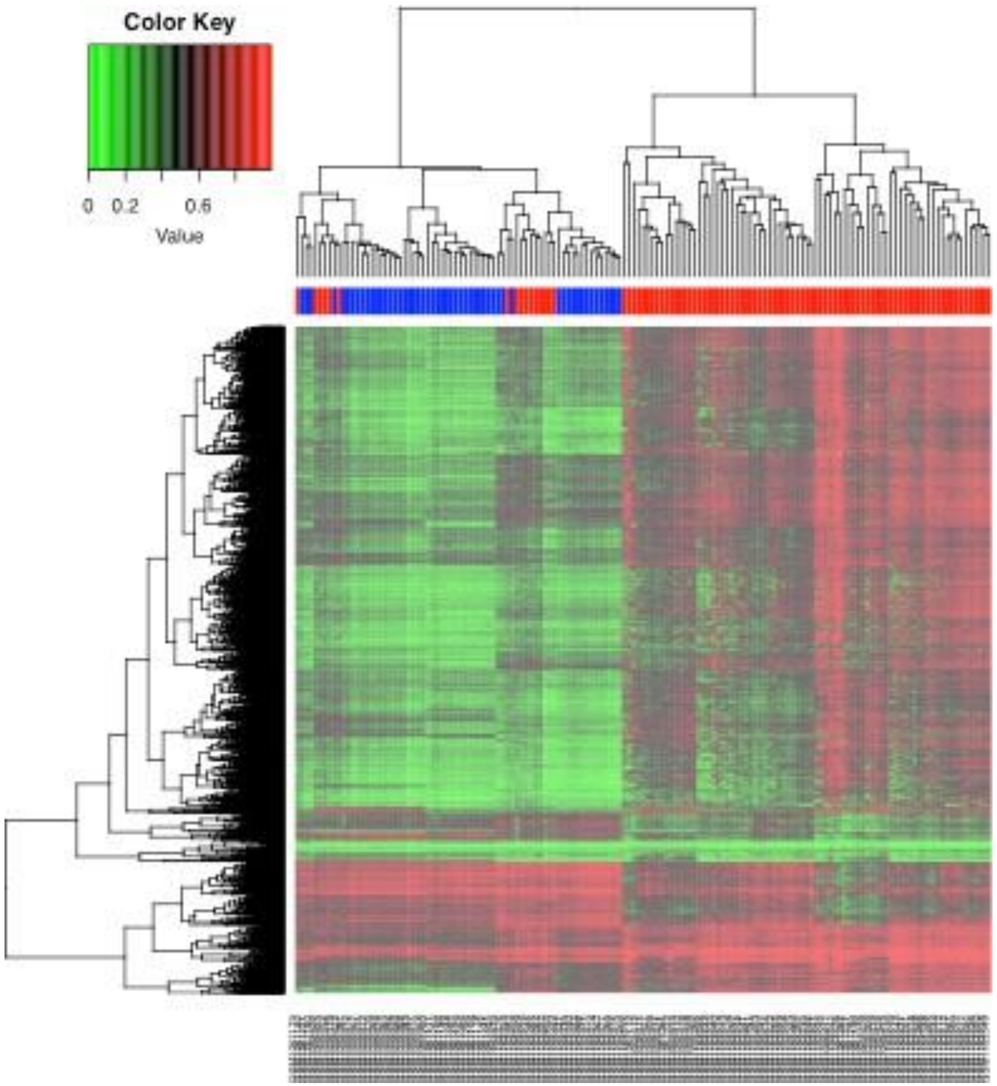


Visualize differential methylation – Manhattan plots



Heatmaps

A heat map is a graphical representation of data where the individual values contained in a matrix are represented as colors.



Heat map + hierachical clustering

Clustering Method

Aggregate similar samples into groups

AGNES agglomerative nesting hierarchical clustering	**DIANA** divisive analysis clustering
bottom-up	top-down
• Starts from single elements and groups them according to the algorythm of choice. • Proceed iteratively by joining similar clusters till all elements are within the same group (ROOT)	• Starts by considering all the elements as belonging to a single group and separates the most different clusters according to the algorythm of choice • Proceeds iteratively untill each group contains a single element (LEAVES)
Good for small clusters	Good for large clusters

Hierarchical clustering

1. Calculate **distances** between samples/probes
2. Search through the distance matrix and find the two most similar clusters according to a **linkage method**
3. Fuse the two selected clusters to produce a new cluster that now contains at least two objects
4. Calculate the distances between this new cluster and all other clusters. There is no need to calculate *all* distances since only those involving the new cluster have changed

ADVANTAGES:

- Simple, easy to understand
- Widely used
- Can cluster both probes and samples

DRAWBACKS:

- Tends to be sensitive to small changes in the data

Hierarchical clustering - distance calculation

$$d_{ij} = \sqrt{\sum_{k=1}^{n}(x_{ik} - x_{jk})^2}$$

Straight-line distance between two points in Euclidean space.
Euclidean distance works well with continuous data

:

$$d_{ij} = \sum_{k=1}^{n}|x_{ik} - x_{jk}|$$

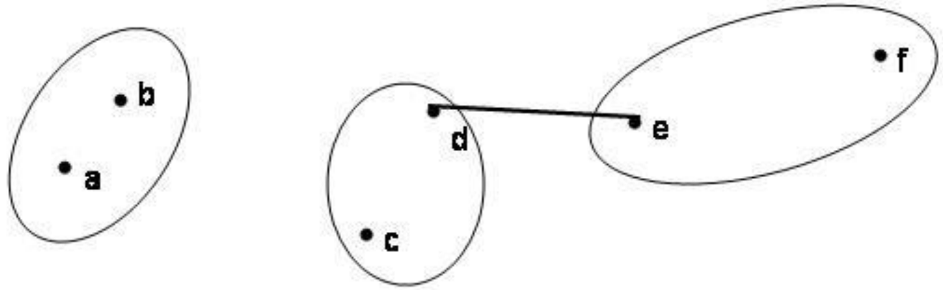
Sum of the absolute distances.
Treats all differences linearly. It less sensitive to outliers, providing a more robust measure in the presence of anomalies.
With high-dimensional data, Manhattan distance tends to be more meaningful than Euclidean distance

$$d_{ij} = \max_k |x_{ik} - x_{jk}|$$

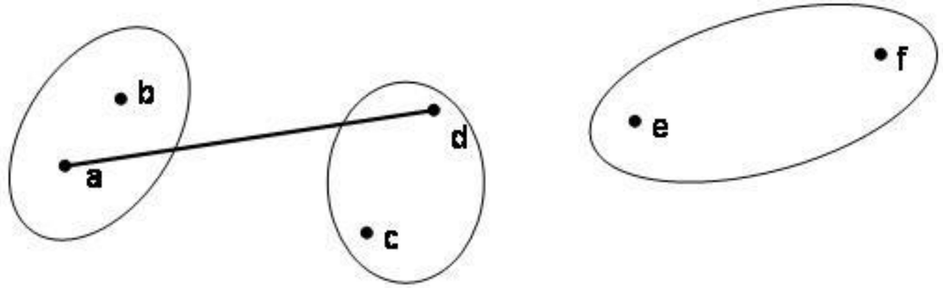
Maximum absolute difference between the coordinates of a pair of points.
Highlights the most remarkable discrepancy between points. It is useful when you're interested to highlight the largest difference between points in a single dimension

Hierarchical clustering – linkage methods

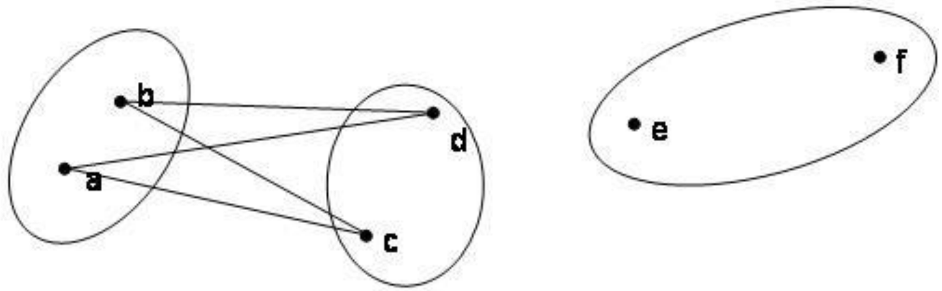
- Single Linkage: join clusters whose distance between closest genes is smallest



- Complete Linkage: join clusters whose distance between furthest genes is smallest



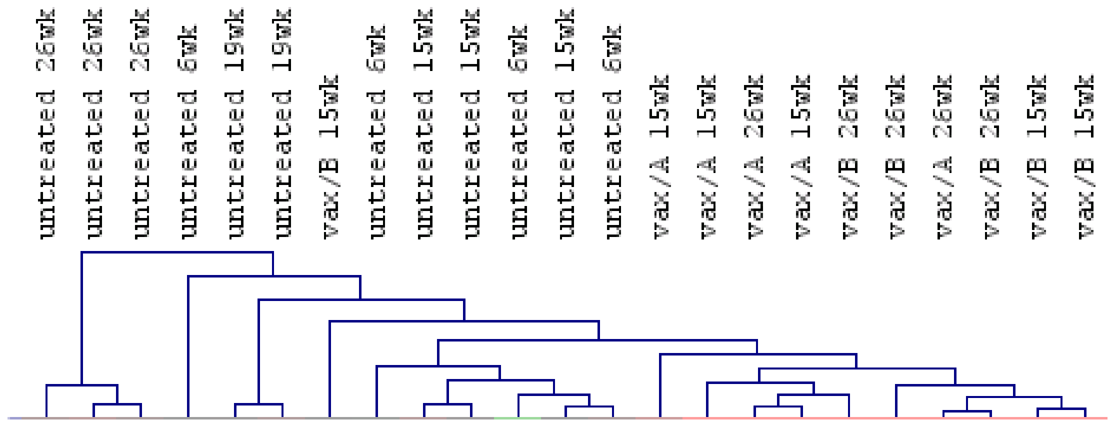
- Average Linkage: join clusters whose average distance is the smallest.



Hierarchical clustering – linkage methods

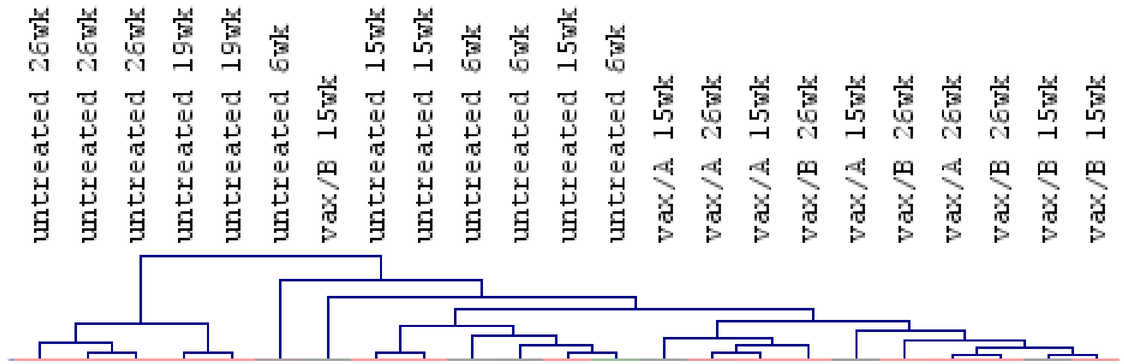
- Single Linkage:

It tends to produce long, “loose” clusters (long chains).



- Complete Linkage:

It tends to produce more compact clusters.



- Average Linkage:

The results tend to be in between the two other methods